Cardinal – Draft v0.1 (Oct 23, 2026)

Version note. v0.1 is the first de facto book draft: the eight Cardinal podcast scripts (Episodes 1–8) converted into book prose, one Part per episode, on the 8-part spine from Draft v1.1. Converted by Gemini 4 Argon (Batch API) on Oct 23, 2026. About 57,009 words. `[NEEDS SOURCE]` marks claims to verify. Contested claims, including Jane Stanford's 1905 death, are framed as contested. Source of record: branch `draft/v0.1`, folder `drafts/v0.1/` in the cardinal repo (with CHANGELOG and OPEN_ISSUES). Older drafts are untouched.

Part 1: The Ledger

1769–1848

The Tall Stick (1769–1776)

Along the lower San Francisquito Creek, where the Stanford campus now sits, lived a community the Spanish records call the Puichon. They were one of roughly forty small, independent tribal groups that outsiders would later lump together as Ohlone, or, in older books, Costanoan, from the Spanish word for coast people. Each group held a small territory, usually a few hundred people in a cluster of villages, connected to neighbors by marriage, trade, and language.

At Jasper Ridge, the biological preserve that Stanford owns in the hills just west of campus, the university's own chronology records more than five thousand years of human occupation. Spanish, Mexican, and American rule together, from 1769 to 2026, is two hundred fifty-seven years. That is about five percent of the human story on this land. If the whole five thousand years were a single day, Europeans would have walked onto that plain at about a quarter to eleven at night.

Before there was a tall stick, there were people who did not need a name for it, because they knew exactly where they were.

Ohlone communities did not plant fields of wheat. Later American newcomers looked at that and called them lazy, or worse, diggers, for the sticks they used to pry up edible bulbs. The settlers got this completely wrong. Malcolm Harris points out what they missed. The Ohlone moved, he writes, "in response to abundance rather than scarcity" (Harris 2023, ch. 1.1). The ripening of acorns, the spawning of fish, the arrival of birds on the bay marshes: short bursts of intense work, followed by long stretches of what Harris calls "social luxury" (Harris 2023, ch. 1.1).

That is not the absence of an economy. It is an economy built on knowledge instead of storage, on knowing which oak drops first and which creek runs in which month. California before 1769

held one of the densest concentrations of human linguistic diversity anywhere on earth. Historians' usual estimate is more than five hundred distinct tribal groups, speaking about a hundred languages. The total population of what the Spanish called Alta California was just over three hundred thousand. About forty thousand of them lived in the Bay Area alone.

Harris opens his history of Palo Alto with this sentence: "To speak of the Ohlone is to speak of the broken link between the Bay Area's indigenous people and the Bay Area itself" (Harris 2023, intro.).

Some places change rulers and keep their people: the language, the place names, the memory of where the water is. Other places rupture. The people who knew the land are separated from it, and the new arrivals start the clock over.

On the peninsula of Baja California, below today's border, the Jesuits had built a chain of missions. Then, in 1767, King Charles III of Spain expelled the entire Jesuit order from his empire. The man sent to carry out the expulsion in Baja was the new governor, Gaspar de Portolá. The Jesuits gathered at Loreto and left for exile in February 1768. A month later, the king's inspector general, José de Gálvez, handed their missions to the Franciscans, under a friar named Junípero Serra.

Junípero Serra was born Miguel José Serra on November 24, 1713, in Petra, on Mallorca. He held a chair of philosophy at the university in Palma and gave it up in 1749. He walked the Mexican and California coast on a leg injured years earlier, on the road up from Veracruz. A year after the Jesuit-to-Franciscan handoff in Baja, in 1769, Gálvez sent Portolá and Serra north, by land and by sea, to occupy San Diego and Monterey before the Russians or the British could.

In October 1769, marching north near today's Watsonville, the friar who kept the expedition's most detailed diary, Juan Crespí, described very high trees of a red color, not known to them, and wrote that because none of the expedition recognized them, they were named redwood from their color. That is the first written mention of the coast redwood.

November 1769. Sixty-four men, by the standard count of the party that left San Diego on July 14, and a string of mules came down onto a wide valley at the south end of a bay no European had ever seen from land. The party held Spanish soldiers, two Franciscan friars, an engineer, Christian Indians from the Baja missions, and their commander, Gaspar de Portolá. They had been sent to find Monterey. They walked past it without recognizing it.

On Saturday, November 4, their scouts climbed a ridge and saw a body of water nobody expected. That ridge is called Sweeney Ridge now. It sits between San Bruno and Pacifica, inside the Golden Gate National Recreation Area, with a discovery-site marker on the trail. The expedition diary records that November 4 was the feast day of San Carlos Borromeo, the saint in whose honor they had come north to found a presidio and a mission at Monterey, and also

the name-day of the king, Carlos III. They said Mass on the beach. They ate mussels off the rocks. Then they turned up into the hills, and from the summit they saw what one of them called an immense arm of the sea.

A day later, on November 5, Portolá's own diary recorded the view flatly. Before them "extended a great arm of the sea, sixteen to twenty leagues" in extent.

On November 6 they made camp beside a creek that runs down from the hills to the bay. They stayed five nights while the scouts went out again. On November 11 a council of officers voted to turn around.

The story goes that they camped beneath one enormous redwood, a tree so tall and alone on the plain that the Spanish called it *el palo alto*. The tall stick.

The story goes means legend. The record shows means a document. And the record on this tree is more interesting than the story.

Even Malcolm Harris, who grew up in Palo Alto, has them camped five days under that tree. The diaries support the five nights by the creek. They do not name this tree.

Portolá went south, returned the next year, founded Monterey on June 3, 1770, sailed away on July 9, and never came back. He later governed Puebla, in Mexico. The story goes that he discovered the bay. The record shows a soldier carrying out Gálvez's plan. In 1770 Serra's headquarters, Mission San Carlos Borromeo, was founded at Monterey and then moved to Carmel.

The first time the tall redwood on San Francisquito Creek shows up in writing is five years after Portolá's camp, in 1774, when another friar, Francisco Palóu, put up a wooden cross near a big tree on the creek and marked it as a possible mission site.

Still, no European ship had sailed into the water the scouts had seen from Sweeney Ridge. Ships on the Manila trade had been passing the California coast for two hundred years. They knew Point Reyes. They knew the Farallon Islands, thirty miles out. They had not known there was a bay behind the cliffs.

August 5, 1775. Night. A small Spanish supply ship called the *San Carlos*—carrying the saint's name from the day of the 1769 sighting—a packet built at the naval base of San Blas, felt her way through a strait that is, at its widest, only about three miles across, and often hidden in fog. The *San Carlos* was the first European keel in the water. Her commander, Lieutenant Juan Manuel de Ayala, had been shot in the right foot on April 3, by a loaded pistol his predecessor left in the cabin. He wrote in his report that he could not do the work in person. His pilot, José de Cañizares, took the launch and drew the first working map of the bay, south to what is now Suisun. Ayala arrived the night of August 5, stayed forty-four days, and called it one of the best

harbors on the coast. They anchored by an island he named Isla de los Ángeles. Today the cove is Ayala Cove; the island is Angel Island. On September 18, rudder damaged, they ran down to Monterey and arrived the next day. Without that chart, the presidio and the mission of 1776 would have been a guess.

In March 1776, the commander Juan Bautista de Anza and Father Pedro Font rode out to inspect the San Francisquito Creek site Palóu had marked with a cross. Font measured the tall redwood with a surveying instrument: fifty varas high, about a hundred and thirty-seven feet. (Some local historians think Portolá's actual 1769 campsite tree was a different one, farther downstream, since cut down.) Then Anza and Font looked at the creek, decided its flow was too unreliable to water mission fields through a dry summer, and moved on.

That year, 1776, the Spanish founded the Presidio of San Francisco on its cliff and Mission San Francisco de Asís, named for Saint Francis of Assisi, the founder of the Franciscan order. Almost everyone came to call it Mission Dolores, after a nearby creek and lagoon the Anza expedition named for Our Lady of Sorrows, Nuestra Señora de los Dolores.

Baptismal Register (1777–1832)

January 12, 1777. About fifteen miles southeast of the rejected creek with the tall redwood, on the bank of the Guadalupe River, near an Ohlone village the Spanish wrote down as So-co-is-u-ka—laurel wood—Franciscan friars planted a cross and put up a shelter for worship. They named the place Misión Santa Clara de Thámien, after Saint Clare of Assisi and after the local people, the Thámien. It was the eighth mission in Alta California, and the first one named for a woman.

The very first plan for a big institution on San Francisquito Creek had been a mission. More than a century later, someone would try again, with a railroad fortune instead of a royal charter. Santa Clara itself kept moving; floods, fire, and an earthquake eventually forced six rebuilds of the church.

Every person who came into a mission was written into a book. A baptismal book. A marriage book. A burial book. The line gave a Christian name, the parents, the village, a godparent, and the minister's signature. That was the shape of the form. The books themselves were kept with real care, which is why historians can count what happened, and why Ohlone families today, including the Muwekma Ohlone Tribe, can trace their ancestors through the baptismal registers of the Spanish missions at San Francisco and Santa Clara. For the people who were here first, the paper trail of their own ancestry begins in a book written by the people who came to change them.

A California mission was four institutions in one.

First, it was a church. Its stated purpose was conversion, to bring Native people to Catholic Christianity. A baptized Native person was called a *neófito*, a neophyte, a new believer. An unbaptized one was a *gentil*, a gentile.

Second, it was an economy. A mission was a farm, a ranch, a workshop, and a warehouse. California was months by sea from Mexico, and supply ships were small, so the friars needed the neophytes to grow the wheat, tend the cattle, weave the cloth, tan the leather, and render the tallow.

Third, it was a military partner. Every mission sat in the district of a presidio, a fort with a small garrison of soldiers. The soldiers protected the friars, and they also went out to bring back people who left.

Fourth, it was a labor system. Once baptized, a neophyte was expected to live at the mission, under the friars' rules, and could not simply walk away. People who left without permission were treated as runaways, and soldiers were sent to bring them back. Santa Clara University's Ohlone heritage project puts it plainly: Native people from dozens of communities were brought to live together at the mission, and, "laboring without pay, these Native Californians built the mission structures and provided the agricultural and industrial workforce to support Spain's colony in California." Neophytes were not bought and sold, and some came, at least at first, for food, trade goods, or safety. What is not in dispute is that they could not leave freely, they were not paid, and they were physically punished.

A rider leaving Mission Santa Clara heading north with a pouch of letters followed a path that had been a footpath long before he was born, an Ohlone trail along the edge of the hills, above the marshes. He passed the creek with the tall redwood. He reached Mission San Francisco de Asís, about forty-five miles up the peninsula. From there, letters went on by ship, or south by horse to Monterey, the capital, and from Monterey down the coast to San Diego, and eventually to Mexico City. Spanish California's long-distance freight came by sea, on supply ships from San Blas, in Mexico, and later on foreign traders. The road was a corridor for friars, animals, tools, letters, orders, and very slow news. There was no single El Camino Real in Spanish times; *camino real* meant royal road, and there were many *caminos reales*, a general term for roads used under the Crown's authority, most of them following older Native trade routes. The road's one hard fact was its slowness. Mexico City was months away, so whoever controlled the landings and the stamps held everyone else.

Junípero Serra served as president of the California missions from 1769 until his death on August 28, 1784, at Mission San Carlos Borromeo in Carmel, when only nine of the twenty-one missions existed. He was buried there. His successor, Fermín Francisco de Lasuén, founded nine more missions between 1786 and 1798, including La Purísima in 1787. By the time all twenty-one missions existed, they stretched from San Diego, founded in 1769, to Sonoma, founded in 1823, about six hundred miles, roughly the distance from London to Milan.

In 1811, north of Spanish California, John Jacob Astor's Pacific Fur Company built Fort Astoria at the mouth of the Columbia River, the first American settlement on the Pacific. Astor had looked west and tried to own it by trade.

Ten years later, in 1821, Mexico won its independence from Spain, and with it, the far-off province of Alta California became Mexican. The year after that, in 1822, an old artilleryman from the Presidio of San Francisco named José Peña began grazing cattle on the ground south of San Francisquito Creek with the mission's permission. Two years later, in 1824, Ohlone people showed a man named Secundino Robles where they dug a red pigment rock south of San José. It was cinnabar, the ore of mercury—quicksilver.

The new Mexican republic had a problem with the missions. They were run by Spanish-born friars loyal to a church still tied to Spain, and they controlled the best land in California, along with most of its labor. Mexican liberals believed in separating church power from the state. Ambitious soldiers and settlers in California wanted that land. And some officials genuinely believed the neophytes ought to be free citizens.

The word was secularization, from the Latin *saeculum*, the world, as opposed to the cloister. A secular priest serves a parish under a bishop. A regular priest, like a Franciscan, lives under a religious order. To secularize a mission meant, on paper, to turn it into an ordinary parish and to release its lands, herds, and people.

It came in steps. On July 25, 1826, California's first Mexican-born governor, José María de Echeandía, issued a proclamation of emancipation that let qualified neophytes in the southern districts leave mission rule and become Mexican citizens.

At Mission Santa Clara, the Native population peaked in 1827 at about 1,464. (In the fall of 2025, Stanford University enrolled 1,866 new first-years; Santa Clara's peak was smaller than one incoming college class living in one compound, working the fields without pay.) Across the mission chain, the death rate for children was especially high. At Mission San Gabriel, one missionary reported that three out of four children died before the age of two. Measles epidemics in 1806 and 1828 swept through the crowded compounds. The missions kept needing new converts because the people already there kept dying.

In 1831, about 18,683 Native people were still under mission control across California, compared with 4,342 soldiers, settlers, and what the count called other classes. Four to one.

By December 31, 1832, at the peak of the system, the missions together held about 151,000 head of cattle and nearly 138,000 sheep, and controlled roughly one-sixth of all the land in Alta California. From 1769 through December 31, 1832, the friars across all the California missions recorded 87,787 baptisms. They also recorded 63,789 deaths. For every ten people baptized, there were about seven burials. Santa Clara was the busiest of all. It recorded more baptisms,

more marriages, and more burials than any other mission in the chain: 8,536 baptisms and 6,809 burials. In 1832, 1,125 Native people lived at Mission Santa Clara.

Disease, Harris writes, rather than extermination campaigns, killed about half of the roughly three hundred thousand Native people of Alta California under Spain and Mexico. The mission was also a ledger: the first written record, for thousands of people, of their own names.

Silver and Hides (1833–1847)

On August 17, 1833, the Mexican Congress passed the decree for the secularization of the missions of California. On August 9, 1834, Governor José Figueroa issued the rules for how it would actually work. Each head of a family, and everyone over twenty, would receive a lot of mission land, not more than four hundred varas square and not less than a hundred. A vara is a Spanish yard, about thirty-three inches, so that plot came to somewhere between about seven and twenty-eight acres. They would also receive half the livestock, shared out, and half the tools, the seed, and the movable goods, or less.

On paper, it was one of the most sweeping land redistributions in North American history.

On paper.

During the mission era, the land was supposed to be held in trust by the friars, for the benefit of the Native people, until they were judged ready to manage it themselves. As Santa Clara University's Ohlone heritage project puts it, the lands "were supposed to be held in trust by the Franciscans on behalf of Native people." A trust is a promise. A title is a deed with your name on it. And when the moment came to turn the trust into titles, the titles mostly went to someone else.

That same August of 1834, a Boston boy at Harvard named Richard Henry Dana signed on as a common sailor aboard the brig *Pilgrim*, bound around Cape Horn for California. Measles had damaged his eyes so badly he could not read, and his family could not afford Europe. The *Pilgrim* was going to California to collect cowhides.

The missions and the ranches ran enormous herds. There was no railroad and no bank, and beef could not reach a market, so the skins were the crop. Boston houses sent ships around South America with manufactured goods and swapped them for hides, which went home to become shoes.

Writing about Monterey in 1835, Dana observed: "They have no circulating medium but silver and hides, which the sailors call California bank notes."

The price of a hide was fixed by custom at two dollars, roughly sixty-five dollars today. Josiah Royce, born in a California mining town in 1855 and writing in 1886, noted that the traders took them at that uniform value. A hide was money. Rancheros paid their debts in hides.

Dana noticed the prices going the other way. "Their hides, too, which they value at two dollars in money, they give for something which costs seventy-five cents in Boston; and buy shoes, like as not made of their own hides, and which have been carried twice around Cape Horn, at three or four dollars," he wrote. Things sold "at an advance of nearly three hundred per cent upon the Boston prices."

A cow was skinned in California. The hide became a shoe in Boston. The shoe sailed back, and the person who raised the cow paid several times the Boston price. In Mexican California, there was one official custom-house, at Monterey. Every trading ship had to enter its cargo there and pay duties before it could trade anywhere on the coast. There were very few safe harbors—Dana counted San Francisco, Monterey, and San Diego as the only ports safe from the winter southeasters—and very few ships. If you lived inland, you bought what the ships brought, at the ships' price, or you went without. At Santa Barbara, where the ship anchored three miles offshore and the town was nearly a mile farther inland, Dana wrote that so far as the crew were concerned, the town "might almost as well have been in the middle of the Cordilleras." One way in, one way out, and whoever held the gate set the price.

On the other side of that trade were the Californios. Royce collected the memories of people who knew them, writing that the white population was made up mainly of Spanish and Mexican colonists "whose chief industry was raising cattle for the hides and tallow, and whose private lives were free, careless, and on the whole, as this world goes, moderately charming and innocent." They were, he wrote, "gay and jovial, full of good fellowship and hospitality." They paid the Yankee traders a little slowly, but generally paid in the end, in hides. Missing from that picture were the Native vaqueros and laborers who did most of the work of raising those cattle. As Royce wrote in the same passage: "The mission Indians had in large part disappeared."

Meanwhile, secular officials called *administradores*—administrators—ran the handover of the missions. Dana saw the system as it was ending and had a sailor's contempt for them. The law declared all the Indians free and independent rancheros, he wrote, but the change in their condition was only nominal; the neophytes were "virtually slaves," as much as they ever were. And the missions' great possessions were given over to be preyed upon by "the harpies of the civil power," the administrators, who usually ended, in a few years, by making themselves fortunes.

In 1835, on the east side of San Francisquito Creek, a neighboring ranch called the Rinconada del Arroyo de San Francisquito—about 2,229 acres—was granted to Rafael Soto, whose father had come north with the Anza expedition. That land is now, roughly, downtown Palo Alto, and the tall redwood stood at its northwest corner.

In December 1835, Dana's ship sailed into San Francisco Bay. Past the Presidio on its cliff, there was a crumbling mission, a few ranches, and a newly begun settlement, mostly of Yankee Californians, called Yerba Buena—good herb, for a wild mint that grew on the hills. A Russian brig from Sitka, in Russian Alaska, was anchored there, loading tallow and grain. Looking at the bay, Dana wrote: "If California ever becomes a prosperous country, this bay will be the centre of its prosperity."

At San Diego, Dana's company had a hide-house built to hold forty thousand hides, and the crew would go in and stare at it, calculating how long it would take to fill. On May 8, 1836, the ship he sailed home on had finally loaded, in his words, "our forty thousand hides, thirty thousand horns," along with barrels of otter and beaver skins. Forty thousand hides at two dollars was eighty thousand dollars—roughly $2.6 million today. One ship. One voyage. Dana reached home in September 1836, finished Harvard, and in 1840 published *Two Years Before the Mast*. For a generation back east, it was the picture of California.

That same year, 1836, secularization began at Mission Santa Clara, and in Monterey a local revolt against a governor sent from Mexico City brought a Californio named Juan Bautista Alvarado to power. Born in Monterey in 1809, Alvarado was about二十-seven when he rose to the governorship.

Now the mission lands moved into a new kind of property. To get a rancho, a citizen petitioned the governor. If the petitioner could not write, a neighbor wrote it. Attached to the petition was a *diseño*, the drawing, a map with no surveyor behind it: a creek bend, a big oak, a hill. The governor asked the local officials and the mission whether the land was free and whether the applicant owned cattle. If they said yes, he issued a *título*, a title. Then a local official walked the bounds with the grantee and put him in juridical possession. The oak was the corner. The neighbors were the survey.

What was missing was a precise line. A grant might say two leagues, more or less. Under Mexican law a governor could grant as little as a third of an acre or as much as eleven square leagues, about fifty thousand acres. Two leagues was most common. In a country with plenty of land and few people, *more or less* was fine. Your neighbors knew where the oak was.

Up the peninsula, where San Francisco International Airport sits today, a Mexican cattle ranch called Rancho Buri Buri covered 14,639 acres—about fifty-nine square kilometers, almost exactly the land area of the island of Manhattan. The name had nothing to do with Spanish; it was traced to an Ohlone village recorded as Urebure, of the Yelamu people, whose families had been brought into Mission Dolores.

In 1839, Alexander Forbes, a Scottish merchant in Tepic, Mexico, published in London what is generally called the first full English book about California, describing a rich, thinly governed province. The outside world was already leaning in.

That same year, 1839, in Monterey, Governor Alvarado granted a former soldier, Antonio Buelna, a ranch on the southwest side of San Francisquito Creek: Rancho San Francisquito, 1,471 acres—almost six square kilometers, nearly three times the country of Monaco. That piece of ground is today the northern part of the Stanford University campus.

By the end of 1839, the Native population at Mission Santa Clara had fallen from 1,125 in 1832 to 291—a drop of about seventy-four percent in seven years. Historian Robert Jackson, working through the baptism and burial registers, found that deaths alone do not explain it. Hundreds left, or were pulled onto the new private ranches. The mission's own census, signed December 31, 1839, by Father Jesús María Vásquez del Mercado, remains in the archives.

Some Ohlone people did get land. Mission Santa Clara issued four land grants to Ohlone men who had entered the mission as children and asked for their own ancestral village sites back, places recorded as Ulistac and Posolmi. Four was the most of any Bay Area mission. But as Santa Clara's historians note, those grants were only a small fraction of the total acreage the mission system controlled. In the Bay Area, nearly all other lands were granted to colonial elites during the 1830s and 1840s. Trust had become title. Just not their title.

In 1841, José Peña—who had gone away to Mexico, come back, and was now teaching school in the little settlement at Santa Clara—received a grant from Governor Alvarado for the ground south of Buelna's ranch: Rancho Rincón de San Francisquito, the corner of the San Francisquito, 8,418 acres. That second, bigger rancho is the southern part of the Stanford campus today.

By 1845, the red pigment cinnabar deposit that Ohlone people had shown to Secundino Robles in 1824 had become the New Almaden mine, which would become one of the most valuable quicksilver mines in North America. Secundino and his brother Teodoro Robles were paid in cash and a share. Mercury is what miners use to pull gold out of crushed rock. The people who knew the pigment did not get the mine. The men who were shown the place did.

In the south that same year, 1845, Governor Pío Pico sold Mission San Juan Capistrano—the whole thing—to his brother-in-law, John Forster, for $710, roughly $25,000 today. An entire mission.

By 1846, the ground under the future campus was divided among Mexican rancheros, held by grants, *diseños*, and memory. Native people who had worked these fields for the mission were now working them for the rancheros, or gone.

On May 13, 1846, the United States Congress declared war on Mexico. Sonoma had not yet heard the news when, on June 14, 1846, armed American settlers rode into town and took Colonel Mariano Vallejo prisoner in his own house. They painted a bear, in berry juice, on a flour sack—what Malcolm Harris, quoting the period, calls "something that they called a bear" (Harris

2023, ch. 1.1). The California Republic lasted twenty-five days. On July 9, Navy lieutenant Joseph Revere raised the Stars and Stripes.

The pioneer John Bidwell, who watched it, called it a "pretense," the "Texas game" (Brechin 1999, ch. 1). General John Frisbie later told a Bancroft interviewer that the real object was San Francisco Bay. Ulysses S. Grant, who fought the Mexican-American War as a young officer, wrote in his memoirs that it was "one of the most unjust ever waged by a stronger against a weaker nation."

On January 30, 1847, the American alcalde of Yerba Buena, Navy lieutenant Washington Bartlett, ordered the name San Francisco used in every official paper, because Yerba Buena was unknown beyond the district. Mariano Vallejo had wanted a rival town named Francisca, for his wife. Bartlett got there first. Vallejo used her second name; the town is Benicia.

That same year, 1847, José Peña sold almost all of Rancho Rincón de San Francisquito to the two brothers, Secundino and Teodoro Robles. And in Mexico City, the war ended with Mexico defeated and its capital occupied. An American diplomat named Nicholas Trist had been sent to negotiate the peace. President James K. Polk recalled him, but Trist ignored the order. On December 4, 1847, Trist wrote to his wife: "Knowing it to be the very last chance and impressed with the dreadful consequences to our country which cannot fail to attend the loss of that chance, I decided today at noon to attempt to make a treaty; the decision is altogether my own."

Nine Days (1848)

Monday, January 24, 1848. Coloma, on the South Fork of the American River. James Marshall, a carpenter from New Jersey, was building a sawmill for John Sutter, a Swiss immigrant whose Mexican grant was eleven square leagues, the legal maximum, nearly fifty thousand acres, bigger than Liechtenstein. Marshall had been running water through the mill's tail-race to scour it deeper. In the morning he walked the channel and saw something shining in the gravel. Royce later warned that the tale is simple in its main incident and tangled in its details; the speeches later put in Marshall's mouth do not agree, but the date the histories use is January 24, and the flake is the part that holds.

Wednesday, February 2, 1848. Two thousand miles away, in the town of Guadalupe Hidalgo, just outside Mexico City, Nicholas Trist sat down with three Mexican commissioners—Luis Gonzaga Cuevas, Bernardo Couto, and Miguel Atristain—and signed the peace treaty.

Article V of the treaty drew the new border. Article XII, in the English of the signed text, read: "In consideration of the extension acquired by the boundaries of the United States, as defined in the fifth article of the present treaty, the Government of the United States engages to pay to that of the Mexican Republic the sum of fifteen million of dollars."

Nine days. The gold was found on January 24. The treaty that handed California to the United States was signed on February 2. Nobody at that table knew.

Mexico sold the land without knowing what was in it. The United States bought it the same way.

The treaty transferred more than 525,000 square miles—fifty-five percent of Mexico, on the order of France, Spain, and Germany put together. It became California, Nevada, Utah, most of Arizona and New Mexico, and parts of Colorado and Wyoming. Half a country, on a paper signed by a man the president had already fired. The price was fifteen million dollars, plus up to three and a quarter million in debts the United States agreed to assume. The Louisiana Purchase, forty-five years earlier, had carried the same fifteen-million-dollar sticker price for a larger piece of the continent. This one came with the harbor Dana had named, and a mill race full of gold.

On March 10, 1848, the United States Senate ratified the treaty, after striking out Article X—the article that had explicitly guaranteed Mexican land grants. The treaty still promised to respect the property of Mexicans who stayed, but the explicit guarantee for the *diseños*, the oak trees, and the *more or less* was gone.

Five days later, on March 15, 1848, the first printed report of the gold at Coloma appeared in a small San Francisco newspaper, *The Californian*.

Nineteen days after the Senate vote, on March 29, 1848, John Jacob Astor died in his house in New York City. A German-born immigrant who had made a fortune in furs and then in Manhattan real estate, he was the richest man in America, leaving an estate commonly estimated at about twenty million dollars—roughly $770 million in consumer prices today, though grocery prices understate a fortune that represented so large a share of a smaller country. His will left $400,000 for a library that later became part of the New York Public Library. Thirty-seven years after his fur company built Fort Astoria on the Pacific, as old Eastern money passed, the West's own fortune was lying in a mill race.

That same spring of 1848, the first Chinese residents of San Francisco arrived, servants brought from China by an American businessman (Chang 2019).

Mexico asked Washington what the deletion of Article X meant. On May 26, 1848, in the city of Querétaro, before the ratifications were exchanged, American commissioners Nathan Clifford and Ambrose Sevier sat down with Mexico's foreign minister, Luis de la Rosa, and signed an explanation: the Protocol of Querétaro. On the deleted article, the American government stated that it "did not in any way intend to annul the grants of lands made by Mexico in the ceded territories." Those grants, the protocol declared, preserved the legal value they possessed, and the grantees could cause their legitimate titles to be acknowledged before the American tribunals. *Legitimate*, it added, meant legitimate under Mexican law in California up to May 13, 1846.

It was a promise that you could prove your paper in an American court.

President Polk did not officially confirm the gold discovery to Congress until December 5, 1848, ten months after Marshall looked into the tail-race.

By the end of 1848, the redwood on San Francisquito Creek stood on the northwest corner of the Soto family's rancho. Across the creek, Antonio Buelna's widow, María Concepción Valencia, held Rancho San Francisquito. To the south, the Robles brothers held Rancho Rincón de San Francisquito. The mission at Santa Clara was a struggling parish. Its fields, which the neophytes had been promised, belonged to rancheros. The Ohlone families who had built that mission were scattered, working for others, or gone. The Native population of California, about three hundred thousand in 1769, was roughly half that. When the United States showed up, as Harris writes, "the rules changed" (Harris 2023, ch. 1.1).

Everyone who held land along the creek held it by paper: a grant signed by a Mexican governor, a *diseño* with an oak tree for a corner, and the memory of neighbors. Two ways of owning land were about to collide. Mexican ranchos worked on memory, on neighbors, on the oak tree. The American system that had just arrived worked on measured lines, filed documents, and courts.

In 1851, the new Catholic bishop of California would transfer the parish at Mission Santa Clara from the Franciscans to the Jesuits—the order Spain had thrown out in 1767 returning eighty-four years later to open Santa Clara College, today the only university campus in California built on the site of one of the twenty-one missions. That same year, 1851, Congress would pass a Land Act requiring every rancho owner in California to prove, in an American court, in English, with American lawyers paid by the hour, that their land was really theirs. More than eight hundred claims would be filed, and though most would be confirmed, confirmation would not be the same thing as keeping the ranch. Buelna's widow would file her claim in 1853 and not receive her final patent until 1868, after squatters had settled on the land and a wealthy San Franciscan named George Gordon had bought out both the squatters and her family; the Robles brothers, too, would lose much of their land to lawyers' fees and debts by the late 1850s.

In 1848, two thousand miles east in Port Washington, a lake town in Wisconsin, a young lawyer from New York was trying to build a practice and not getting far. His brothers would leave for the California gold fields ahead of him, and after a fire burned out his office in 1852, he would follow them to the mining country and open a store, where the men digging for gold bought shovels and flour from him—because the shovel is a surer business than the gravel. He did not know that yet.

His name was Leland Stanford.

Part 2: The Toll

1848–1869

To everybody. Keep quiet. When the last spike is driven at Promontory Point, we will say done. Do not break the circuit, but watch for the signals of the blows of the hammer. — W. N. Shilling, telegraph operator, May 10, 1869

Panama (1848–1850)

September 1, 1849. Monterey. Forty-eight delegates [NEEDS SOURCE] met in Colton Hall, a two-story schoolhouse of local stone above the old Spanish presidio, to decide what California was.

The United States had owned the country, on paper, since the Treaty of Guadalupe Hidalgo the year before. It had not organized it. There was no territorial governor sent from Washington. There was a military man, a flood of gold miners, and a question.

A territory is the waiting room. Congress governs it. The president appoints the governor. The people do not elect senators, and they do not vote for president. A state writes a constitution, elects its own officers, and gets two seats in the Senate no matter how large or small it is. Most of the West sat in that waiting room for decades: Oregon waited more than twelve years from acquisition to statehood, Nevada nearly seventeen, Utah forty-eight, New Mexico and Arizona sixty-four. California walked past it in two years and seven months, about twenty-five times faster than New Mexico and Arizona.

On September 5, 1849, the delegates voted on whether to ask Congress for a territorial government or to write a state constitution and demand admission. The clerk's count was twenty-eight to eight, for a state. They were not asking permission to wait. They were writing the document and daring Washington to refuse it. The constitution they signed on October 13 banned slavery. On November 13, 1849, the voters ratified it, 12,061 to 811, a landslide in a place that, a year earlier, had almost no American electorate at all.

Three forces arrived together. Gold, which by the end of 1849 had dumped a population into a place that had almost no American government. The refusal, at Monterey, to wait. And the Compromise of 1850, the national bargain that let California in as a free state, with no slavery, and postponed the breakup of the Union for a decade. Millard Fillmore, who became president when Zachary Taylor died, signed the admission bill on September 9, 1850. Congress had spent the year bargaining. The men in Colton Hall had not.

A free state on the Pacific was a prize both sides of that fight thought they understood. They did not yet understand what it would do to a railroad. In a few years the question of where to lay the tracks would be the same question as where slavery was allowed to grow.

In 1849, Collis Potter Huntington was twenty-eight, a hardware merchant from upstate New York, and he wanted California.

There was no railroad across the continent and no practical wagon road for a shopkeeper in a hurry. The sea route around Cape Horn was eighteen thousand miles. One army officer's ship, William Tecumseh Sherman's, took 202 days. Looking back at that isolation, the historian Stephen Ambrose wrote that it must be doubted whether so desirable a place had ever been so cut off.

Huntington took the other way, the Isthmus of Panama. His steerage ticket cost more than $250, roughly $10,000 today. The ship that should have met him on the Pacific side was not there. For two months he was stranded in the mud and the rain. He wrote to his brother: "Death is carrying off the Americans most fearfully. There are eight to ten deaths a day."

Cholera, dysentery, malaria, yellow fever. Huntington did not die. He went into business on the beach, selling medicines and food to men who could not leave. He later said he made $3,000 at it—roughly $120,000 today—earned because other people were stuck. He remembered the walk across the isthmus plainly: "It was only twenty-four miles. I walked it." (Accounts differ on the distance; the biographer Richard Rayner puts his tramp at thirty-nine miles.) He walked it at least twenty times, buying and selling, while the fever worked through the camps.

When a place is hard to reach, the money is not in the thing at the end of the path. It is in the passage. Huntington did not know the last step yet. He knew the first. Do not dig. Sell to the people who are digging, and charge them for the crossing.

Before any of these merchants met in California, there was a story the Stanford family liked to tell, and it has to be marked as a story. According to a biographical sketch in the Senate's memorial addresses after Leland Stanford's death, a "Mr. Whitney" spent a night at Josiah Stanford's farm, Elm Grove, on the road between Albany and Schenectady, when Leland was thirteen, and talked about a railroad to the Pacific. The boy never forgot it. Leland was born on March 9, 1824, so thirteen was about 1837. The record problem is the name. There were two Asa Whitneys. One made car wheels and worked on the Mohawk and Hudson Railroad. The other was a China merchant who came home convinced that the United States should be the road to Asia, went to China only in 1842, and began his public campaign in 1844, years after a thirteen-year-old Leland would have sat at that table. The memorial says only "Mr. Whitney." Gordon Chang's history of the Chinese railroad workers identifies the visitor as the China merchant, though the dates argue against it.

What is not legend is the merchant's own book. In 1849 Asa Whitney, the promoter, published *A Project for a Railroad to the Pacific*. He wrote that the change of the route for the commerce with Asia had, since before the time of Solomon, changed the destinies of empires, and that its march had always been westward: "Through us must be the route to Asia, and the change to our continent will be the last, the final change."

Senator Thomas Hart Benton, in the same season of prophecy, imagined Chinese workers building that West, and said they must talk together, and trade together, and marry together. The China dream and the China panic belonged to the same decade. The storekeepers would get rich on both.

The Store and the Fort (1850–1861)

June 1850, on the trail west. Charles Crocker was not yet a railroad man. At the Green River crossing, another party moved its animals onto ground Crocker's men had picketed. Crocker drove the animals off.

"That ground is ours," Crocker said.

They threatened to pull his stakes. He pulled a pistol.

"You do, if you dare."

They backed down. Ambrose records this as one of several times Crocker used the pistol, and he quotes Crocker's own rule: a man well assured of his own position, who shows a bold front, need not fear anybody. On that same journey Crocker found his horse-guards playing cards, called for the deck, burned it, and told them it would not do; they had to have their wits about them. The weapon the record gives him, on a riverbank in 1850, is the pistol. An ax handle would show up later, in another man's hand, used to discipline workers, not to hire them.

In the early 1850s, Sacramento was a levee town that flooded, burned, and rebuilt, selling tools to men on their way to the gravel. Four shopkeepers found each other. Huntington and Mark Hopkins, the quiet partner with the ledger, were neighbors at 52 K Street, in the hardware trade. Crocker was a big man, around 250 pounds, in dry goods and a freighting business, who would become the foreman. Leland Stanford, a lawyer who had burned out of a practice in Wisconsin, sold groceries. Shovels and flour. The miner's lottery was the customer's problem. The margin on the shovel was the store's.

Downriver in San Francisco, the merchants were learning a different trade.

On May 14, 1856, at the corner of Montgomery and Washington in San Francisco, James King of William, editor of the *Evening Bulletin*, was shot by James P. Casey, a city supervisor whose

criminal record King had been printing. King died on May 20. On May 22 the Committee of Vigilance hanged Casey, and with him a gambler named Charles Cora, from the windows of its headquarters at 41 Sacramento Street. The building had sandbags piled against the walls. People called it Fort Gunnybags.

This was the second committee. The first, in 1851, had been merchants and vigilantes hanging men in a gold-rush city that did not yet have a working court. The 1856 committee was larger, more military, and more sure that it was the government. Its members turned out under arms: a fort on a commercial street. California had been a state for less than six years. When the official state was new, slow, or captured, the people with warehouses decided they would be the police and the court. Sometimes they were punishing a murderer. Sometimes they were deciding who counted as the public.

The state already had a habit of building institutions beside the law and then asking the law to bless them. The vigilance committee was one. The railroad company, chartered by the state and subsidized by Congress, would be the lasting one. The committee hanged two men and disbanded. The company stayed, and it charged rent on the only modern road out.

Through the 1850s, that road could not be chosen, because choosing it chose slavery's direction.

Jefferson Davis, as secretary of war, sent out the Pacific railroad surveys of 1853 and wanted the thirty-second parallel, from New Orleans to San Diego. No free-state politician was ready to charter or fund a road that would help slavery expand. Free-soil men wanted Chicago, or St. Louis, or Minneapolis. No slave-state politician was willing to give the road to them. In 1854 the United States paid $10 million in the Gadsden Purchase for the strip that made Davis's southern route possible on American soil, and the Army experimented with camels for the desert. The camels were a detail. The blockade was the fact.

Theodore Judah, a young railroad engineer, put it in a pamphlet in 1857, reaching for the Bible a year before Abraham Lincoln's "House Divided" speech made the verse famous. The Pacific railroad, Judah wrote, was a house divided against itself: it could not be done until the route was defined, and if the route was defined, the opposing interest was powerful enough to defeat it. In January 1859 he wrote that President James Buchanan favored the extreme southern route and would veto a bill for any other.

Seven months later, on August 13, 1859, it was a hot day in Council Bluffs, Iowa, on the east bank of the Missouri River, looking at Nebraska. Abraham Lincoln was in town. He was not yet the nominee. He was a full-time politician, a year past the debates with Stephen Douglas, trying to become the Republican candidate for president. That evening he spoke at Concert Hall. The crowd was large; the local paper, the *Nonpareil*, had told the town to go and hear Old Abe.

He had not taken the easy way in. At St. Joseph, Missouri, he could have ridden the only railroad across the state and gone home to Illinois. Instead he boarded a stern-wheel steamboat and came nearly two hundred miles up the Missouri to this bluff. The morning before the speech, he walked to the top with the Puseys, the family he was staying with. Lewis and Clark had stood on that same bluff in 1804. Lincoln looked at the river and the opposite shore and told his friend Pusey that not one, but many railroads, would center here.

The next day, August 14, a group of citizens gathered on the big porch of the Pacific House hotel to hear him answer questions. When he finished, the crowd thinned. W. H. M. Pusey recognized a young man still standing there. Grenville Mellen Dodge was twenty-eight, a railroad engineer. Pusey pointed him out and said Dodge knew more about railroads than any two men in the country.

Lincoln's head came around. He studied Dodge for a moment, and went over to him. He sat, crossed his long legs, swung a foot, put his hand on the younger man's forearm, and went straight at it.

"Dodge, what's the best route for a Pacific railroad to the West?"

Dodge answered at once: from this town, out the Platte Valley. Lincoln asked why. Dodge said the forty-second parallel was the most practical and economic route, which made Council Bluffs the logical place to begin, because of the railroads already building west from Chicago to this point. Lincoln kept asking. Years later Dodge described the conversation in a soldier's phrase: "He shelled my woods completely and got all the information I'd collected."

There was a money angle under the geography, and it was small, and it was real. Lincoln was about to accept seventeen lots in Council Bluffs as collateral on a loan of $3,500—roughly $130,000 today—to a fellow lawyer, Norman Judd. The lots were in the Riddle Tract, on the floodplain, the same tract where Dodge was buying. The future president was also looking at his collateral. He was, in that year, a politician and a railroad lawyer running for president.

The railroad was already his subject. At twenty-three, in his first campaign, for the Illinois legislature, Lincoln had told voters that no other improvement could equal the utility of the railroad. He finished eighth of thirteen. He lost, and he kept the subject. In a tax case for the Illinois Central he billed $2,000. The railroad answered that this was as much as Daniel Webster himself would have charged. Lincoln re-billed $5,000, sued, and won. Ambrose calls him one of the foremost railroad lawyers in the West. In 1857 he had defended the first railroad bridge across the Mississippi, the Rock Island bridge, after a steamboat called the *Effie Afton* hit it and burned. Lincoln went to the river and measured the currents himself. He argued for travel from East to West, whose demands, he said, were not less important than those of the river. The jury hung. A hung jury was a win for the bridge. The boats did not get to own the crossing. Rayner

quotes him in that case on the manifest destiny of the people to move westward. The destiny was a legal argument about a bridge.

The Iowa railroad men did not forget the porch. At the 1860 convention they helped nominate him, and they got John Kasson to write the railroad plank of the Republican platform. It said that a railroad to the Pacific Ocean was imperatively demanded by the interests of the whole country, and that the federal government ought to render immediate and efficient aid in its construction. Still, in 1860 Judah reported that the debate on slavery, and on other matters, had left Congress no time to consider the Pacific railroad bill.

In the fall of 1860, a druggist in Dutch Flat, California, named Daniel Strong wrote to Theodore Judah. Strong knew the old emigrant road over Donner Pass, the one abandoned after the Donner party, and he thought a railroad could use it, because there the Sierra crested only once.

Judah rode up with him. Anna Judah, later, remembered that the two men nearly got snowed in and had to break camp in the dark. Back in Strong's store, Judah stretched his paper on the counter and would not sleep until the figures were down. Then he told Strong he would survey the Donner route above every other.

The ridge was not gentle. At Cape Horn, above what was then called Illinoistown, the old wagon road hung over a drop of about fifteen hundred feet, roughly the height of the Empire State Building. People who had stood there said a railroad was impossible. Judah said it was the line.

He brought the map to the Sacramento storekeepers. In Anna Judah's later account, he told them that they were tradesmen in Sacramento, that their property was there, and that if they would help him survey, he would make them the Central Pacific. With a congressional bill passed, he said, they would have the control of business that makes a fortune. If nothing more, they could own a wagon road from Dutch Flat over the mountains.

They took both. Quietly they formed the Dutch Flat and Donner Lake Wagon Road, a toll company on the line the railroad would follow. Crocker said later that none of them knew anything about railroad building; they were enterprising men, and they wanted the road to come to Sacramento.

Meanwhile, on the train east to his inauguration in early 1861, Lincoln took a turn at driving the locomotive. Dodge, in the audience for the inaugural address, wrote to his wife that Old Abe had delivered the greatest speech of the age: it was backbone all over. Then Dodge added the practical line. It looked as though they could get all their measures through.

In San Francisco, three street signs hold the memory of that moment. Union Square is a shopping square now, a column in the middle, Geary and Powell at the northeast corner. California Historical Landmark 623 records that the name originated in 1860, from public meetings held on that ground in support of the federal Union in the year before the Civil War broke—the Union that California's gold was helping to pay for, and that a Pacific railroad was supposed to pin to the coast. Sanchez Street runs south through the Mission and Noe Valley; the appendix to *The Beginnings of San Francisco* records that it was named for the family of José Antonio Sánchez, a soldier of Juan Bautista de Anza's expedition of 1776 (Rancho Buri Buri, down on the peninsula, carried a Sánchez name too, whether one soldier or a family of them). And Judah Street, in the same appendix, was named for Theodore D. Judah. The map looks neutral. It is a minute of the meeting.

In the spring of 1861, the Deep South left the Union, and its members left Congress. On June 28, 1861, the four Sacramento merchants incorporated the Central Pacific Railroad.

Their combined wealth, Ambrose writes, was $159,000, roughly $6 million today. The federal subsidy they were about to chase paid $16,000 a mile on flat land. Ten of those miles cost $160,000. Their whole fortune, pooled, was about one flat mile short of ten miles of easy track. And a simple shopping-basket translation of that fortune understates it: in 1861 those dollars were a larger share of a smaller country than groceries can show. Even so, they were not rich enough to build a railroad. They were rich enough to hold the franchise while the United States built it with them.

The Elephant (1861–1865)

On September 4, 1861, Leland Stanford was elected governor of California. The Democrats were split, and a Republican who had been a grocer a few years earlier won with a plurality.

He took the oath on January 10, 1862, in the middle of the Great Flood. Sacramento was a lake. The new governor came to the oath by boat, through the streets, and the capitol waiting for him was only temporary. He served one term, to December 10, 1863. For that whole term he was also president of the Central Pacific, an office he held from 1861 to 1890. The house he worked from was the governor's office and the railroad's office. The man who signed state paper for the railroad was the railroad's president. His partners would call him lazy and vain. He would insist, for the rest of his life, on being called Governor. He was good at being the face.

In the assembly chamber he read his inaugural address from a prepared text, and he told the state what kind of settler he wanted. Gordon Chang records the words: "The settlement among us of an inferior race is to be discouraged."

He said that Asia, with her numberless millions, sent to their shores "the dregs of her population." He called the Chinese "a degraded and distinct people," exercising a deleterious influence upon what he called the superior race. The chamber applauded. These remain, Chang

notes, his best-known words about the Chinese. On the campaign stump, separately, his phrases had been shorter: "the dregs of Asia," "that degraded race." The inaugural was one occasion; the campaign was another. They agreed with each other.

The same year, in the governor's house, Jane Stanford fell seriously ill with a lung infection. Western doctors did not help her. The family's cook, Moy Jin Kee, brought a doctor from Sacramento's Chinese quarter, Yee Fung Cheung, who treated her with herbs. She recovered. Jane became so attached to Jin Kee's teenage brother, Jin Mun, that she talked of adopting him. When the boy left their service she gave him a gold ring engraved with his name. He wore it for seventy years.

Public speech, private kitchen. The governor who wanted the Chinese discouraged owed his wife's recovery to a Chinese physician, and his company was about to decide it could not lay the track without Chinese crews.

In Washington, the votes that had blocked every northern route were no longer in the chamber.

Chang's history states it in one turn: a divided Congress could not agree, and secession let Washington authorize a northern route the South had opposed. The historian Robert Russell adds the cold institutional point: because the Southerners were gone, Congress could pass a Pacific railway law several years earlier than it otherwise would have. California's gold and Nevada's silver were helping pay for the war just begun. There was loose talk that the state might follow the South out of the Union. The railroad directors, one of them the governor, would have none of secession. They wanted the coast held, and they wanted the subsidy that holding it would justify. Dodge's account of Lincoln's motive was military and political at once: the road had to be built not only as a military necessity, but as a means of holding the Pacific Coast to the Union.

Judah went back to Washington and, at his wife Anna's suggestion, turned a room in the Capitol into what he called the Pacific Railroad Museum, with maps, surveys, and Anna's paintings of the Sierra on the walls. Members came to look. Many of them asked the obvious question: how do you get across those mountains?

Aaron Sargent, a California congressman and Judah's ally, spoke for the bill on January 21, 1862. He called it a military necessity. He said trade with Japan and China would jump, that California's loyalty would be assured, that no foreign army would dare attack the coast. Judah's complaint was that Sargent spoke before an empty hall. Judah himself was everywhere the bill was. He served as secretary of the Senate's Pacific railroad committee and as clerk of the House subcommittee, with the privilege of the floor in both houses. The engineer was also the staff.

Thaddeus Stevens of Pennsylvania, who owned a foundry, wrote in a rule that the iron be American-made. He said the Southerners would be back someday, with the same arrogant, insolent dictation which the North had cringed to for twenty years, forbidding any road that did not run along the southern border. Justin Morrill of Vermont grumbled that the nation could hardly afford both guns and railroads.

The House passed the bill on May 6, 1862, seventy-nine to forty-nine. The Senate passed it on June 20, thirty-five to five, and the House concurred in the Senate's changes. Lincoln signed it on July 1, 1862, the same day as Malvern Hill, the last battle of the Seven Days on the Virginia Peninsula. Ambrose's account gives George McClellan's army about a thousand dead and three thousand wounded in that fight, and a country depressed by it. In Washington, a pen. In Virginia, a battle the Union won that did not feel like a victory.

The law that resulted was, in Ambrose's phrase, complicated to an almost incomprehensible degree. The shape of it was simpler.

Two corporations would build toward each other: the Union Pacific west from the Missouri, the Central Pacific east from Sacramento. Each got a right of way two hundred feet on either side of the track. Each got land, in a checkerboard. A section is one square mile, 640 acres. The grant was five alternate sections per mile on each side of the line—every other square, ten sections in all, or 6,400 acres for each mile of track. The squares the government kept in between became valuable because the railroad was there.

And the government loaned its credit, in bonds: $16,000 a mile on flat ground, $32,000 in the foothills, $48,000 in the mountains. It was a thirty-year loan at 6 percent—a loan, not a gift. The bonds were a debt the companies were supposed to repay. In practice the debt would outlive the builders and land on their heirs, while the land grant was the part that behaved like a gift. A mountain mile paid triple a flat mile; in today's prices, that mountain mile was roughly $1.6 million. The Central Pacific had to build fifty miles within two years, and finish its line by July 1, 1876, a few days before the nation's hundredth birthday. So it mattered, enormously, where someone said the mountains began.

From 1861 to 1864, prices in the East jumped, because the Union was printing paper money, the greenbacks. California's economy still ran on gold coin. Sacramento's dollar and Washington's dollar were not the same dollar in those years.

Judah wired his partners in Sacramento: "We have drawn the elephant. Now let us see if we can harness him."

An elephant, in the American phrase of that century, was a prize you had won and might not be able to feed. They had the franchise. The Sierra was still there.

The next day, July 2, 1862, Lincoln signed a second land law: Justin Morrill's act, the one Morrill wanted even while he doubted the country could afford guns and railroads both. It granted each loyal state 30,000 acres of federal land for every senator and representative it had under the census of 1860, to endow at least one college of agriculture and the mechanic arts. New York had thirty-one representatives and two senators, thirty-three seats: 990,000 acres, on the order of 120 times the size of Stanford University's 8,353-acre campus today. Cornell University, chartered in 1865, was the school New York built with that grant. Stanford is not a land-grant college. It would be founded later, privately, by a railroad man, on a fortune that week's other law helped create. Same week, same president: public land turned into a college by a statute, and public land turned into a railroad subsidy.

The triple rate for mountain miles was an invitation, and Charles Crocker accepted it.

He drove the state geologist, Josiah Whitney, out to Arcade Creek, about seven miles east of Sacramento. This was not Asa Whitney the promoter, and not the car-wheel Whitney; Josiah Whitney mapped mountains for a living. The ground at the creek was flat. What Crocker showed him was a fan of reddish earth washed down from the hills. Whitney agreed that the Sierra Nevada, for the purposes of the subsidy, began there. On September 29, 1862, Governor Stanford telegraphed Washington about evidence to fix the "western base of the Sierra Nevada." Aaron Sargent, no longer in Congress but still in the capital, carried the geologist's report to Lincoln. Lincoln said it seemed about right.

The mountains moved about fifteen miles west, on paper. Foothill miles paid $32,000. Mountain miles paid $48,000. The difference was $16,000 a mile. Across fifteen miles, that was $240,000 in extra bonds—in 1863 prices, roughly $6.5 million today. The United States had been talked into paying mountain rates for a fan of red dirt on a valley floor.

Sargent told it as a joke at Lincoln's expense, and at his own: "Here you see, my pertinacity and Abraham's faith moved mountains." (Another report puts a variant of the joke—"Here is a case in which Abraham's faith has moved mountains"—in Lincoln's own mouth.) Faith, in the joke, meant a signature. The mountains did not move. The category did.

The track still had to be laid, but the extra $240,000 was not payment for a steeper grade. It was payment for a definition. Judah refused to sign the affidavit. He told Strong he could not, because "the foothills do not begin here." He wrote to Anna that he had brought his partners a franchise and laid it at their door: "Rightly used it gives them unlimited credit throughout the world, and they would beggar it."

He also told her he could not make these men appreciate the elephant on their shoulders. They would not do what he said they must; there would be trouble in Congress if they went on this

way. He could not find the money to stop them. He was out. On November 2, 1863, Theodore Judah died. He was thirty-seven.

In the spring of 1863, while Judah was still fighting his partners, the war and the railroad sat in the same room again. General Ulysses S. Grant ordered Grenville Dodge to Washington. Dodge was alarmed; he thought he was about to be called to account. Instead, Lincoln wanted to talk about the Missouri River, about towns fifty miles up and down the bank that all wanted to be the place the Union Pacific started, even as Robert E. Lee was preparing to march into Pennsylvania. Lincoln remembered the porch in 1859. Dodge's verdict was short: he found Mr. Lincoln "well posted."

A year later the companies came back for more. Lincoln signed the amending act on July 2, 1864. The land grant doubled, to ten alternate sections on each side, about 12,800 acres a mile—land equal to about one and a half Stanford campuses for every mile of finished track. The companies were permitted to issue their own first-mortgage bonds, equal to the government bonds, which pushed the United States into second place in line if the railroad failed. And the government would advance bonds after twenty miles of track instead of forty. Representative Elihu Washburne described the galleries, when it passed, as a tempest of the wildest disorder, lobbyists male and female. The Union Pacific's man, Thomas Durant, had a lobbyist, Joseph P. Stewart, spreading $250,000 in Union Pacific bonds, $20,000 of it to Charles T. Sherman, the general's brother. (Years later, in his epilogue, Ambrose would sum up the partners' Washington method even more bluntly: Huntington lied, handed out bribes, and got the president to accept a fictitious map.)

The men who held the statute also decided they should hold the contract to build. The Sacramento partners awarded the construction to Charles Crocker and Company, a firm they owned. Ambrose calls it an almost identical device to the Crédit Mobilier. In the East, Thomas Durant and a promoter named George Francis Train bought a sleepy Pennsylvania corporation, the Pennsylvania Fiscal Agency, and renamed it the Crédit Mobilier of America. The Union Pacific paid that company, which the Union Pacific's own directors owned, to build the Union Pacific. The profit was in the building, whether or not a ticket was ever sold. Among the insiders was Congressman Oakes Ames of Massachusetts, whose family made the shovels that Huntington, Hopkins, and Crocker had sold to miners. The store and the subsidy had the same supplier.

On March 18, 1865, while the war was still not over and Lee had not surrendered, Abraham Lincoln signed patents in Washington that conveyed Mission San Juan Capistrano and Mission San Luis Rey to Bishop Joseph Sadoc Alemany of California. The patents confirmed a decree the Land Commission had issued in 1855, on a claim Alemany had filed in 1853; a later Supreme Court case, *Foster v. Mora*, measured the San Juan Capistrano parcel—the buildings and the gardens that a Mexican governor had once sold for $710—at 46.74 acres. Lincoln did

not return all the missions on that day (San Gabriel's patent was November 19, 1859; San Diego's was May 23, 1862; La Purísima waited until 1874). What he did was narrower. The American patent was how the church got its parcel back, through the same federal title office that was confirming Mexican rancho grants, and that, in the same decade, was giving railroads millions of acres. One office, three kinds of land: a mission church, a Mexican ranch, a checkerboard along a track. The railroad was simply the applicant that arrived with a statute written for it.

The Summit (1864–1867)

James Harvey Strobridge was the Central Pacific's construction boss. Irish, over six feet tall, thirty-seven when he took the work. At Bloomer Cut a black-powder blast cost him his right eye. The Chinese who learned English called him the "one-eyed bossy man." Foremen clapped a hand over an eye to make a crew think he was coming. He said that men generally earned their money when they worked for him.

Strobridge had not wanted Chinese workers hired. His line, in Ambrose, was plain: "I will not boss Chinese." He said they could not do the work. Then he did boss them, because the white crews were leaving for the mines and the mountain was not going to wait. The story goes that Crocker answered him by pointing to the Great Wall of China. The payroll is plainer, and earlier: Chinese crews under contractors including Hung Wah and Ah Toy were already on the Central Pacific's books in January and February of 1864, a year earlier than the usual telling puts the decision.

By Chang's account, the company needed a workforce it could barely imagine, and it printed thousands of copies of a woodcut flyer to circulate in California and in China. No copy survives. Crocker admitted to Huntington that no one on the company side had translated the flyer back into English. The men who believed they were buying labor could not read their own advertisement. Chinese workers became about 90 percent of the Central Pacific's labor—in Chang's judgment, by far the largest single workforce in American industry to that date.

When there was trouble among the Chinese crews, Ambrose writes, Strobridge could pick out the leaders with a glance. He confronted them, usually with an ax handle, and they gave way. The handle was Strobridge's; Crocker was the man who hired him. Crocker, for his part, later claimed he quarreled with Strobridge about the way he spoke to the men. "They are human creatures," Crocker said. Strobridge answered that "you have got to do it," and that they were "about as near brutes as they can get." Crocker said he found out that it was true. Strobridge's wife, Hanna Maria, lived in a passenger car fitted out as a house at the end of track, with an awning and a caged canary by the door. The men called her the "Heroine of the Central Pacific," and Ambrose notes she was the only white woman who saw the work through from beginning to end. The canary was real. It did not soften the ax handle.

Meanwhile, the Dutch Flat wagon road was already charging tolls before the rails reached the pass. In February 1866 Hopkins wrote Huntington [NEEDS SOURCE] that snow stopped the mountain work five or six months a year, and that getting freight over the pass meant paying teamsters a dollar and a quarter a ton. Even then, he said, the men would rather stay on the old Pacific Road—more taverns, more waiting girls and barmaids, a familiar road. Hopkins's same letter put the construction in a builder's arithmetic: reach the desert this season and it was a six-month job; miss it, and it was eighteen months. On the flat, beyond the snow, a day of track was a day of government bonds.

The winter of 1866 into 1867 brought snow so deep, and so bright, that men went snow-blind. E. B. Crocker, Charles's brother and the company's lawyer, wrote to Huntington that they had had to buy up all the cheap goggles in the market. Think of that, he wrote: what an item in railroad building.

Inside the summit tunnel the work moved a few inches at a time. Crews of three stood at the granite face, one man holding the drill, two swinging sledges of fourteen to eighteen pounds. They drilled, they packed the hole with black powder, they ran, they waited for the blast, and they walked back into the smoke and did it again. Ambrose's account of that facing is six to twelve inches in a day, with shifts going around the clock. A month of that work moved the mountain less than the length of a room.

On Christmas Day, 1866, the *Dutch Flat Enquirer* reported that a gang of Chinese workers had been covered by a snow slide and that four or five died before they could be dug out. Near the summit tunnel, an avalanche carried away some twenty Chinese, whose bodies were found after the spring thaw. How many died on the line, the company did not publish. The historian Thomas Chinn's judgment is that the loss of life was heavy. (In June 1870, a train would pass through Sacramento carrying, by one newspaper report [NEEDS SOURCE], the bones of perhaps 1,200 Chinese workers, boxed to be sent home.)

On Monday, June 24, 1867, near the solstice, in the middle of the building season, three thousand Chinese workers put down their tools at once, from Cisco to Truckee, across nearly thirty miles of mountain.

The demand, as the company understood it, was parity with white workers: $40 a month instead of $35 (in today's prices, $35 a month was roughly $840 a month, or about $10,000 a year), and a day of ten hours instead of eleven. Five dollars, and an hour. The strike was not a demand to get rich. It was a demand to be paid like the other men on the same rock. A newspaper [NEEDS SOURCE] called it the greatest strike the country had known.

Mark Hopkins wrote to Huntington with the fear of a bookkeeper: if they succeeded in this demand, then they controlled. E. B. Crocker wrote the most honest line: "The truth is they are

getting smart." His remedy was to inundate California and Nevada with laborers—freedmen, Chinese, Japanese, all kinds.

Charles Crocker went up the mountain. The strike leaders told him, in words he later reported as theirs, "Eight hours a day is good for a white man, all the same good for a Chinaman." In that telling they asked for eight hours, not the ten the company wire had named. The hours shifted with the teller; the pay did not. Crocker answered in a pidgin of his own making: "John Chinaman no make laws for me; I make laws for Chinaman. You sell for thirty-five dollars a month, me buy; you sell for forty dollars and eight hours a day, me no buy."

Then the food stopped. Crocker claimed he cut it off. The labor agents stopped supplying provisions. E. B. Crocker wrote that none of them went near the strikers for a week; they did not want to exhibit anxiety. After eight days most of the men went back to work.

Histories that follow Charles Crocker's version call it a defeat. Chang reads the partners' own letters and sees men who were shaken, noting that pay for skilled Chinese workers rose over the following months. What the principals wanted next was not a negotiation. It was a flood.

"We want a hundred thousand chinamen here," E. B. Crocker wrote, "so as to bring the price of labor down."

Huntington went further: it would be all the better for them and the state, he wrote, if half a million came over in 1868. The governor who had called them the dregs now loved the supply.

On August 3, 1867, Chinese crews holed through the summit tunnel and let the air in.

Done (1869)

April 28, 1869. Seven in the morning, a whistle in the Utah desert.

The Central Pacific's crews set out to lay more track in a day than anyone had ever laid. Charles Crocker had bet Thomas Durant $10,000—roughly $251,000 today—that they could do ten miles. Ambrose writes that Durant, reportedly, was sure they could not, and took the bet.

By seven that evening they had laid ten miles and fifty-six feet, up the slope toward Promontory, bending rails around curves by hand. Almost a mile an hour, for twelve hours. The celebratory accounts named eight Irish rail handlers. They did not name a single Chinese worker. Several thousand men did the day; the story kept eight names. (At the end of construction, Strobridge did seat his Chinese foreman, most likely the labor contractor Hung Wah, at the head of a dinner table, and a paper ran the scene under the heading "Chinese Laborers at Table." The honor was private. The pay packets were the public fact.)

A few days after the ten-mile day, some of those Chinese workers, perhaps including Hung Wah, climbed a bluff near Monument Rock with a brown-glazed jug of rice wine that had come from the Pearl River delta. They could see the line, and beyond it the Great Salt Lake. They drank, left the jug, and went back to camp. Archaeologists found it on the outcrop a century and a half later. Chang writes that the bottle, though mute, had something to say. It was the celebration that was not on the program.

May 10, 1869. A dry flat in the Utah Territory, north of the Great Salt Lake, a place called Promontory Summit. Water had to be hauled in. There was no band and no stage. Two locomotives faced each other across a gap of a couple of rail lengths. One was the Central Pacific's *Jupiter*. The other was the Union Pacific's *No. 119*.

The ceremony was two days late, for a reason that was not smoothed over at the track: Union Pacific workers, owed wages, had detached Thomas Durant's private car, set a guard on it, and held him until he wired for the money and paid them. The vice president of the road had been a hostage to his own unpaid crews.

Leland Stanford, governor of California turned railroad president, was the only one of his four Sacramento partners who came. In his hand was a silver-headed maul. In the tie beneath him, polished California laurel, a golden spike sat in a hole already drilled so the precious metal would not be scarred. Reporters had been telling the country that the blow would be heard farther than any sound a mortal man had made. A telegraph operator named W. N. Shilling beat his warning down the wire to every office that was asking when: "To everybody. Keep quiet. When the last spike is driven at Promontory Point, we will say done. Do not break the circuit, but watch for the signals of the blows of the hammer."

The engineers, Booth and Bradford, pulled the whistles. A veteran in the crowd said later they all yelled like to bust. Stanford's speech ran less than three minutes. No one at the center of the flat said the name of a worker. A squad of Chinese men in clean blue frocks had already carried one rail into the gap, set it, and stepped back. An Irish squad had carried the other. The laurel tie was in place. The spike was seated in its hole, so the gold would not have to take a real blow.

Stanford swung. He missed, and the maul hit the rail.

It did not matter. The operator closed the circuit anyway. The wire went out across the continent: *done*.

The officials announced that the two railroads together had laid 1,776 miles of track, as in the year of independence. When the dignitaries left, other workers pulled the gold and the laurel out, put ordinary iron and ordinary wood back in, and drove the spikes that actually held. (The

laurel tie would burn in the San Francisco fire of 1906; the golden spike sits today in the art museum at Stanford.)

Andrew J. Russell's photograph, *East and West Shaking Hands*, two engines nose to nose, became the image of the railroad. Chang is careful: there is no evidence that Russell arranged the crowd to keep Chinese workers out of it. Look closer, and a ghostly figure with his back to the camera, blurred because he was moving, stands in the middle of the frame. From his clothes, his hat, and his stature, he is Chinese. He is in the most famous picture of the road he helped build, and he is a smear. A century later, in 1969, a cabinet secretary named John Volpe would stand at the centennial and ask the crowd, "Who else but Americans could drill tunnels in mountains thirty feet deep in snow?"

Four years after Promontory, in February 1873, when Representative Jeremiah Wilson's congressional committee put Collis Huntington on the stand and asked for the construction accounts, Mark Hopkins carried fifteen leather volumes down to the basement furnace. Huntington came back and told the committee the records were lost: "My partner Hopkins is a peculiar man. He considered the paper no longer worth saving."

Doctors had already been selling California's climate before the rails met—in 1852 James Blake wrote that there was not a day in the year when the mind or the body was weakened by heat or numbed by cold, and J. Praslow, a physician in San Francisco from 1849, wrote that the climate was so healthy that illness was rarely found. From 1840 to 1900, one estimate puts illness behind about a quarter of the new arrivals, and the great health rush would begin after 1870, when a state report called California the "sanatorium of the world." Soon enough the railroad would carry the coughing rich, and a boy.

In May 1869, the road was finished. Now the four partners owned it. Everyone who wanted to move a bushel of wheat had to pay.

Part 3: The Octopus

1874–1886

Friends (1874–1876)

November 30, 1874. New York. Collis Huntington sat down to write a letter to San Francisco.

The four railroad partners did not particularly like each other, and they liked Leland Stanford least. Friendship, in the Gilded Age, was not a feeling. In the history of the transcontinental railroads, it was an arrangement (`[@white-railroaded]`). Measures in California lived or died on who would stand with whom when the vote was called.

Some of the arrangements were with Congress, and they survived on paper because a man named David Colton saved his mail. Colton was the partners' man in San Francisco, a junior associate who wrote constantly to Huntington in New York. Huntington wrote back in his own voice: dry, busy, sure of the price of things. Between the middle of the 1870s and Colton's death in 1878, the letters piled up. On November 30, 1874, looking ahead to the Congress about to sit, Huntington wrote to Colton: "I think this coming session of Congress will be composed of the hungriest set of men that ever got together."

Friendship with a newspaper worked the same way as friendship with a senator. You did not always buy the printing press. You bought the pipe the news came through.

In the 1870s the Associated Press was that pipe. A paper that was not in the association did not get the national wire, which meant it did not get the country, not in time to print it. Whoever stood at the gate decided which stories moved.

Huntington understood the gate. He wrote to Mark Hopkins that if the Associated Press agent in Washington were seen to, a great deal of hostile copy could be kept back ([@white-railroaded]). His lobbyist, Richard Franchot, once got that Washington reporter onto the Central Pacific payroll. It worked until the Associated Press editor, James Simonton, noticed that the tone of the stories had changed, and suspected why ([@white-railroaded]).

In 1875 the partners tried something larger. The *San Francisco Chronicle* had begun ten years earlier, in January of 1865, as a theatrical handbill called the *Daily Dramatic Chronicle*, started by two brothers, Charles and Michael Henry de Young. Charles was the editor who treated scandal as a product. By the 1870s the handbill had become a daily, and for a time the railroad had a hand on it. In 1875 the partners wanted an exclusive Associated Press membership for the *Chronicle*, and they offered double the going rate ([@white-railroaded]). Double. The point of paying double was not generosity. It was to cut two other papers, the *Bulletin* and the *Call*, off the wire—papers the railroad hated. The attempt failed. A news monopoly was worth more than a fair price, because a news monopoly decided what a fair price sounded like in print.

On January 17, 1876, Huntington was back on the rails. A rival, Tom Scott of the Pennsylvania Railroad, was pushing a bill in Congress to help his own Pacific line, the Texas and Pacific, a southern route to the coast ([@white-railroaded]). Huntington took the night train to Washington.

"It cost money to fix things so that I would know his bill would not pass," Huntington wrote to Colton. "I believe that with $200,000 I can pass our bill."

Two hundred thousand dollars in 1876 is roughly $6.3 million now, as the cost of one bill.

Twelve days later, on January 29, 1876, and again on March 23, Huntington used the sentence he reached for whenever he wanted to sound clean: "I keep on high ground." The second time he finished the thought. He kept on high ground so that no investigation could hurt them.

Lobbying is arguing for a bill in public, with your name on the argument. Bribery is paying an official to use the office as a lever. Huntington's letters live in the gap. He did not write that he handed a senator an envelope. He wrote that two senators were switched this week and could be switched back with the proper arrangements ([@white-railroaded]). The high ground was not innocence. It was a method: pay in ways that look like friendship, so that if a committee asks, the paper smells like business.

Nineteenth-century Americans were not naive about this ([@white-railroaded]). They were shocked less by a bought newspaper or a greedy congressman than by what it added up to: a republic whose decisions were for sale. The word they used was corruption, and they meant it the old way, as rot in the structure, not as one man's bad manners.

And it was not a California invention. While Huntington priced a bill to fight Scott, farther north Jay Gould was gathering lines. Charles Perkins of the Chicago, Burlington and Quincy said the Union Pacific's life depended on high rates more than any other company's ([@white-railroaded]). John Murray Forbes, whose Boston family had made its first fortune in the China trade, said Gould had as many aliases as a London professional thief ([@white-railroaded]). Stanford's face was Western. The technique was national.

Nob Hill (1876–1878)

In 1876, up a steep street in San Francisco, the hill did not yet have its fame. It had a view, wind, and sand. The men who had just finished a railroad decided the view was the point.

Leland Stanford built first, at California and Powell. The house was finished in 1876. It was a mansion of more than fifty rooms, on about two acres, inside a wall thirty feet high (Rayner 2008). The front doors were mahogany and rosewood, seven inches thick. A glass dome over the hall sat seventy feet up. The carved framework of the conservatory cost $30,000. A custom library ladder, three rungs, cost $150. The whole production came to $2 million in the money of the 1870s (Rayner 2008)—roughly $63 million now.

In 1899 an economist named Thorstein Veblen would give this a name: conspicuous consumption, spending whose job is to be seen. The mansions came first. The theory arrived to describe them. Stanford's house was not a machine for living. It was a machine for being looked at, from the bay and from the offices of men who paid the railroad's rates. That same year, 1876, the Stanford family also bought a farm south of the city in Santa Clara County, about 650 acres, a place called Mayfield Grange (Harris 2023).

California wrote its history on signposts and then forgot the ink. Nob Hill is one of those names. *Nob* is not a saint. The Spanish named towns for Francis, Clare, Joseph, and the rest, and those names are still the mission chain. *Nob* is short for *nabob*. And *nabob* is an English mangling of a word from India, *nawab*, a deputy of the Mughal emperor, a man who governed in someone else's name and got rich doing it. The British brought the insult home for merchants who made a fortune in the East and spent it loudly in London. San Francisco picked it up for a different east: Sacramento grocers and Nevada silver men who hauled their money up one steep ridge and built houses the rest of the city had to crane its neck to see.

So the hill is a joke that froze. Catholic California named places for heaven. Railroad California named a hill for showing off.

Even before the hill was finished, the crowd found the address. In July 1877, the country was four years into the long depression that had started in 1873. In San Francisco, unemployed men gathered on a vacant lot near the city hall, a sandlot, and the meeting that was called against the railroads turned, fast, against the Chinese (Harris 2023).

They marched up the hill on Leland Stanford's mansion (Harris 2023). The man at the front of the Workingmen's Party of California was Denis Kearney, an Irish-born drayman who hauled freight for a living, and a speaker who had found the sentence that would move a crowd.

"I will give the Central Pacific just three months to discharge their Chinamen," Kearney told the sandlot in 1877, "and if that is not done, Stanford and his crowd will have to take the consequences" (Harris 2023).

Three months. That was the number in the threat. Kearney's slogan, the one painted on banners, was four words: "The Chinese Must Go." Kearney was better at being a racist than at being a labor leader (Harris 2023). It was easier to make Chinese workers pay for the depression than to make Stanford pay. Stanford had Kearney arrested (Harris 2023). Then Stanford did what the very rich have often done when a crowd learns the address. He left town for Mayfield Grange, the 650-acre farm south of the city (Harris 2023). The crowd had marched on a mansion. The owner was becoming a country gentleman.

Scapegoating is not just meanness. It is a political technology. The anger in the sandlot was not imaginary. Wages were down. The railroad really was a monopoly. A man who hauled goods for a living had reason to hate a company that set the price of moving goods. Kearney took that anger and nailed it to a people who could not vote, who had built the railroad, and who were still in the company's workforce because the company wanted them there. In his 1862 inaugural address, read from a text, Stanford had said Asia sent "the dregs of her population," and he had called the Chinese "a degraded and distinct people" [NEEDS SOURCE]. On the stump the line had been shorter: "the dregs of Asia" [NEEDS SOURCE]. By the time the tracks were finished he wanted more of them, because they kept the wage down. In 1877 his neighbors wanted them gone, and he was happy to let the neighbors do the shouting.

The workers and the owner were not on the same side of the door. The crowd was outside the mansion. The men the crowd wanted fired were the railroad's labor, paid by the month, living in camps the mansion never had to see. A monopoly can survive a riot if the riot aims at its workforce instead of its rates. That is what the sandlot gave the Big Four: a villain who was not them.

Up on the hill, the building went on. Mark Hopkins let his wife talk him into a house a block above Stanford's, at California and Mason (Rayner 2008). He was the quiet partner, the ledger. Even after he was a multimillionaire he kept a small cottage, kept no carriage, and took no vacations (Rayner 2008). The palace went up anyway.

He never lived in it. On the night of March 28, 1878—some notices date the death the twenty-ninth—a private railroad car stood on a siding at Yuma, on the Colorado River, where California meets Arizona (Rayner 2008, ch. 12). Hopkins was sick. He had not wanted the trip. Huntington wanted eyes on the ground the Southern Pacific was reaching for, out past the river, and Hopkins knew that Huntington's schemes usually had something behind them. He got off his sickbed, into his own company's car, and went to look. He went to sleep in the car. He did not wake up (Rayner 2008, ch. 12).

He was the partner the others could stand: the bookkeeper, the man who had answered panic, in the building years, with as few words as possible. He died in the one house he actually used, a car on a company track, at the edge of the map.

"My old friend Mr. Hopkins has gone over the river," Huntington wrote after the death. "His death has made me very sad. I was very fond of Mr. Hopkins. We had been friends from the day of our first meeting, and partners in business, as you know, for many years" (Rayner 2008, ch. 12).

Huntington did not come west for the funeral. He was too busy (Rayner 2008, ch. 12). He added that there might be other men as good as Hopkins, but that he would not live long enough to form such a friendship again (Rayner 2008, ch. 12). The affection was real. So was the arithmetic. A partnership is a machine that expects a man to be replaced, and the man who ran the machine from New York was not going to miss a vote in order to stand by a grave.

Two weeks after Hopkins died on the siding, on April 10, 1878, a new cable car line opened up California Street. Getting up the hill was its own advertisement. Cable cars already climbed Clay Street; Andrew Hallidie's line had started in 1873, and it stopped two blocks short of Stanford's door. So Stanford and his railroad friends built another one, up California Street itself, past the new palaces. A railroad president built a vehicle so the public could reach his front steps, and so he would not have to walk the last two blocks.

A block away, Charles Crocker wanted a whole square: California, Sacramento, Taylor, and Jones (Rayner 2008). One lot on it, at 1100 California, belonged to a man named Nicholas

Yung, an undertaker, comfortable but not rich, who liked the view and would not sell at Crocker's price. The offers climbed—$6,000, then more, and at some point Yung is said to have named $12,000 (Rayner 2008). Crocker stopped bargaining and built a fence.

The story goes that the fence was thirty feet high. The record is plainer and a little different: press accounts described a wooden fence forty feet high around three sides of Yung's house, so the sun barely got in, though some later retellings shrink it to thirty (Rayner 2008). It was four stories of boards, on three sides of a family's windows. The *San Francisco Examiner* was writing about it by August 17, 1878, under the name the city gave it: a spite fence (Rayner 2008).

The cost, in the newspaper accounts that survive in later write-ups, was about $3,000—roughly $100,000 now. Crocker could lose that sum the way another man loses a cufflink. What he bought was a lesson. The price of refusing the railroad was darkness at noon.

The story goes that Yung answered by putting a coffin on his roof, skull and crossbones turned toward Crocker's windows (Rayner 2008). What a *Chronicle* report preserves is the threat: a coffin above the fence, as an advertisement for the undertaking business and a reminder of mortality. Threat or object, the picture stuck. The man who buried the dead, boxed in by the railroad. Yung eventually moved the house to another lot he owned, on Broderick Street, and refused to sell the empty ground.

Two months after the *Examiner* named the spite fence, on the evening of October 8, 1878, a carriage stopped outside David Colton's house on Nob Hill (Rayner 2008). He was unconscious inside it. The company said a horse had thrown him. He died two days later. The rumor in the street was a stabbing. The truth of that night stayed dark (Rayner 2008).

Four men had owned the Central Pacific in equal shares and let a construction company they also owned pay them for building it. By the fall of 1878 one of the four had died in a car on a siding, and Colton, the fifth man they had let partway in, was dead on Nob Hill in a story nobody could finish. The survivors inherited the dead men's leverage, and they used it on the widows. Colton's widow, Ellen, was walked through a story in which her husband had been an embezzler, and she was pressed to give the shares back (Rayner 2008). Grief was not a pause in the business. It was an opening in the cap table.

The Wheat Field (1879–1880)

On March 1, 1879, in Washington, President Rutherford B. Hayes vetoed a bill passed by Congress to cut Chinese immigration hard, down toward fifteen passengers on a ship. His reason was a treaty. The United States had promised China, in the Burlingame Treaty, a freer movement of people. Hayes did not veto because California's workers had changed his heart. He vetoed because he did not want to tear up a diplomatic bargain in public. Then he sent a minister to renegotiate the bargain.

In California that same year, the party Denis Kearney fronted did not stay in the street. Voters adopted a fresh state constitution in 1879, over the railroad's opposition ([@white-railroaded]). Kearney's Workingmen helped write a streak of anti-Chinese clauses into it, and the same constitution created a state railroad commission—three elected men whose job was to watch the rates ([@white-railroaded]).

Also in 1879, a San Francisco newspaperman published the most famous economics book of the century to explain the rise in the land. Henry George's *Progress and Poverty* argued that a railroad, a ditch, a town—all of that improvement—raises the value of the land around it, and the rise flows to whoever holds the title, not to the people who did the digging. He did not need a wheat field to draw the picture. He already had the railroad.

"A railroad company approaches a small town as a highwayman approaches his victim," George wrote. "The threat, 'If you do not accede to our terms we will leave your town two or three miles to one side!' is as efficacious as the 'Stand and deliver,' when backed by a cocked pistol."

Build through the town, or build past it and let it die. After the road is built, charge what the town cannot refuse.

In San Francisco, the Workingmen's Party ran a Baptist minister named Isaac Kalloch for mayor in 1879. The *San Francisco Chronicle*—where the railroad had held influence—went after him personally. That grip had not made the paper gentle; it had made the paper a faction. Kalloch answered from the pulpit and, by the paper's own later account, said things about Charles de Young's mother that a man of that temper would not let stand. In August of 1879, Charles de Young shot Kalloch. The minister lived. The sympathy of the shooting helped elect him.

On the evening of April 23, 1880, at about half past seven, Mayor Kalloch's son, Milton Kalloch, walked into the *Chronicle* office and shot Charles de Young dead at his desk. The *New York Times* ran the wire the next day, plain as a bill of lading: Charles de Young killed; Mayor Kalloch's son; the *Chronicle* office; half past seven.

The son was tried and acquitted. One juror's reported view was that he would have done the same [NEEDS SOURCE]. Michael Henry de Young, the surviving brother, took the paper.

A newspaper can be a public record. It can also be a political weapon owned by a side. De Young had not shot Kalloch because of a disagreement about grain rates. He shot him because the paper and the candidate had become personal enemies, and in that San Francisco the personal was how politics was conducted. The railroad did not order the shot. It did not have to. It had spent years teaching the city that a paper was a piece of property, to be captured, starved of wire service, or aimed. Once that is normal, editors start carrying the same tools as the men they cover.

Eighteen days after Charles de Young died at his desk, the fight reached the valley.

May 11, 1880. A Tuesday morning in a wheat field, two and a half miles north of a new town called Hanford, in the San Joaquin Valley of California (Rayner 2008, ch. 13). The valley is the long flat trough that runs down the middle of the state, between the coast mountains and the Sierra Nevada. In spring the wheat stands high enough to hide a man on a horse.

In Hanford that morning, families had come in for a picnic (Rayner 2008, ch. 13). The children, at least, expected to have fun. The adults had heard who got off the train a few hours after midnight. His name was Alonzo W. Poole. He was a United States marshal, and he was carrying eviction notices (Rayner 2008, ch. 13).

This was not empty land, and it was not a picnic that went wrong for no reason. In 1877 the Southern Pacific had finally received its federal patents to land in this part of the valley, along a branch line toward the Mussel Slough of the Kings River (Rayner 2008, ch. 13). Settlers were already on it. Many of them had brought water to ground the county histories call barren, and had made a crop. Railroad agents then announced they would sell at prices from $17 an acre to $40 (Rayner 2008, ch. 13).

The company itself had advertised the land at "$2.50 an acre upwards" (Rayner 2008, ch. 13). Seventeen dollars is almost seven times that advertised floor. Forty dollars is sixteen times it. The difference was not the dirt. The difference was who got paid for the work. And the Pacific Railway Act of 1862, the law that gave the railroad its land in the first place, had named a different ceiling in Section 3: grant land still unsold three years after the road was finished was to be open to settlers at not more than $1.25 an acre, paid to the company.

Poole rode out of Hanford in a wagon with a Southern Pacific land man beside him (Rayner 2008, ch. 13). Two neighbors who were buying the disputed ground from the railroad met them in a buggy. Their names were Mills Hartt and Walter Crow. In the buggy they had two shotguns, a Spencer rifle, and three pistols (Rayner 2008, ch. 13). Hartt had been bragging in town about what he would do to settlers who would not move. Crow was known as one of the best shots in California. He had knocked wild geese out of the air with that rifle (Rayner 2008, ch. 13).

They stopped at a homestead where nobody was home, put the family's belongings in the road, and lined shotgun shells up on the porch (Rayner 2008, ch. 13). Then they went on to a place held by Henry Brewer and John Storer, part of which the railroad had sold to Walter Crow. It was about half past ten (Rayner 2008, ch. 13).

At the picnic, word arrived. Riders came out through the wheat and the alfalfa. Poole later said he saw forty or fifty. Other witnesses said about fifteen (Rayner 2008, ch. 13). Testimony cannot settle the number of men on horseback. A legend will.

Poole knew the neighborhood, and the neighborhood liked him. A settler named James Patterson asked him to wait until the court appeals were heard. Poole said he was doing his job.

Someone told him to give up his pistol. He would not. Then a nervous horse struck him and knocked him down (Rayner 2008, ch. 13). He was the most reliable witness, and he was in the dust when the shooting started.

Nobody agrees who fired first. In less than a minute, twenty or thirty shots were fired (Rayner 2008, ch. 13). Walter Crow did most of the killing. James Harris. Iver Knutson, forty-nine, a father of nine, a man Crow knew. Daniel Kelly. Archibald McGregor, armed with a penknife. John Henderson. Hartt was hit too, and took more than a day to die (Rayner 2008, ch. 13).

There was a pause. Smoke hung over the wheat. Crow backed the others off with a pistol and said, "One at a time" (Rayner 2008, ch. 13). The marshal told him to take a horse and go to the timber. Crow would not. He walked into the wheat he had bought from the railroad, the field he had told his friends he would harvest that year. A mile and a half on, armed settlers met him. He raised his pistol. He was shot in the back—with his own rifle, according to one report—and he fell dead in a ditch (Rayner 2008, ch. 13).

"As has long been anticipated, the conflict."

That was how the *Visalia Times* opened its account four days later, on May 15, 1880 [NEEDS SOURCE]. The paper named seven dead: five settlers, and two men who had come to take the land. An eighth man, Edwin Haymaker, had only a scalp wound that morning. He cleaned it at the well. He developed pneumonia, and he was dead within the month (Rayner 2008, ch. 13). Count eight if you count Haymaker. The next legend of the American West, the gunfight at the O.K. Corral in Tombstone, a year and a half after this, killed three (Rayner 2008, ch. 13). The movies kept Tombstone. The wheat field is the one that tells you how California actually worked.

The story goes that this was honest farmers against a monster. The record is meaner. Some of the settlers had fought for the Confederacy (Rayner 2008, ch. 13). Their league drilled like a militia. Masked men had burned a homestead, and each side blamed the other (Rayner 2008, ch. 13). Crow knew the men he killed. Nobody rode out with a settled plan to start a slaughter, and drawn guns made one likely (Rayner 2008, ch. 13). Both things are true. The railroad wrote the price. The men in the field pulled the triggers.

The settlers at Mussel Slough had improved the ground. The railroad held the patent. The patent won. In the end the courts sided with the company. Some families left. Some paid the higher price. The Southern Pacific kept the land (Rayner 2008, ch. 13).

Put the spring of 1880 on one calendar. April 23, a son kills an editor in a newspaper office on the bay. May 11, eighteen days later, seven men die in a wheat field, and an eighth is dying of a scalp wound he does not yet know will kill him. Same state. Same railroad sitting under both stories, not as the finger on every trigger, but as the condition both fights were fought inside. A city and a valley that had come to expect the conflict.

That same year, 1880, Nicholas Yung died, while Charles Crocker's forty-foot fence still stood on Nob Hill (Rayner 2008). A miner named George Hearst picked up the *San Francisco Examiner*, a weak Democratic daily, third in the city. And the federal census takers counted the people Denis Kearney had pointed at from the sandlot: 75,132 Chinese people in California, in a state of about 865,000. Nearly one Californian in twelve. Of all the Chinese people in the United States—105,465 in the 1880 census—seven in ten lived in that one state.

Exclusion and the Octopus (1881–1884)

On February 19, 1881, the *San Francisco Wasp* printed a cartoon titled "In the Hollow of His Hand" (`[@white-railroaded]`). In it, Crocker and Huntington stand on the railroads and hold Stanford up, and Stanford holds the state railroad commission in his hand, a commissioner small enough to close his fingers on. Two years after the constitution of 1879 created three elected watchdogs to watch the rates, a humor magazine drew the watchdogs sitting in the palm of the man they were supposed to watch.

A public creates a guard to stand between a monopoly and its customers. The monopoly then hires the guard, or elects him, or tires him out, until the guard's reports describe the monopoly's wishes as the public's safety. You do not have to repeal the law. You inhabit it.

Six weeks later, on April 2, 1881, the *Wasp* printed a poem by its editor, Ambrose Bierce (Rayner 2008). He had the Mussel Slough dead in mind, and he aimed the last stanza at a man who had not been in the field:

> These mounds are green where lie the brave who fought and fell at Hanford. O, point me out, I pray, the grave of Leland Stanford.

Bierce had a nickname ready: "Stealand Landford" (Rayner 2008).

In Washington, the renegotiated treaty with China had come back. It let the United States restrict Chinese immigration, so long as the restriction was "reasonable." *Reasonable* is a lawyer's word. It means whatever the next Congress can pass and the next president will sign.

In 1882 Congress tried for a twenty-year ban on Chinese laborers. President Chester A. Arthur vetoed it as more than the treaty allowed. Congress came back with ten years. On May 6, 1882, Arthur signed the Chinese Exclusion Act.

It took five years from Kearney's three-month threat on the sandlot, and it arrived wearing the language of a treaty. It was the first significant federal law restricting immigration, and the first time federal law barred an ethnic working group because Congress claimed that group endangered the good order of certain places. It was not the first unfriendly statute—the Page Act, in 1875, had already made it much harder for Chinese women to enter the country, under the stated target of prostitution and forced labor. The 1882 law was wider. It shut out Chinese

laborers, skilled and unskilled, for ten years. It made Chinese people ineligible for citizenship. It required papers of the Chinese who were allowed to come and go: merchants, teachers, students, travelers. A workforce that had been recruited, by this same railroad, with handbills the railroad's own officers could not read, was now a people the federal government would not admit and would not naturalize.

Seventy-five thousand Chinese people in California—one in twelve of the state. The men who spoke for the state described an invasion. The count describes a minority, concentrated because the jobs were concentrated: on the railroad, in the fields, in the laundries and the factories of one coast. Exclusion did not break the railroad monopoly. It fed it a story. The monopoly remained. The people who had built it were barred at the door.

Stanford's own record is the cleanest exhibit. As governor he had called them a degraded people. As railroad president he had wanted them by the hundred thousand, to bring the wage down. The inconsistency is not a mystery and it is not a private paradox. It is a sequence. Use a people when you need their hands. Describe them as a danger when their hands make your neighbors angry. Sign nothing yourself if other men will sign it for you.

That August of 1882, a drawing in the *Wasp* called *The Curse of California* put the railroad on the page as an octopus, tentacles on the farmers, the miners, the whole state (Rayner 2008). And that fall, California voters tried to break it. They elected an antimonopoly governor, George Stoneman, a Democrat, and a legislature that looked, on paper, like the railroad's worst day ([@white-railroaded]). The new constitution already promised a commission and a harder hand on rates. Stoneman wanted taxes, including back taxes, and a bill to outlaw the special contracts by which the railroad charged one shipper a quiet price and another shipper a punishment ([@white-railroaded]).

Then, in 1883, Ellen Colton's lawsuit against the railroad partners brought her late husband's mail into open court (Rayner 2008; [@white-railroaded]). Huntington's private letters from the 1870s—the "hungriest set of men," the $200,000 bill, the "high ground"—were read aloud and reprinted in the pamphlet *How Members of Congress Are Bribed*. What the letters did, five years after Colton died on Nob Hill, was turn the private voice of the railroad into a public document. Huntington's defense, for the rest of his life, was sarcasm and a bad memory.

In 1884 Governor Stoneman called a special session of the California legislature to do railroad business and nothing else ([@white-railroaded]).

Count the votes he thought he had: a Democratic majority of sixty-one to nineteen in the assembly, and thirty-two to eight in the senate ([@white-railroaded]). The bills sailed through the assembly. In the senate, bill after bill ended in a tie, and the lieutenant governor, a Democrat, broke the tie against them ([@white-railroaded]). Nine of ten Democratic senators from San Francisco voted the railroad's way. They were the lambs of a man the city called the Blind Boss ([@white-railroaded]).

His name was Christopher Buckley ([@white-railroaded]). Buckley had publicly supported the antimonopoly bills. Then his senators killed them. The reason was not that Buckley loved Stanford; it was that a fight between a railroad and a reform wave was an opportunity only a fool would waste ([@white-railroaded]). The railroad could taste defeat, or it could sweeten Buckley's pot. Nineteenth-century slang for the dollar was *simoleons*. Buckley liked them ([@white-railroaded]).

The Pacific Railway Commission, looking back, estimated that the Southern Pacific spent about $700,000 on politics that year—roughly $26 million now ([@white-railroaded]). It did not all go to one envelope. It went into a stew: railroad money, kickbacks, payments from anyone who needed a franchise or a contract, protection, the ordinary boodle of a city boss, mixed until a senator could not have told you which dollar had started in a freight tariff ([@white-railroaded]). Once the money is in the stew, every dollar looks like every other dollar.

"The political marriage of Leland Stanford and Christopher Buckley gave birth to the Octopus," Richard White writes ([@white-railroaded]).

The public had the image in the *Wasp* before White had the marriage certificate. What the 1884 session added was a date and a method: not a metaphor floating free, but a boss, a senate, a special session, and a pile of money large enough to turn a sure majority into a tie. The voters had built a commission and elected a hostile governor. The railroad did not repeal the commission. It bought the senate votes that kept the commission's teeth in a drawer. The law stayed on the books, where it could be pointed at. The practice stayed in Buckley's pocket, where it could be used. A government that looks like it is governing the railroad, and a railroad that is governing the government: that is the octopus. Not the animal. The arrangement.

While the legislature sat in California, the Stanfords were in Europe.

March, 1884. Florence, Italy. Leland Stanford was a railroad president and a former governor. The house on Nob Hill had a dome seventy feet high. He and his wife Jane had one child, Leland Stanford Jr., born on May 14, 1868 (Harris 2023). He was fifteen. In a hotel in Florence, the boy had a fever. David Starr Jordan would later call the illness Roman fever; university histories call it typhoid [NEEDS SOURCE].

On March 13, 1884, the fever killed him, two months short of his sixteenth birthday (Harris 2023). His mother would keep his room as he left it. His father had spent twenty years turning a franchise into a government.

The Caption (1885–1886)

In March 1885, a year after his son died in Florence, Leland Stanford—with Christopher Buckley's help—bought his way into the United States Senate ([@white-railroaded]). Senators were still chosen by state legislatures, not by the voters. A boss who owned a city's delegation owned a piece of the Senate.

A year later, Stanford helped put a second man there. George Hearst had been lucky in mining on a scale that looks like fiction: the Comstock, the Homestake, the Anaconda, claims that became the backbone of American metal (Rayner 2008). He wanted a Senate seat, and in those years a rich man campaigned by feeding legislators. Hearst got there in 1886 with the support of Leland Stanford, who chipped in because Hearst's opponent was Aaron Sargent, a man in Huntington's orbit (Rayner 2008). Hearst was appointed to a short vacancy on March 23, 1886, and later chosen for a full term beginning March 4, 1887, the same day he would hand the masthead of the *San Francisco Examiner* to his twenty-something son, "W. R. Hearst, proprietor." William Randolph Hearst would hire Ambrose Bierce as the first of his high-salaried writers, and Bierce would use the *Examiner* to call Huntington's head a dromedary, give him one leg in the grave, one arm in the treasury, and one eye on the police, and write, when Huntington told a committee he could not recall, that he took his hands out of his pockets long enough to be sworn (Rayner 2008). Stanford's feud with his partner financed, at one remove, the most widely read attack on Huntington.

That same year, 1886, a memo prepared for the president of the Union Pacific ran down the New York papers one at a time, the way a quartermaster runs down boots ([@white-railroaded]). One sheet existed to blackmail. One would print what it was handed. And the *Star*, the memo said, did "as Huntington directs" ([@white-railroaded]). As White wrote: "Nineteenth-century Americans were not shocked by the corruption of the press" ([@white-railroaded]). What frightened people was the use. A railroad that can raise a rate and then rent the description of the rate is not only a carrier. It is an editor.

And in January of 1886, the Southern Pacific's long tax fight reached the Supreme Court of the United States (Winkler 2018).

California had tried to tax the railroad's property without letting the company deduct its mortgages the way an ordinary landowner could (Winkler 2018). The Southern Pacific had refused to pay, turning the refusal into test cases that ran for years while counties spent time and money they did not have. The judge who had heard the cases on the federal circuit in California was Stephen Field, a justice of the United States Supreme Court riding circuit, which justices still did (Winkler 2018).

Leland Stanford had a close relationship with Justice Stephen Field (Winkler 2018). When the railroad decided to bring the test cases, Field recommended a lawyer, a law professor named

John Norton Pomeroy, who admired the way Field wrote opinions (Winkler 2018). Recommending a lawyer to a friend is not, by itself, a crime. Field then did the thing modern judicial ethics would forbid: he presided. He did not step aside. And during the litigation, he shared with the railroad's lawyers confidential memoranda about the case written by other justices (Winkler 2018).

In the circuit opinions, Field had said corporations were persons protected by the Fourteenth Amendment (Winkler 2018). The amendment had been ratified in 1868 for a different person: the formerly enslaved, and anyone a state might try to crush. Its promise was that no state shall "deny to any person within its jurisdiction the equal protection of the laws." Field's move was to slide the railroad into the word *person*. Treating the company's property differently from a widow's property was, he wrote, "the very essence of tyranny" (Winkler 2018). He offered a comparison meant to shock: a tax law that changed its rules according to whether the owner was a white man or a Black man would be an outrage; so, he said, was a tax law that changed its rules because the owner was a corporation (Winkler 2018).

The amendment had been written because states were denying rights to Black Americans. Field used that sentence to protect a railroad's balance sheet.

Now, in January 1886, *Santa Clara County v. Southern Pacific Railroad* was before the full Supreme Court (Winkler 2018). The question on the table was Field's question: does the Fourteenth Amendment cover corporations? Before the argument, Chief Justice Morrison Waite told the lawyers the Court did not want to hear about it (Winkler 2018).

When the decision came down on May 10, 1886, the Court ruled for the railroad on the narrowest ground available (Winkler 2018). The county had included the value of the fences along the tracks in the tax bill. California law did not allow that. Because the case could be decided on that ground, the opinion said it was "not necessary to consider any other questions," including the grave constitutional questions decided below (Winkler 2018).

The Supreme Court did not hold, in 1886, that corporations are persons under the Fourteenth Amendment. It said it did not need to reach the question. It decided a fence.

Then the court reporter went to work. His name was J. C. Bancroft Davis (Winkler 2018). His job was to write the syllabus and the headnotes, the summary at the front of a published opinion in the *United States Reports*, so a busy lawyer could see what was decided without reading the decision. Headnotes are not the law. They are a caption. Davis's caption said this:

"The defendant Corporations are persons within the intent of the clause in section one of the Fourteenth Amendment to the Constitution of the United States, which forbids a state to deny to any person within its jurisdiction the equal protection of the laws" (Winkler 2018).

Above the opinion, Davis printed a remark he attributed to the Chief Justice, from before the argument:

"The Court does not wish to hear argument on the question whether the provision in the Fourteenth Amendment which forbids a state to deny to any person within its jurisdiction the equal protection of the laws applies to these corporations. We are all of the opinion that it does" (Winkler 2018).

Davis had checked with Waite before he published. Waite's terse private reply survived: "I leave it with you to determine whether anything need be said about it in the report inasmuch as we avoided the constitutional question in the decision" (Winkler 2018).

Davis printed it anyway. He had once been president of a railroad himself, the Newburgh and New York Railway (Winkler 2018). A reader who stopped at the headnote would walk away believing the Court had held that corporations were people. The Court had not.

Three years later the caption would become a citation. In 1889, in *Minneapolis & St. Louis Railway Co. v. Beckwith*, a railroad that had killed three hogs in Iowa and been ordered to pay $24 in damages—roughly $900 today—would spend tens of thousands of dollars taking the case to the Supreme Court, where Stephen Field would write the opinion upholding the statute, state that corporations are persons under the Fourteenth Amendment, and add one sentence: "It was so held in Santa Clara County v. Southern Pacific Railroad" (Winkler 2018). Field had written, three years earlier, an angry note that the Court had dodged the constitutional question (Winkler 2018). He had not forgotten. He cited the dodge as if it were a decision. And by 1912, when a former congressional and Supreme Court librarian named Charles Wallace Collins counted every Fourteenth Amendment case since 1868—604 in all—only 28 would involve African Americans, who usually lost, while 312, more than half, would involve corporations, turning the amendment into what Collins called the "Magna Charta of accumulated and organized capital" (Winkler 2018).

Mussel Slough made the papers for a week. The headnote stayed in the bound volume. The railroad did not need to win every case; it needed the word *person* to be available the next time a state tried to tax a fence. And back in California, grief had already given the fortune a new shape. Grief does not launder a fortune, but it can change what the fortune is for. The octopus had an heir, and the heir would be made of stone, and scholarships, and a clause that said forever.

Part 4: The Boy and the Farm

1884–1905

The Room Across the Hall (1884–1889)

March 13, 1884. Florence. A boy of fifteen died two months short of his sixteenth birthday, and his parents decided the fortune would outlive them.

Leland Stanford Jr. had been born on May 14, 1868. Back in the house on Nob Hill in San Francisco, his bedroom sat across the hall from his mother's. Jane Lathrop Stanford turned the tutor's room next door into a study where she worked with her secretary, Bertha Berner. She kept the boy's bedroom as it was when he died. She later described what she did there: going "into dear Leland's room and kneel down and tell all my sorrows to the blessed Jesus. I feel my darling boy and my dear husband are pleading with me." (White 2022)

The sentence that came out of the night in Florence has two wordings, and they are not the same sentence. David Starr Jordan wrote in his memoir that after a long and dreary night the father woke with these words on his lips: "The children of California shall be my children." Jordan said the father told him the story in 1891. Malcolm Harris, in *Palo Alto*, prints a different pronoun, having Leland tell Jane, "The children of California shall be our children," when they decided to build on the farm (Harris 2023). It is a remembered sentence, not a diary line. *My children* makes the university a father's replacement son. *Our children* makes it a couple's.

Herbert Nash, the boy's tutor, wrote a biographical sketch that put Jane at the center: young ladies and the conventions of society had no attractions for the boy, but toward his mother he would display the most attentive gallantry. Jane made Nash the university librarian. The sketch became the life that was not to be questioned.

Six weeks after the boy died, on April 24, 1884, in Paris, Leland Stanford made his will, looking to the endowment of the university. Jane made her will the same way, and copies were sent to America.

In the fall of 1891, after the university opened, a story went around that the school had been founded under spiritualist influence, and that a medium named Maud Lord Drake had been the guiding hand. In 1892 Leland and Jane sat down with Jordan and dictated a statement they wanted kept. Maud Lord Drake, the statement said, had been unknown to them until they met her at a séance with Ulysses S. Grant and his wife in October 1884. About that time, the statement added, Mrs. Drake was detected in fraud.

Jane added a sentence of her own: "No spiritualistic influence affected the decision. Mrs. Drake had no more to do with it than a babe unborn."

No record survives of what was said in that parlor. Jordan added in his printed statement that both Stanfords were, for a time, deeply interested in a spiritualism that seemed to offer a demonstrable life after death. With the Grants they attended several séances in Washington.

Jordan wrote that they never got, through mediums, any evidence they found convincing. The wills came first, in Paris. The Drake claim came after, and they spent a page denying it.

The séances the documents describe at the Nob Hill mansion were conducted with the Reverend John Philip Newman and his wife, friends of the Grants, trying to reach the boy (White 2022, 45). Mrs. Newman was the chief medium. Amid luminous clouds, the boy did not appear. A different child did, a spirit the Newmans called Ensign, and the medium reported that heaven rejoiced in the decision to dedicate the fortune to God and the church. That was a séance claim, not a deed. Bertha Berner later wrote that after a trip to Australia, where the dead did not answer, Jane never again attended a séance.

The Grants were in the story for a more ordinary reason: patronage, the trade in favors inside a circle that does not have to call them favors. A railroad pass. A pulpit. Later, an embassy.

On June 2, 1884, from New York, Ulysses S. Grant wrote to Leland Stanford. Before the failure of the Wall Street firm Grant and Ward, Grant's youngest son, Jesse Root Grant, had sent his wife, child, and nurse to California and had expected to follow. His ready money had been in the firm.

"He deposited with the firm on the 2nd of May $86,000," Grant wrote (White 2022).

Grant asked whether it was too much, under those circumstances, to request a pass from New York out and back for the son and the family. He told Stanford to refuse if it was inconvenient. Eighty-six thousand dollars in 1884—roughly $3.2 million today—had been wiped out in a Wall Street partnership that carried a former president's name. On July 2, an invitation to Long Branch followed. The circle held. Five years later, in 1889, Frederick Grant, the general's older son, wrote to Stanford about his appointment as American minister to Austria. A pass, then a post. The office was public. The ask was private.

The pulpit was louder. The Reverend John Philip Newman had been the Grants' pastor in Washington and then in New York. In 1884 he was between jobs. The Stanfords brought him west to preach the boy's memorial. Richard White writes that they reportedly promised him $10,000, equal to his yearly salary (roughly $372,000 today), though the payment is not in an account anyone has produced (White 2022).

The boy traveled west in a funeral car draped in black. While they were bringing the body home, the Stanfords looked at Cornell, Harvard, the Massachusetts Institute of Technology, and Yale. What Jane said about Yale was an insult, and it is on the page. She said graduates of Harvard and Yale were of benefit to no one, including themselves. Her husband, she said, employed "six of them as car-conductors on the Market Street Line now." (White 2022)

Back in California, publicity claimed the memorial featured the greatest floral displays on the West Coast, supposedly costing $20,000 (roughly $744,000 today), a figure White marks as a

publicity claim rather than a bill. White writes that if it were not for the parents' grief, the praise of a rich dead boy the preacher had never known would have been risible. Newman's sermon drew charges of blasphemy for comparing the boy to Christ, and set off a church fight over whether a minister who was not an Episcopalian should have been in an Episcopal service.

A dead child, a borrowed Christ, a fee the size of a president's year. The university had been decided in April, in Paris. The performance was in the winter. Grief was real. So was the stage.

The Dead Hand (1885–1891)

On March 9, 1885, the California legislature passed an enabling act so a private deed of trust could found a university. The title began: "An act to advance learning, the arts and sciences, and to promote the public welfare." It was written as a general law. It had one institution in mind.

On November 11, 1885, Leland Stanford and Jane Lathrop Stanford signed the Founding Grant. By then Leland was a United States senator; senators were still chosen by state legislatures, and he had taken the seat that March. The deed stated their desire to found, on their estate known as the Palo Alto Farm, in San Mateo and Santa Clara counties, "a University for both sexes," with colleges, schools, museums, galleries, and everything necessary to a university of high degree. They conveyed the land to a board of trustees and to their successors forever.

Two clauses in the instrument formed the lock:

"the principal thereof shall forever remain intact, and that the rents, issues, and profits thereof shall be devoted to the foundation and maintenance of the University hereby founded and endowed."

And, inside the rights the founders reserved for themselves while they lived:

"but this reservation does not include the right or power to sell or encumber any of the real property granted."

A trust is property held by people who do not own it for themselves. The trustees spend the income on the purpose named in the deed. The principal stays. Lawyers call a clause that keeps ruling after the giver is dead a dead hand. The same lock can freeze a mistake. The clause that stopped a sale of the land was the clause that would later stop Jane from turning the school into a boys' college.

One name on the trustee list carried over from the railroad's legal wars: Stephen J. Field, a justice of the Supreme Court of the United States. The same man in whose Ninth Circuit courtroom a railroad tax case had grown a caption treating corporations as persons now sat on the board of the railroad president's university. The deed did not blush.

The 1885 grant did not forbid tuition; it left the question to the trustees. What the founders told David Starr Jordan was a promise of a fortune. The ultimate endowment, they said, would be about $30 million. Jordan had fought to get Indiana University's budget to $50,000 [NEEDS SOURCE]. He was offered thirty times that, someday.

Someday was not a deposit. Before Leland died, he gave the university three ranches totaling 85,000 acres. The Vina ranch never turned a profit, and its vines died of phylloxera. Gridley's grain was in decline. The Palo Alto stables usually lost $7,000 a year, most of Jordan's $10,000 salary. Trustee George Crothers said the ranches brought "nothing but taxes." White notes that taxes did exceed the revenue, and the founding papers prohibited a sale (White 2022).

Frederick Law Olmsted, the landscape architect who had made Central Park, was hired to site the campus. He wanted the buildings in the foothills. The Stanfords wanted flat land. They were building a monument you could march through. Olmsted suggested, and Charles A. Coolidge of Shepley, Rutan and Coolidge—the successor firm of the architect H. H. Richardson—designed, the low buildings of the inner quadrangle, Encina Hall for the men, and a long drive from the railroad station planted in Mexican fan palms and Canary date palms: Palm Drive. Flat land. A quad. An arch. A church at the center.

In *Imperial San Francisco*, Gray Brechin describes the result: a symmetrical composition of sandstone, continuous arcades, and a Romanesque chapel loosely modeled on Trinity Church in Boston, the whole roofed in red tile because the Stanfords wanted the buildings to resemble California's missions (Brechin 1999). Brechin's word for the place is a "supermission."

At the entrance stood a triumphal arch of San Jose sandstone, more than a hundred feet tall, second in size only to the Arc de Triomphe—a "Babylonian eruption," White writes, in the middle of the mission-style quiet (White 2022). The frieze on the arch was called *The Progress of Civilization*. The donors put themselves in it, on horseback, ahead of a locomotive entering the Sierra. Jane suggested the outline of that story and gave herself and her husband featured roles. Augustus Saint-Gaudens, the most famous sculptor in the country, turned her suggestions into the program.

Leland wrote to Jordan a sentence Brechin quotes: our graduates are to be "[missionaries] to spread correct ideas of civilization" (Brechin 1999).

The old Franciscan missions had sat on crown land the friars held, in the official story, for the Native people they baptized, before secularization in the 1830s scattered that claim. The families who had passed through the baptismal books at Mission Dolores and Mission Santa Clara were the Muwekma Ohlone; the campus sat in their watershed. Under the missions lay Native labor. Under the quad lay a railroad fortune graded in large part by Chinese workers paid less than white crews, as the Central Pacific's own managers had admitted under oath (Chang 2019). Adobe and red tile had become sandstone and red tile.

Jordan explained in his memoir the names chosen for the roads in front of the quad. The road the quadrangle faced they called Serra Avenue, for Father Junípero Serra, who had founded the first of the California missions. Lasuen Street was named for Fermín Lasuén, Serra's successor, the contained man after the impulsive one, who had carried the mission chain farther, including La Purísima. Other campus roads took the names Cabrillo, Portolá, and Crespí. The founders wanted the mission era in the directory, and they got it.

A few miles from the farm, in the same Santa Clara Valley, another widow was building.

Sarah Lockwood Pardee Winchester signed her name Sallie. She was the heir to a repeating rifle. Her husband, William Wirt Winchester—the son of Oliver Winchester, the New Haven manufacturer who had turned a shirt business into the Winchester Repeating Arms Company and had given Yale the land for an observatory in 1871 (built in 1882)—had died of tuberculosis in New Haven on March 7, 1881. In 1886, as historian Mary Jo Ignoffo established from the house's papers, Sarah bought an eight-room farmhouse near San Jose and named it Llanada Villa (later 525 South Winchester Boulevard).

The Stanfords laid their cornerstone on May 14, 1887, the day their son would have turned nineteen. Two grieving households, both rich, started monuments about fifteen miles apart, within roughly a year of each other.

The tourist story has the two widows backward. The story goes that a Boston medium told Sarah Winchester that the rifles had stacked up so many dead that the spirits would haunt her unless the hammers never stopped—producing stairs to nowhere and a window in a floor. The record does not show that medium. The name later books give him, "Adam Coons," has no trace in her lifetime; it first appeared in 1967, in Susy Smith's *Prominent American Ghosts*. Sarah's companion, Henrietta Severs, said she was not superstitious. She had been raised Baptist, and after 1908 she attended St. Paul's Episcopal Church in Burlingame. A carpenter added the number thirteens to the house after she was dead; the first printed mention of them was in 1929. The documented spiritualist in the valley was the railroad widow on the farm. And there is no evidence Jane Stanford and Sarah Winchester ever met or wrote to each other. The person who connected them was a lawyer in San Jose named Samuel Franklin Leib, who was a Stanford trustee and also Sarah Winchester's legal and financial adviser.

In 1893, nearly a decade after Leland Jr. died, curators from the university museum measured every piece of furniture in the boy's Nob Hill bedroom, noted where it stood, and moved his things into the campus museum. They told Jane they had arranged the room "just as it was in your city home" (White 2022).

Two Suits and a Bicycle (1891–1896)

October 1, 1891. The inner quadrangle was still being finished on the morning of the opening ceremony.

The university's public history counts 555 students in that first year, about 130 of them women, roughly one in four. Richard White's history counts 559, mainly from California, which made it the largest college in the far West (White 2022). Four students separate the two counts. David Starr Jordan stood before the pioneer class.

"It is hallowed by no traditions; it is hampered by none," Jordan said. "Its finger posts all point forward."

A school with no traditions had not yet had time to lie about itself.

In the crowd was Herbert Hoover. He had been orphaned at nine in Iowa—his father, a Quaker blacksmith, died first, and then his mother—and relatives had passed him along until he ended up in Oregon with an uncle who did his duty without much tenderness (Whyte 2017). A visiting engineer had told the boy that mining engineers were scarce and well paid. The East Coast schools were beyond his means, and the new college in California was not going to charge tuition.

In the summer of 1891 Hoover went to Portland and sat the entrance exams. He came up short. Joseph Swain, a Quaker who would teach mathematics at Stanford, gave the exams and watched him attack plane geometry.

"I observed that he put his teeth together with great decision and his whole face and posture showed his determination to pass the examination at any cost," Swain recalled (Whyte 2017).

Swain admitted him on the condition that he bone up. On August 29, 1891, a few weeks after he turned seventeen, Hoover stood at the Salem station with a ticket for San Francisco, two suits in a satchel, and a bicycle. He boarded the train. At the edge of the campus he studied, failed English, and was waved through again on the promise of effort. He was given the first room assignment under the red tile of Encina Hall, and for the rest of his life he liked to say he had been Stanford's first student. There was no electricity and no hot water for weeks after classes began.

Free, in the grant, meant the college would not send a bill for instruction. It did not pay for the train, the room, or the year of geometry a boy had missed.

Less than two years later, on June 21, 1893, Leland Stanford died.

The Panic of 1893 was already on, the start of the deepest depression the country had been through. The Southern Pacific, the basis of the fortune, was missing dividends and could not pay full interest on its bonds. Within two years of opening, White writes, the new college was in danger of collapsing (White 2022). Leland's will left the university $2.5 million in cash (roughly $93 million today). It left about $7 million (roughly $260 million today) largely to his relatives, not

to Jane's. The bulk of the estate, whatever it was actually worth, went to Jane, and the university lived on her largesse. The three ranches still could not be sold.

Then the federal government sued. In 1894 the Justice Department went after the Stanford estate for $15 million—roughly $580 million today—and froze the assets, including the bequest to the university.

The money was the old government construction loans to the Central Pacific, come due. The Central Pacific was a California corporation, and California law made stockholders personally liable for debts the company did not pay. Stanford and his partners had drained money out of the Central Pacific into their own pockets, sold the hollowed-out railroad to British investors, and had the Southern Pacific, which they still controlled, lease it back. Because the British investors were afraid of California's stockholder-liability law, the Associates had kept the Central Pacific stock in their own names. In 1894 the government came to the names.

"People think that Governor Stanford left me a very rich woman," Jane told a visitor. "I thought so myself, but it now seems that I was left a legacy of debt, trouble and worry." (White 2022)

She kept the university's doors open on the household allowance the probate court allowed her, on an insurance policy, and on the sale of personal belongings. She cut faculty salaries ten percent. The faculty, White writes, would have left if there had been anywhere to go in a depression. Jordan stayed and defended the Stanfords, because the alternative was a dead college. He called it the cause of higher education in America. That year, the cause was a widow selling furniture.

Jane's lawyer, Russell Wilson, hired Joseph Choate to argue the case at the Supreme Court. In 1896 the Court ruled that the stockholders were not personally liable, even if California law had said so when the railroad was incorporated; the government would have to collect from the Central Pacific itself. Official university histories treat that ruling as the end of the money trouble. White writes that it solved nothing. Just before the decision, Choate wrote to Jane that a victory might put "perhaps $40 million" in her hands—a guess White calls a "considerable exaggeration." Choate's advice was to rewrite her will and fix a maximum amount the university should be allowed to hold. Her public promise had been to pour the fortune in. Her lawyer was telling her to pour less.

The boy was ten years dead. The university was five years old. The fortune was a lawsuit with a palm drive.

The Expanding Peoples (1898–1900)

February 25, 1898. Washington. The Secretary of the Navy, John D. Long, had gone home for the afternoon. Theodore Roosevelt, the assistant secretary, was acting secretary for those hours. He cabled Commodore George Dewey in Hong Kong, marking the message secret and

confidential: "Keep full of coal. In the event of declaration of war Spain, your duty will be to see that the Spanish squadron does not leave the Asiatic coast, and then offensive operations in Philippine Islands."

The next day Long wrote in his diary that "the very devil seemed to possess him yesterday afternoon." Long left the order standing.

Ten days earlier, on February 15, 1898, the battleship *Maine* had exploded in Havana harbor, killing 266 men (Navy inquiries in 1898 and 1911 blamed a mine; a 1976 study led by Admiral Hyman Rickover concluded the likeliest cause was a coal-bunker fire next to a magazine, and the question remains debated). President William McKinley asked Congress for war, and Congress declared it in April. As Navy historians note, the story that Roosevelt engineered the war single-handedly has largely been refuted; an 1896 Navy war plan already pointed at Manila. Gray Brechin writes that the Navy Department had planned the seizure of the Philippines "two years before Gridley was ready to fire" (Brechin 1999). What Roosevelt did on one February afternoon was aim the fleet.

On May 1, 1898, Dewey's squadron slid past the old fort at Corregidor into Manila Bay. "You may fire when you are ready, Gridley," Dewey told his captain. Six hours of shelling sank the Spanish fleet.

Hawaii came into the American empire the same summer, annexed by joint resolution on July 7, 1898, while the war was still on. The resolution carried the name of Representative Francis G. Newlands, an heir by marriage to the William Sharon silver fortune who had laid out Burlingame and Hillsborough, the suburbs just up the peninsula from the Stanford farm. In October, his brother-in-law, Hall McAllister Jr., wrote that keeping "the entire Philippine group" would prove of incalculable benefit to San Francisco (Brechin 1999).

On December 10, 1898, in the Treaty of Paris, Spain ceded Puerto Rico and Guam, gave up Cuba, and sold the Philippines to the United States for $20 million (roughly $800 million today). Fighting between American and Filipino forces around Manila began on February 4, 1899. Two days later, on February 6, the United States Senate ratified the treaty by a vote of 57 to 27, barely the two-thirds required. The $20 million went to Spain.

Americans called the war that followed an insurrection. At the peak, 75,000 U.S. troops were in the islands (Brechin 1999). Both sides committed atrocities: according to the State Department's Office of the Historian, U.S. forces at times burned villages, reconcentrated civilians, and used torture, while Filipino fighters tortured captured soldiers and terrorized civilians who cooperated with the Americans. General William Shafter told the *Chicago News* that it might be necessary to kill half the Filipinos so the remaining half could be advanced to a higher plane of life than their present semi-barbarous state (Brechin 1999).

David Starr Jordan campaigned in public against annexing the Philippines. His argument was not a sermon on consent. As Harris records, Jordan argued that the colonies were "incapable of civilized self-government" and "not habitable by Anglo-Saxon people," adding, "The home cannot endure in the climate of the tropics" (Harris 2023; White 2022, 104).

Across the China Sea, the Empress Dowager Cixi had held the real power of the Qing throne since 1861. In 1898 she confined the reforming Guangxu Emperor under house arrest and moved to stop foreign businessmen from treating her realm as a buffet (Whyte 2017, 76). Western newspaper reports said she had "the soul of a tiger in the body of a woman." In the summer of 1900 she backed the Boxers, the martial, magical movement that meant to drive the foreigners out.

In June 1900, Herbert Hoover—twenty-five years old, the orphan who had come to Encina Hall with two suits and a bicycle—was a mining engineer in the foreign settlement at Tianjin. On June 14 a Boxer force, described as 25,000 strong, hit the Chinese city and burned the French cathedral (Whyte 2017, 77–78). About 2,000 foreign soldiers held the settlement, protecting 700 foreign civilians—300 of them women and children—and thousands of Chinese Christians. Imperial troops sent to scatter the Boxers joined them and turned their guns on the compound. Hoover helped build barricades at the gates out of sacks of sugar and grain, fought fires, and carried food on his bicycle, chewing nuts without shucking them. His wife, Lou Henry Hoover, refused to wait in the basement and ran her own errands with a .38 Mauser on her hip.

Rail and telegraph lines were cut. Back at Stanford, faculty and students read a newspaper report that some of their best alumni were probably dead: "It is almost certain that they have fallen victims to the savage ferocity of the Mongolian hosts, unless they managed to make their escape previous to the bloody massacre, which is not likely." (Whyte 2017)

They had not fallen; casualties inside the foreign compound were light. And the coal was the point.

Hoover considered the Kaiping mines, northeast of Tianjin, the only successful mining venture in China by Western standards (Whyte 2017, 75–80). Nine thousand men worked three collieries directed by Hoover's employer, Chang Yen-mao. Chinese law kept the mines with the emperor, and Chang could not sell them to foreigners. During the siege the Boxers stopped the coal, the share price fell, and Russian and Japanese troops were said to be planting flags on the pits. The foreigners in the settlement arrested Chang on suspicion that he was negotiating with the Boxers; Whyte notes there was no firm evidence, but they locked him up anyway. From confinement Chang signed a power of attorney telling Gustav Detring to put the company under the British flag on the best terms he could get.

Hoover and Detring wrote the deal and backdated the papers so the reorganization would look as if it had been agreed to before the uprising. The new British firm was capitalized at one million shares of one pound each, of which existing Chinese shareholders were to receive

375,000; what the London promoters would take for themselves was not written down. Hoover refused to let the deed go straight to his London boss, Charles Algernon Moreing. Having already told Moreing that without absolute control the game was not worth the candle, Hoover had the deed made out to himself as trustee.

Whyte calls the takeover "the biggest mining transaction in the history of China" (Whyte 2017, xi). In *Palo Alto*, Harris quotes the historian Ian Phimister: the previous Chinese owners "were defrauded of their property" (Harris 2023, ch. 2.3).

The Dowager Queen (1899–1904)

In California, Jane Stanford was rearranging her own empire.

In 1899, afraid that a campaign to amend the state constitution in Stanford's favor might fail and pull her gifts down with it, she deeded the entire university endowment—$25 million, roughly $1 billion today—with no reservations and no trust clauses, to a single man: the San Jose lawyer and Stanford trustee Samuel Franklin Leib (White 2022, 79). The idea was that if a court ruled her gifts to the university illegal, Leib could deed them back to her. George Crothers later called it a strange procedure; had Leib died or acted on the deed, the university might never have recovered the money. It was the same Samuel Leib who handled the property, investments, taxes, and will of Sarah Winchester down the road in San Jose. One lawyer's filing cabinet held the paperwork of the railroad fortune and the paperwork of the rifle fortune.

On May 31, 1899, on the advice of Russell Wilson, Jane sold her Southern Pacific stock for a reported price of $11 million (roughly $440 million today). Crothers later called the sale one of the great disasters to befall the university: it was $7 million (roughly $280 million today) less than the Speyer banking house got when it sold the same stock the following year to E. H. Harriman, the man who already ran the Union Pacific.

On the same day she sold the stock, May 31, 1899, Jane signed what Crothers called the Great Deed, conveying most of her estate back to the university. Crothers called it a great but defective grant. A clause inserted by Mountford Wilson gave Jane the power "to grant, bargain, sell, convey or lease" all the property she had just given. She kept so much control that Crothers thought the gift had, in effect, not happened; she treated it lightly enough, he said, to pull millions back out. The dead hand of 1885 and the key to unlock it were in the same folder.

On November 6, 1900, California voters passed the constitutional amendment 137,667 to 67,737, about two to one. It confirmed the founding trusts and let the legislature exempt Stanford's lands and trust funds from taxation while they were used for education. In exchange, the amendment stated that California students were not to be charged tuition unless the legislature later authorized it (a later education code did). The yes votes subsidized a private university by not collecting property tax from its land.

Five days after the polls closed, on November 11, 1900, David Starr Jordan accepted a letter of resignation that the economist and sociologist Edward Ross had written back in June. Jordan had held it until after the election, hoping the situation would change. He accepted it, he said, "with great reluctance," praising Ross as a teacher and a man.

Ross and others pointed out that if the dismissal had become news before November 6, the tax-exemption amendment might have failed. With the vote safely counted, Ross called in the San Francisco reporters and blamed Jane Stanford. She disliked his politics: he had spoken publicly against "coolie" labor—the era's slur for Chinese workers—and in favor of municipal ownership of utilities rather than private franchises.

On the morning of November 14, 1900, the *San Francisco Chronicle* ran the headline: "Dr. Edward Ross Forced Out of Stanford University" (White 2022).

Jordan had not yet read Ross's statement when reporters reached him, and his on-the-record praise of Ross ran next to Ross's accusation. Ross publicly thanked his "great chief"—what White calls a Judas kiss that placed the firing squarely on Jane. Telling the story of the Ross affair, White writes that "Jane Stanford played the dowager queen" (White 2022, 95). It was her university. The dismissal was her due. Ross lost his job and won the headline; the tax exemption was already in the constitution.

During the same years, Jane moved to close a different door. By 1899, according to the university's own magazine, women made up about two in five Stanford students. The 1885 grant had stipulated "a University for both sexes." Crothers stopped Jane from issuing an instruction that would have told the trustees to break the grant outright, warning her that breaking it risked the entire trust. He got her to settle for a cap of 500 women. In 1904 she still called coeducation an unsettled experiment.

In the center of the quad she built Memorial Church, which she called her "Koohinoor." The facade read: *Erected to the Glory of God and the Loving Memory of My Husband Leland Stanford*. She chose the wall inscriptions herself, painting a line she attributed to her husband: that the best form of religion is trust in God and a firm belief in the immortality of the soul. She wanted nondenominational Christianity with the dogma stripped off; White's verdict is that the church "spoke mainly to a sect of one" (White 2022). Meanwhile, in Australia, her brother-in-law Thomas Welton Stanford funded a Stanford fellowship in psychic research and donated his collection of "apports"—objects he believed spirits had delivered into séances.

From 1899 to 1905, George Crothers—who worked for Jane without a salary so she could not treat him as an employee to be dismissed at will, and who became a trustee in 1902 in a seat some had expected to go to Herbert Hoover—feared she might hand the university away altogether. Catholic Archbishop Patrick Riordan had her ear and had tried, and failed, to stop her from signing the Great Deed in 1899. Jane toyed with turning Stanford into a boys' school, giving it to the Catholic Church, dividing the fortune among her heirs, or giving the endowment

to Columbia—a cruel dig at Jordan, since Leland had first offered the Stanford presidency to Columbia's Nicholas Murray Butler. By 1902 she was a catechist, taking instruction to enter the Roman Catholic Church. Crothers argued that a hierarchical church grown rich on the sacraments would betray her hope of a purer Protestantism. She postponed the conversion, though she still summoned Crothers in the middle of the night with rumors of students in the campus's shady lanes—stories that came back false—and worried that women students endangered the men's celibacy. She even told Crothers it might be better if she resigned as surviving founder before she made some foolish change. Crothers answered the midnight summonses and kept the deed intact.

Outside the farm, the Philippine war ground on. On November 27, 1900, Major Edwin F. Glenn subjected Tobeniano Ealdama, the town *presidente* of Igbaras on Panay, to the "water cure"; court-martialed and convicted, Glenn received a one-month suspension and a $50 fine. On September 28, 1901, at Balangiga on Samar, townspeople attacked a company of the Ninth U.S. Infantry, killing dozens of soldiers while about 28 villagers died in the attack. Afterward, Brigadier General Jacob H. Smith told Major Littleton Waller he wanted no prisoners: "Kill and burn. The more you kill and burn, the more you please me." Asked the age limit, Smith said, "Everything over ten," ordering Samar turned into a "howling wilderness" where "even the birds could not live." Waller partly countermanded him—"we are not making war on women and children"—and Smith was court-martialed, admonished, and retired by Roosevelt.

Mark Twain, a vice president of the Anti-Imperialist League from 1901, asked whether the United States would go on conferring its civilization upon "the peoples that sit in darkness." Roosevelt declared the war over on July 4, 1902, though fighting continued; in all, according to the State Department, more than 4,200 U.S. combatants, more than 20,000 Filipino combatants, and as many as 200,000 Filipino civilians died.

On May 13, 1903, in San Francisco's Union Square, President Roosevelt dedicated the Dewey Monument, a Roman victory column with the names of Dewey's ships cut into the base, *Olympia* first. He told the crowd that the proper place of Americans was "with the great expanding peoples, with the people that dare to be great" (Brechin 1999).

Room 120 (1905–1906)

January 14, 1905. At the Nob Hill mansion in San Francisco, seventy-six-year-old Jane Stanford drank from her bedtime bottle of Poland Spring water, tasted something bitter, and forced herself to vomit. A pharmacist analyzed the water and found a massive, lethal dose of strychnine, the alkaloid sold as rat poison (White 2022).

Nine days later, on January 23, Jane went down the peninsula to the Hotel Vendome in San Jose—the same town as Samuel Leib's law office and Sarah Winchester's Llanada Villa—to rest, see old friends, meet with trustees, and fight with Jordan over the university budget. Then she decided to sail for Hawaii.

On February 15, 1905, she boarded the steamship *Korea* in San Francisco, bound for Honolulu and Yokohama (the passenger list spelled Bertha Berner's name as "Benner"). The dock was crowded and military bands played for a fellow passenger: Major General Arthur MacArthur, commander of the Department of the Pacific, who had served as military governor of the Philippines from May 1900 to July 1901 before clashing with William Howard Taft. Jane had packed $75,000 in jewels—roughly $2.8 million today. On the voyage she spoke freely about the poisoned Poland Spring bottle to General MacArthur, his wife, and a Chicago grain merchant named Charles Sidney Dole (White 2022, 167–168). Nothing in the record shows that anyone later asked the general what she said.

Three days after the *Korea* sailed, on February 18, 1905, in the Blue Room of the White House, the Chinese minister Liang Cheng presented President Roosevelt with a portrait of the Empress Dowager Cixi, painted by the American artist Katharine Carl for the 1904 St. Louis World's Fair. Roosevelt accepted it for the United States. The other dowager's face was a diplomatic gift; the Palo Alto dowager was on a ship in the Pacific, talking about a bottle.

Jane docked in Honolulu looking, in the phrase of the record, "well and hearty." The governor of the Territory of Hawaii was George R. Carter, whom Roosevelt had appointed in 1903 after consulting, among others, David Starr Jordan (White 2022, 168). On February 21, Jane checked into the Moana Hotel at Waikiki, which had opened on the beach in 1901. She took Room 120. Bertha Berner and Jane's maid, May Hunt, shared Room 122.

Tuesday, February 28, 1905. Jane spent the day being happy. A carriage took her slowly along the windward coast because she wanted the ride to last; May Hunt remembered that she chattered and sang. Around one o'clock they picnicked in a grove. As Berner later told the *Hawaiian Star*, the hotel had packed sandwiches of graham bread, Swiss cheese, tongue and lettuce, hard-boiled eggs, gingerbread, oranges, chocolate candy, and cold coffee. Jane ate heartily of everything except the eggs. They were back at the Moana by four.

At half past six the three women went down to dinner. On the verandah Jane told a guest named Mrs. Grinbaum about the pleasant day. Dinner was a bowl of soup; it took ten minutes. She sat outside until eight or a quarter past, then went upstairs with Berner and asked her to set out her medicine.

Berner put a cascara sagrada capsule on the bed stand; cascara was an over-the-counter laxative whose capsules in that era often contained a trace of strychnine on the theory that it hurried the herb. Then Berner took a small bottle of bicarbonate of soda, two to two and a half inches tall with a mouth wide enough for a teaspoon, from a medicine basket. She asked May Hunt for a spoon, measured the half teaspoon Jane wanted, and left it beside the capsule.

"Give it time to get out of my stomach," Jane said, referring to the glass of water she would drink before taking the medicine (White 2022). She urged Berner to take some of the soda too. Berner declined.

Berner and Hunt left around nine. Jane called Hunt back because she was having trouble with the door lock; Hunt showed her how it worked and went to bed. Berner thought her employer fell asleep without taking the dose, then woke later and swallowed it.

Hours later, a voice filled the hallway. A guest across the hall, Adam Heunisch, opened his door and found Jane Stanford in her doorway, her arms thrown up.

"Oh I am so ill," she cried. "Get me a doctor." (White 2022)

Heunisch ran for the elevator. Dr. Francis Humphris lived in the hotel; Berner told the elevator boy the doctor should not even stop to dress. May Hunt tried to get Jane back to bed, but she would not go, gripping the door casing. She said a spasm had thrown her out of bed. When Humphris arrived, she told him she had been poisoned again and asked for a stomach pump. When he asked how she knew, she said she had been poisoned before and turned to Berner. Berner said the earlier Poland Spring bottle had held arsenic, then corrected herself: strychnine.

The stomach pump did not arrive in time. Unlike the massive dose in January, the Honolulu dose was a small, precise amount of pure strychnine alkaloid. Her jaws set. Humphris felt her muscles lock and took the glass from her hand. Berner watched the last spasm.

"Oh God, forgive me my sins," Jane said, her jaw stiffening. "Is my soul prepared to meet my dear ones?"

Then came the words that Richard White notes she would rarely if ever have spoken in life, having spent twenty years calling death a gentle passage into a higher realm:

"This a horrible death to die." (White 2022, 173)

Her body went rigid. The soles of her feet turned inward toward each other, insteps arched, toes pointing forward, knees wide. Her eyeballs protruded and her pupils dilated. Her jaws fixed, her fingers contracted with the thumbs dug into the palms, and her breathing stopped. From the moment Dr. Humphris entered Room 120 to the final spasm, ten minutes had passed.

Chemists in Honolulu found strychnine in the bicarbonate of soda and in her body; the physicians testified that a trace from a cascara capsule could not explain the death. On the evening of March 9, the coroner's jury returned its verdict, printed in the newspapers on March 10:

"Jane Lathrop Stanford came to her death at Honolulu, Island of Oahu, Territory of Hawaii, on the twenty-eighth day of February, A.D. 1905 from Strychnine poisoning, said strychnine having been introduced into a bottle of bicarbonate of soda with felonious intent by some person or persons to this Jury unknown and of the contents of which bottle Jane Lathrop Stanford had partaken." (White 2022)

During the inquest, Judge William F. Stanley, representing the estate, sat beside the deputy attorney general and whispered questions. The day before Bertha Berner testified, it became public that Jane's will left her $15,000 (roughly $560,000 today). Jane's brother Ariel Lathrop called Berner a medium. No transcript of Berner's inquest testimony survives; White describes her as bold, flirtatious, charming, quick, and calculating, while noting that none of those adjectives is a conviction. Newspapers also chased a Chinese servant, Ah Wing. Hawaiian officials concluded the poison had been put into the soda in San Francisco and declined to pay for a prosecution. High Sheriff Arthur M. Brown and his deputy Henry told the press that David Starr Jordan, trustee Timothy Hopkins, and their private detectives had come to Honolulu to cover up a crime, not to solve one.

The night before the funeral, Jordan and Hopkins drew up a statement attributing Jane's death to natural causes: advanced age, unaccustomed exertion, a surfeit of unsuitable food, and the unusual exposure of the picnic, perhaps aggravated by strychnine in the laxative capsule and a small amount in the bicarbonate through a "pharmacist's error." No bicarbonate containing strychnine was ever found on the market, and Jordan never released the report he claimed held his evidence. Doctors recruited in San Francisco diagnosed angina pectoris, a spasm of the heart; Dr. Clifford Wood in Hawaii called that theory "simply rot" (White 2022).

Jordan's university inherited the fortune if the death was ordinary and the will stood; a murder investigation threatened to embarrass the estate, invite a contest over the will, and reopen the mishandled January poisoning. A century later, Malcolm Harris writes that it is no longer particularly controversial to suggest Jordan was part of a conspiracy, most likely with Bertha Berner, though no one has produced definitive proof (Harris 2023). The chemists found strychnine. The president found a picnic.

The Southern Pacific—now controlled by E. H. Harriman—found a newspaper. W. H. Mills, an old political operative for the Associates, paid off a wide stretch of the California press, including the *San Francisco Post*, on the railroad's behalf (White 2022). Presumably at the request of Russell Wilson and Ariel Lathrop, the *Post* denounced the murder stories and praised Jordan and Wilson for rejecting them—before spoiling the favor slightly by observing that private detectives are more often hired to hush a scandal than to solve one.

At the funeral on the farm, the congregation sang "Nearer, My God, to Thee" and the choir sang the Stanford alma mater. The casket was placed inside a steel chest. The next day, in the family mausoleum, the chest was slid into the marble sarcophagus between her husband and her son, and sealed. For twelve years the inscription had waited with a blank space: *Jane L. Stanford.*

Born in Mortality, August 2, 1828. Passed to Immortality—. On March 16, 1905, a workman chiseled the date: February 28, 1905 (White 2022).

Fourteen months after the workman cut the marble, on the morning of April 18, 1906, the San Andreas Fault broke. The spire of Jane's Memorial Church and the triumphal arch came down. Five days after the earthquake, in Beijing, the Empress Dowager Cixi issued an edict sending 100,000 taels of silver "as a mark of friendship" for the sufferers in San Francisco, plus 40,000 taels to the Chinese minister in Washington for the city's Chinese residents; because the Roosevelt administration refused relief gifts from foreign governments, the American minister William Woodville Rockhill returned the 100,000-tael note with assurances that no slight was intended, while the 40,000 taels for Chinatown stayed on its errand. Six months after that, on October 11, 1906, the San Francisco school board ordered the city's Japanese schoolchildren into the segregated Oriental School where Chinese children had been sent since 1885.

In 1906, the year after Jane died, David Starr Jordan became the inaugural chair of the Committee on Eugenics of the American Breeders' Association. In 1909, Sarah Winchester sent $300,000—thirty years of Jordan's salary—through funds entrusted to Yale to endow the William Wirt Winchester Hospital for tuberculosis in New Haven. More than a century later, on September 13, 2018, Stanford's trustees voted to rename Serra Mall, the Serra dormitory, and Serra House while keeping Serra Street; on October 7, 2019, the university's main address changed from 450 Serra Mall to 450 Jane Stanford Way. In October 2020, because of Jordan's eugenics work, the trustees stripped his name from Jordan Hall (Building 420) and removed the statue of Louis Agassiz from its facade.

The dead hand in the Founding Grant said the land could not be sold. It did not say the sandstone could not crack.

Part 5: The Engineer's Century

1891–1945

The Orphan and the Mine (1891–1905)

In the summer of 1891, before opening day, Herbert Hoover arrived at a campus that was not finished. The Iowa orphan—born in 1874, raised a Quaker, and sent to Oregon after his parents died—had been let into Stanford on a conditional admission after he passed mathematics and failed the rest of the entrance examination (Whyte 2017). He lived on the farm while the sandstone was still going up. He ran a laundry agency and a lecture bureau. He wrote a friend that he was working awful hard and had schemes for making more (Whyte 2017). The hunger was not a joke.

Earlier on that same ground, in 1878, Eadweard Muybridge had set a row of cameras on Leland Stanford's stock farm to see whether a horse ever had all four hooves off the ground (Harris 2023, ch. 1.3). It did. Those frames were the ancestor of the motion picture, paid for by the railroad president. Four years earlier, down the coast, on January 31, 1874, Thomas Balch Elliott and his Indiana neighbors had formed the colony that became Pasadena. Many of them had come for a milder climate, sick with asthma or tuberculosis. In 1882 Emma Bangs kept a Pasadena boarding house the town called the busted lung brigade; a parrot there had learned to cough exactly like a lunger (Wood 1917, 120–121; Baur 1959; Kilston 2018). By the 1890s, when the university opened on the stock farm, the state was already drawing people who needed a new climate or a new start.

At the start of Hoover's second semester, a geologist named John Casper Branner arrived to chair the department. Hoover could type. Branner hired him as an office assistant at $25 a month—roughly $1,000 a month in modern money—and Hoover kept the job until he graduated (Whyte 2017, 36). It was not a living. It was a door. Hoover's own rule, told to a classmate, was simple: do your work so they notice it, and always be on the job (Whyte 2017).

A guest lecture by Branner in Monterey captured another student, Lou Henry (Whyte 2017, 43). She talked her parents into letting her study geology, the only woman in the program. When she first met Hoover in Branner's laboratory, she expected someone brawny. What walked in was speechless, hair uncombed, tanned from summer work. She did not look away (Whyte 2017, 43).

Hoover graduated in the pioneer class of 1895, in geology, with a degree and almost no money. His first job was pushing ore carts in a Nevada mine (Whyte 2017). The degree had not delivered a career. It had delivered a person a London banker could be talked into trusting.

In the 1890s, investors in the City of London—the financial district, not the city of churches—were putting money into holes in Western Australia, South Africa, and China. They could not see the holes. Somebody had to go and look, price the risk, buy the machinery, hire the men, keep books an auditor in London could read, and talk to whatever government claimed the ground. The scarce thing was not dynamite. It was a report a stranger would believe. A degree from a new university on a horse farm was, for a few years, a strange but usable proof.

The recommendation that moved Hoover was not an application he filled out. Louis Janin, a San Francisco mining engineer, passed him to the London firm of Bewick, Moreing and Company. In 1897, at twenty-three, they sent him to Western Australia (Whyte 2017).

Coolgardie and Kalgoorlie were inland, hot, short of water, and full of money. Hoover told the firm which holes to buy. He recommended the Sons of Gwalia. The name was Welsh. The mine made his reputation (Whyte 2017).

Lou Henry took her geology degree in 1898. On May 1 of that year, Commodore George Dewey's squadron destroyed the Spanish fleet in Manila Bay in a morning. The war was short. The empire was not. By 1899, Bewick, Moreing had sent Hoover to China. In February 1899 he married Lou Henry in Monterey, and they sailed the next day—two people who could read the same rock (Whyte 2017, 43).

On May 12, 1901, President William McKinley broke ground in San Francisco's Union Square for a monument to Dewey's victory at Manila Bay. Union Square was already named, for pro-Union rallies on the eve of the Civil War. The granite shaft rising there was not about that war. It was about an ocean.

Two years later, on May 14, 1903, President Theodore Roosevelt dedicated the finished monument—a granite shaft with a bronze Victory on top, wreath in one hand, trident in the other [NEEDS SOURCE: script gives May 13, 1903; review log gives May 14, 1903]. Five years had passed from the battle to the column; a morning in Manila had become a stone in a shopping square before the uniforms were out of fashion. James D. Phelan gave the shaft to the city, and Mayor Eugene Schmitz accepted it. Roosevelt said Manila had opened the Pacific to American commerce. The shots that count, he said, are the shots that hit (*Morning Oregonian*, May 15, 1903).

While Roosevelt was in the city, the Union League Club gave him a farewell banquet with covers for eight hundred. The *Morning Oregonian* printed the toast list: "The Army, General MacArthur. The Navy, Secretary Moody. California, Governor Pardee. The Citizens' Committee, M. H. de Young" (*Morning Oregonian*, May 15, 1903, 2). General Arthur MacArthur—Douglas MacArthur's father—had commanded in the Philippines. He toasted the Army at the banquet for a naval victory in the islands he had helped to hold, while M. H. de Young of the *Chronicle* toasted the citizens. A lot of people standing under the column in Union Square think it is for General MacArthur. It is for Dewey; Arthur MacArthur was inside at dinner.

That same year, 1903, Jane Stanford opened Memorial Church at the center of the Stanford Quadrangle, topped by an eighty-foot, twelve-sided spire on a crossing that was not tied hard enough to the roofs around it.

On October 17, 1904, in a remodeled saloon in San Francisco's North Beach, Amadeo Pietro Giannini opened the Bank of Italy (James and James 1954). Born in San Jose in 1870, Giannini had left school at thirteen; he did not go to Stanford, and he did not go to Berkeley. The Bank of Italy had Italian-speaking tellers, took small deposits, and made loans to tradesmen. By the end of 1905, its assets were above $1 million (James and James 1954).

Across thePacific in 1905, eight years after Hoover first went down the Sons of Gwalia, the parliament of Western Australia appointed a Royal Commission on the Ventilation and Sanitation of Mines (Whyte 2017, 100). A royal commission is not a royal visit; it is a government inquiry with the power to compel testimony and write a judgment. This one was

hard on most of the operators in the goldfields. Workers, the record of the inquiry said, were unduly exposed to dust and fumes and falling rock. At the Sons of Gwalia, the commission found the air "notably bad" (Whyte 2017, 100).

The same parliament was launching six or eight royal commissions a year, some of them on questions as small as stone or stucco for the new law courts (Whyte 2017, 100). A government that inquires into everything will inquire into you. That took some of the sting out. It did not take the finding out. Hoover was thirty-one when the commission spoke. The question of how a young engineer got his results followed him.

In France in 1905, Alfred Binet built a test to find children who were falling behind in school, so they could be helped. Binet warned against using it as a measure of inborn worth. In Palo Alto that same year, a house was built at 367 Addison Avenue for Dr. John Spencer, a physician who would be mayor, with a twelve-by-eighteen-foot garage in the back. And on February 23, 1905, the *San Jose Mercury* announced that Jane Stanford had appointed William James—the Harvard philosopher, the most famous philosopher in America, and chair of the American branch of the Society for Psychical Research, who went to séances looking for one honest medium, what he called a white crow—to teach at Stanford (White 2022) [@white-jane-stanford]. Less than a week later, she was dead in Honolulu.

The Spire and the Plank (1906–1913)

William James came to Stanford anyway, arriving in January 1906, nearly sixty-four and tired of lecturing (White 2022, 281–282) [@white-jane-stanford]. He told his wife Alice why: "If I can only get through these next 4 months and pocket the $5,000 I shall be the happiest man alive" (White 2022, 281) [@white-jane-stanford]. Five thousand dollars for one term was roughly $186,000 in modern money—more than sixteen years of the $25-a-month wage Hoover had earned typing for Branner. James was being paid a young engineer's youth for four months of talk.

His only class was a history of philosophy. Three hundred students enrolled, and another hundred and fifty visitors sat in. He wrote that they did not know what hypothesis and concept meant, and that he found it hard to aim too low. One of his few mentions of the woman who had wanted him there was a thank-you: the memorial holiday for Jane Stanford's death let him cancel a class (White 2022, 281–282) [@white-jane-stanford].

On April 18, 1906, at 5:12 in the morning, the United States Geological Survey marked a foreshock. The main break came twenty to twenty-five seconds later and tore nearly three hundred miles of the San Andreas Fault. William James was in a campus bedroom. He timed his room at 5:13.

"My room was shaken like a rat by a terrier," James wrote home that week (White 2022) [@white-jane-stanford]. The wooden houses held. The chimneys came down. The higher university buildings, he wrote, were "largely piles of ruins" (White 2022) [@white-jane-stanford].

At the center of the Quadrangle, the crossing of Memorial Church moved by itself. The eighty-foot spire fell through the chancel roof. The mosaic on the north face blew out into the Quad. The apostles commissioned for the altar were smashed. Hundreds of feet of arcade lay flat. The spire was not put back.

The same morning, in San Francisco, a different kind of monument burned. The golden spike from Promontory sat in the museum at Stanford, but the polished laurel tie it had been driven into burned in the San Francisco fire that followed the quake (Chang 2019). The object that marked the ceremony survived. The object the workers built toward did not.

On Nob Hill, the Stanford mansion went the next day, April 19. Ah Wing, the family's old cook and the caretaker on the hill, locked the door, walked to Jones Street, looked back, and saw the house in flame. The fire finished what the fault had started.

The other man moving through the city that morning was thirty-six-year-old A. P. Giannini. The story goes like this: Giannini was thrown from his bed in San Mateo, borrowed a produce wagon, drove into the city, pulled about $2 million in gold, coins, and securities out of the Bank of Italy—roughly $74 million today—hid the money under vegetables, and rolled past the looters. The next days, while other bankers waited for their superheated vaults to cool, he set a wooden plank across two barrels on the Washington Street wharf in North Beach and lent money on a face and a signature. *Time* magazine and UC Berkeley's own history for Giannini Hall have both told that version.

The record is messier. The Office of the Comptroller of the Currency, the federal office that charters national banks, tells the same morning with a different number: $80,000 in coins—roughly $3 million today—hidden under crates of oranges, and then loans on a handshake from a wharf. The scholarly biography by Marquis James and Bessie Rowland James says Giannini used two of his stepfather's produce wagons, loaded with all the cash, gold, and records they would hold, and set up a plank on two barrels on the Washington Street wharf within two days (James and James 1954). The plank, the wagon, and the loans to people with nothing left to pledge are in every serious account. The $2 million is the legend. The $80,000 is the comptroller.

A bank is a gate if it only lends to people who already have collateral. Giannini did the opposite, and he gave it a shape. Character lending is a loan on the person, not the pledge—a face, a signature, a handshake. Branch banking is one bank with many offices, instead of a separate little bank in every town.

Back at Stanford, the earthquake let William James go. He walked into San Francisco with a colleague looking for the colleague's sister. A whole population stood in the streets with whatever they had pulled from houses about to burn. James called the fires "most beautiful in the effulgent sunshine" (White 2022, 282) [@white-jane-stanford]. By early May he was back in Cambridge. Before he left, he wrote trustee Horace Davis that the faculty did not trust the people governing the university, and that Mrs. Stanford's death had been a chance for a new start which nobody had taken. President David Starr Jordan answered that James was not speaking for the responsible members of the faculty (White 2022, 282) [@white-jane-stanford].

Jordan was a great ichthyologist, the first president of the university, a pacifist at real cost, and a eugenicist. He loved the word *bionomics*—the claim that one science could cover heredity, history, economics, and ethics, and then tell a society what to do (Harris 2023, ch. 2.2). Eugenics was the idea that a nation should breed human beings the way a farmer breeds livestock, encouraging the births it calls fit and discouraging or preventing the births it calls unfit. In those decades this was mainstream progressive science. University presidents held it. So did public health officers. If poverty and what they called feeblemindedness were mostly inherited, then the hierarchy on the ground was nature.

California wrote it into law. In 1909 the state authorized the medical superintendents of state institutions to operate on inmates when, in the superintendent's judgment, it would improve the person's physical, mental, or moral condition [NEEDS SOURCE: primary citation to 1909 statute]. No consent from the patient was required, and no court order. Over the following decades California sterilized more people under that authority than any other state; later counts put the state's share at roughly a third of the recorded sterilizations in the United States, though the counts do not agree with each other because the people it was done to were not carefully counted [NEEDS SOURCE].

That same year, 1909, W. Cameron Forbes—of the Boston Forbes family that had fought western railroad men over rates in the 1870s—became the civilian governor-general of the Philippines, serving until 1913. Boston, Manila, and the column in Union Square were one circuit.

In 1910, a psychologist named Lewis Madison Terman arrived at Stanford with his wife, Anna, and two small children, Fred and Helen. Terman's young adulthood had been marred by a long and debilitating course of tuberculosis, requiring the move to California's mild climate in the first place (Harris 2023, ch. 2.2). Fred, the son, came west because his father could not stay east and breathe. At Stanford, the father set out to measure minds.

Down the coast, another group of outsiders was arriving for the sun and the payroll. On October 27, 1911, the Nestor Film Company opened the first permanent studio in Hollywood, at the northwest corner of Sunset Boulevard and Gower Street. The building had been the Blondeau Tavern before a local liquor law hurt the trade. David Horsley, who had been making pictures in Bayonne, New Jersey, leased it. Behind the tavern stood a wooden stage about twenty by forty

feet, with muslin overhead to soften the sun. The actors brought their own clothes. The carpenters brought their own tools.

Horsley and the other independents had left New Jersey in part because Thomas Edison's lawyers were winning. Edison and a handful of licensed companies held the patents on cameras and projectors through the Motion Picture Patents Company, the Trust. The independents went west to get far from the process servers. Nestor's first California picture was a one-reel western, about 950 feet long, called *The Law of the Range*, released on December 13, 1911. The advertisement called it a "rousing Western foto-drama… taken in the West." The trade-paper reviewer wrote: "The story is interesting, but the photographs are not very good" (*Moving Picture World* 1911).

In the autumn of 1912, Herbert Hoover joined the board of trustees of Leland Stanford Junior University. A trustee legally holds an institution for somebody else—at Stanford, written down in 1885 as the children of California. The board hires the president, watches the endowment, and decides what the founder's forever clauses still mean. As biographer Kenneth Whyte wrote, "He joined the Stanford board of trustees in the autumn of 1912 in circumstances as cloudy as the awarding of his geology degree seventeen years earlier" (Whyte 2017, 124).

The cloud was a residency rule. Trustees were supposed to live close enough to do the work. Hoover escaped it by claiming he had immediate plans to move to San Francisco, though he knew he was at least a year away from leaving London. He still called the university "the best place in the world" (Whyte 2017, 124). He wanted his hand on it. Once he was on the board, he said the place was deficient in a very great proportion of the departments, and that too many of the professors were not good enough (Whyte 2017, 124). Unlike William James, who had come for $5,000 and told the truth on the way out, Hoover came onto the board while still living in London and stayed for the rest of his life.

In 1913, David Starr Jordan left the Stanford presidency, though the bionomics he had sheltered did not leave with him. That same year, California passed the Alien Land Law, telling first-generation Japanese immigrants—*issei*, born in Japan, distinct from their American-born citizen children, the *nisei*—that they could not own farmland.

In December 1913, Cecil B. DeMille got on a train heading west to shoot *The Squaw Man*. The plan was Flagstaff, Arizona. The story DeMille later told was that Flagstaff was colder than when he left New York, so everybody got back on the train and rode all the way to California. Public television's *Lost LA* notes that this version is widely taken to be apocryphal. A different memory, kept by the American Film Institute's historians, has DeMille saying he got to Flagstaff, saw high mountains, and wanted plains with mountains in the distance; the star, Dustin Farnum, said they ought to go where the other companies were. They reached Los Angeles on December 20. On Christmas Eve, DeMille signed a lease on a barn at Selma and Vine. Shooting started December 29. To save $3, DeMille played a faro dealer himself.

Los Angeles was an open-shop town, where men worked without joining a union. When the studios needed carpenters and electricians to build a street that looked like Wyoming, the wage was the point. On *The Squaw Man*, DeMille noted that carpenters and other hands cost 25 to 50 percent less than on the East Coast (Sklar 1994). A carpenter in Los Angeles in 1913 worked at three-quarters of a New York wage, or at half.

Bionomics and Bread Lines (1914–1928)

By the summer of 1914, Herbert Hoover was a partner on London Wall, two doors from the rooms where he was about to try to feed a country (Whyte 2017). On June 28, 1914, a Serbian nationalist killed the Archduke Franz Ferdinand in Sarajevo. Hoover, in London, told himself it was just another Balkan lapse into barbarism and went back to work (Whyte 2017).

On the afternoon of August 4, 1914, a few hours after German troops crossed into neutral Belgium, Hoover walked a little more than a city block to the American consulate (Whyte 2017). Belgium had about 7.5 million people and grew only about 20 percent of the food it ate. The German army took the harvest, and the British blockade, meant to starve Germany, also closed the occupied country. Between them, a nation that imported four pounds of every five it ate began to beg (Whyte 2017).

Hoover's first errand was small. An American engineer named Millard Shaler had 1,500 tons of cereals stuck on a dock. Hoover saw that 1,500 tons would not touch a country. When the crisis hit, the skill that mattered was the mining engineer's list: survey, account, price the risk, move food, talk to governments that hate each other. He had practiced it underground. He built the Commission for the Relief of Belgium, the CRB—a private, neutral machine in London, buying food around the world, landing it at Rotterdam, moving it by canal, and handing it to Belgians to distribute. He wanted, in his own words, the strongest centralization and an effective monopoly: absolute command of his half, no knitting bee (Whyte 2017).

That autumn, Hoover sent a clipped cable from London: "Over one million people on bread line in Belgium at present moment with supplies estimated to last from one to three weeks. We shall require upwards of twenty thousand tons foodstuffs monthly" (Whyte 2017).

He also ran a campaign he named himself: "enormous propaganda" (Whyte 2017, 146). He browbeat American newspapers and governors, hinting to one state that another state had already given more. He told reporters to go see the famine and to play up what he called the "detailed personal horror stuff" (Whyte 2017, 146). He sent a motion-picture crew to film the food lines in Brussels. He hired famous writers. George Bernard Shaw sent a sentence: "one Belgian goes without a full meal whilst thousands of lapdogs are living in luxury from Canterbury to San Francisco" (Whyte 2017, 206).

The first gift ship, carrying 3,500 tons of beans, bacon, flour, and rice—a sixth of a month's requirement—reached Holland on November 21, 1914 (Whyte 2017). Hoover got two great

British shipping lines to carry the food free, along with the piers and the telegraph. He treated charity as a logistics firm.

While Hoover was moving grain into Rotterdam, Lewis Terman was finishing his book at Stanford. In 1916 he published *The Measurement of Intelligence* (Terman 1916). Inside it was the test Binet had built in France, which Terman and his Stanford team had translated, reset to age norms for American schools, and forced into a curve whose average would always be 100. Mental age divided by actual age, times 100: a single number for a whole child, which Terman called the Stanford-Binet intelligence quotient, or IQ. A child could finish in about fifty minutes. Terman believed general intelligence was an "original endowment" dictated by genetics alone, unchanged by school or hard work (Terman 1916).

In 1917, when the United States entered the war, Terman helped design the Army's intelligence tests. More than 1.7 million draftees took them. A campus instrument became a government instrument in a year.

That same year, President Woodrow Wilson made Hoover the United States Food Administrator. His name became a verb: to *Hooverize* meant to clean your plate, to skip the meat, to waste nothing (Whyte 2017). Reporters expected a giant and found an unobtrusive man at a desk. In California, by 1918, Giannini's Bank of Italy had become the first bank with branches across a whole state.

In 1919, with the war over, Hoover was in Paris advising Wilson at the peace conference. While feeding Belgium, he had conceived of gathering the paper of the war. In June 1919 he sent a telegram to Stanford's president, Ray Lyman Wilbur—a cable the Hoover Institution still calls its founding document. Hoover offered $50,000, roughly $964,000 today, and told Wilbur to send Professor Ephraim D. Adams to Paris to start collecting.

Adams went, joined by Ralph Lutz and Frank Golder. They gathered pamphlets, newspapers, posters, and government documents across Europe, adding them to the files of the Commission for the Relief of Belgium, the United States Food Administration, and the American Relief Administration. That same year, 1919, a student named John Steinbeck enrolled at Stanford.

In 1921, Terman began his long study of gifted children, *Genetic Studies of Genius*, starting with a core of 643 California children who scored at or above 135 on his test [NEEDS SOURCE: exact year 1921 from outline/study]. He pledged not to publish their names. He enrolled his own son and daughter, Fred and Helen.

He missed one. The study passed over the only child of a Stanford engineering professor: William Shockley. Shockley scored 129, six points under the line Terman was using for genius—a score that ranked him, in Malcolm Harris's phrase, "slightly subgenius" (Harris 2023, ch. 2.2).

In 1922, Hoover's collection was renamed the Hoover War Library. That year it took in 40,000 documents, housed inside the Stanford library. In 1925, John Steinbeck left the campus without a degree, having used the place as a library before going to ranch work. And in 1928, A. P. Giannini gave $1.5 million—roughly $29 million today—to found the Giannini Foundation of Agricultural Economics at UC Berkeley and build Giannini Hall. The man who had left school at thirteen and lent on a face put his name on the public university, not on Stanford.

Hooverville and the Garage (1928–1939)

In November 1928, Herbert Hoover was elected president of the United States. He had been secretary of commerce. He had been the man who Hooverized. He believed, with a mining engineer's faith, that a problem of production and distribution could be administered.

The stock market broke in the autumn of 1929. What followed was a collapse of work. The Bureau of Labor Statistics did not begin its modern monthly unemployment survey until 1940, but its later retrospective estimate for 1933 put unemployment at 24.9 percent: 12,830,000 people, about one worker in four, with March 1933 as the worst month and farmers losing their land often not even in the count (Bernstein 1969). United States Steel had about 225,000 full-time employees in 1929; by April 1, 1933, the full-time number was zero (Bernstein 1969). Whoever was left was part-time.

In November 1930, Giannini renamed the Bank of Italy the Bank of America. That same year, the word *Hooverville* first showed up in print, credited to Democratic National Committee publicity director Charles Michelson. People who had lost their rooms built camps of cardboard, tar paper, tin, and packing crates, and named them for the president.

In October 1931, an unemployed logger named Jesse Jackson and about twenty men put up shacks on the tidal flats south of Pioneer Square in Seattle, where a shipyard had been. The camp grew to as many as 1,200 people on about nine acres—a few city blocks, named for a man who had moved 20,000 tons of food a month. The city tried to burn it. The men rebuilt it, and the camp lasted until 1941, when the war wanted the land back for ships. Belgium could be treated as logistics, because the food existed somewhere else. A depression in the country that grew the food was a question of who owned the work.

On July 28, 1932, the limit had a photograph. World War veterans—the Bonus Army—had camped in Washington asking for early payment of a bonus the law had promised them later. Troops under General Douglas MacArthur, the son of the general who had toasted the Army in San Francisco in 1903, drove them out with cavalry and tanks, burning the camps. How much of the order to cross the Anacostia River Hoover had given, and how much MacArthur decided once he was over the bridge, remains disputed. The record shows the eviction, the general, and the president who took the blow.

In November 1932, Hoover lost the presidential election to Franklin Delano Roosevelt, the grandson of Warren Delano Jr., a partner in Russell and Company, the American firm that had made a great part of its China fortune in opium. After the defeat, Hoover went back to Palo Alto. He sat in his study handling as many as 1,500 letters a day, walking over to the Hoover War Library to watch the anti-Hoover literature accumulate on shelves he had paid for (Whyte 2017). He spoke of the relief of no longer making ten vital decisions a day (Whyte 2017). The archive held the verb and the insult in one building.

While Hoover watched his library fill, Lewis Terman's son Fred was teaching electrical engineering across the campus. In 1934, two of Fred Terman's students, William Hewlett and David Packard, finished their degrees. Packard—six foot five, a high school class president, an athlete, the kind of man David Starr Jordan had liked to imagine as the Stanford type—went to General Electric in Schenectady, New York. Hewlett, who had impressed the admissions office less, went on to graduate work at Stanford and at MIT, and then came back. Terman took both. He told them to make a run for it in California instead of becoming permanent employees of a firm back east.

In 1937, Giannini's Bank of America lent Walt Disney the money to finish *Snow White and the Seven Dwarfs*, and the bank stayed with the studio through *Fantasia*. In the late 1930s, at Stanford, brothers Russell and Sigurd Varian worked with physicist William Hansen to build the klystron, a vacuum tube that generated high-power microwaves. In 1938, Fred Terman arranged a Stanford fellowship so David Packard could return from Schenectady to work on vacuum tubes with Russell Varian.

Packard took a leave from General Electric. He and his wife, Lucile, drove west with a used Sears drill press in the car. In September 1938, the Packards moved into the downstairs flat at 367 Addison Avenue in Palo Alto, paying $45 a month in rent—roughly $1,100 a month today. Hewlett moved into a shed behind the house, while the landlady, Ione Spencer, lived upstairs in 369. They chose the house because it had the twelve-by-eighteen-foot garage in the back. Lucile Packard went back to work in the Stanford registrar's office, and her paycheck was the steady money; in the evenings, she kept the books.

Hewlett and Packard began part-time work in the garage with $538 in working capital—roughly $12,800 today—made up of cash plus the value of the drill press. Their first real product was an audio oscillator, the Model 200A, a box that made a clean tone. In 1938, a sound engineer from Walt Disney's studio heard it and asked for changes. The modifications produced the Model 200B, and Disney ordered eight of them, at $71.50 each, to test the sound system for *Fantasia*. On January 1, 1939, the two men formalized their partnership in the garage. A coin toss decided the order of the names: Hewlett before Packard.

Total War and the Lease (1941–1945)

On December 7, 1941, Japanese planes attacked Pearl Harbor. Three weeks later, Frederick Terman was called east to Cambridge, Massachusetts, to run the Radio Research Laboratory at Harvard (Harris 2023, ch. 2.4).

The Harvard laboratory was the other half of radar: not how to see the enemy, but how to keep the enemy from seeing you. While the Varians' klystron tube gave Allied radar the microwaves to see in the dark, Terman's teams worked with British scientists on "window," or "chaff"—strips of metal foil dropped from airplanes so a radar screen filled with ghosts (Harris 2023, ch. 2.4). Terman traveled to England to coordinate the teams. At Normandy, Allied planes dropped enough chaff and jammed enough signals to paint ghost fleets on German screens (Harris 2023, ch. 2.4).

In 1942, Philip Morse called the other Stanford son—William Shockley, who had gone through Caltech and MIT to Bell Laboratories—into anti-submarine operations research as research director (Harris 2023, ch. 2.4). Shockley solved his first problem in days. Ship-dropped depth charges were set to explode seventy-five feet down, which worked if a submarine had time to dive. The same charges, dropped from airplanes onto boats still near the surface, exploded too deep to hurt them. Shockley had the airplane charges reset to thirty-five feet. Within two months, successful anti-submarine attacks rose by a factor of five (Harris 2023, ch. 2.4).

Seventy-four days after Pearl Harbor, on February 19, 1942, President Franklin Roosevelt signed Executive Order 9066. The order let the secretary of war mark military areas and exclude any or all persons from them. It did not say the word *Japanese*. It did not have to. Lieutenant General John DeWitt drew the zones and ordered the removal, and Congress passed Public Law 503, making disobedience a crime punishable by up to a year in prison and a $5,000 fine—roughly $100,000 today.

About 122,000 people of Japanese ancestry on the West Coast, most of them American citizens, were forced into guarded camps the government called relocation centers. Not one person of Japanese ancestry in the United States was convicted of a serious act of espionage or sabotage during the war. One widely repeated estimate puts the property loss—orchards, nurseries, shops sold in days—at about $400 million then, roughly $8.2 billion today [NEEDS SOURCE].

In Palo Alto, local historians counted about 144 Japanese Americans. In March 1942, they registered their property at the police station on Bryant Street. In April, an 8:00 p.m. curfew was imposed on them alone. Before the buses came, Arthur Okado, president of the Palo Alto Japanese American Association, published a statement in the *Palo Alto Times*: "Without reservation, we are loyal to this, our country, the United States of America" [NEEDS SOURCE:

primary *Palo Alto Times* citation]. The association had bought about $2,700 in defense bonds—roughly $55,000 today—and seven of their men were already in the Army.

On May 19, DeWitt ordered Palo Alto's Japanese Americans out. On May 26, 1942, 144 people boarded buses for the Santa Anita racetrack, renamed a reception center, taking only what they could carry and no pets, while friends passed candy and oranges. By the middle of June, not one Japanese American remained in Palo Alto. Ten weeks after Okado's statement, the FBI arrested him; he was later cleared and came home in February 1945. Most of the Palo Altans were sent on from Santa Anita to the camp at Heart Mountain, Wyoming, including fifty-nine students from the Palo Alto schools. Cherry Ishimatsu, of Palo Alto High School, later remembered the scene inside Heart Mountain: "Here we are, all prisoners behind barbed wire fences and a military guard surrounding us, and here we have young people doing a parade with their American flags flying" [NEEDS SOURCE: primary Densho/interview citation].

In 1943, Lewis Terman retired from teaching at Stanford. In July 1945, as the war neared its end, Vannevar Bush delivered his report *Science, the Endless Frontier* to the president, arguing that because basic research is where practical power comes from, the federal government should fund it permanently through universities, with scientists deciding where the money goes (Bush 1945).

The war in Europe had ended in May 1945; the war in the Pacific ended in August, with the surrender signed in September. Late in 1945, rising into 1946, Frederick Terman came home to Palo Alto (Harris 2023, ch. 3.1).

He returned to a small department and a university that had spent the war lending its best people to other men's buildings. MIT had the radar lab; Harvard had Terman; Berkeley had a hand in the atomic bomb. Stanford had individuals. Terman had learned in Cambridge how federal money moved, what a proposal had to look like, and how to run classified rooms where most of the staff could not discuss most of the work. His diagnosis was that adequate at many things meant invisible when the contracts were written.

Four years after Terman returned, in 1949, A. P. Giannini died; by then the Comptroller of the Currency could call Bank of America the largest bank in the world, with more than five hundred branches and about $6 billion in assets, roughly $84 billion today. In 1956, Lewis Terman died—the same year the boy his study had ranked slightly subgenius, William Shockley, shared the Nobel Prize in Physics for the transistor.

When Fred Terman stepped off the train in Palo Alto at the end of the war, the land was waiting, and the land had a lock on it. In 1885, Leland Stanford's Founding Grant had told the trustees they had no power to sell or convey the farm's acres. For sixty years that clause had frozen a fortune in dirt—horses, then orchards, then a war that barely touched the ground.

Terman looked at the Founding Grant's sentence and saw a door. The grant said the land could not be sold. It did not say the land could not be leased.

Part 6: The Park

1945–1970

01. The Yards and the Stalls (1937–1946)

July 7, 1937. The Marco Polo Bridge outside Beijing. The war that the postwar peninsula inherited did not begin at Pearl Harbor. It began in Asia, with the Second Sino-Japanese War—not the first war of the 1890s—and the fighting had already run four years before the United States entered it. Rana Mitter, a historian of China's war, puts the Chinese dead in a range of fourteen to twenty million (Mitter).

When the United States entered the war, the San Francisco Bay built the ships. Henry Kaiser's four yards at Richmond turned out 747 vessels, 519 of them Liberty ships, according to the National Park Service. Richmond grew from a town of about twenty thousand people to more than one hundred thousand. One hull, the SS *Robert E. Peary*, went from keel to launch in four days, fifteen hours, and twenty-nine minutes, from November 8 to November 12, 1942. Four days for one hull. Across the water, Bechtel's partner yards and Bechtel's Pacific bases, stretching from Alameda out across the islands, were the work of a different company on the same bay (Denton 2016).

On January 3, 1942, on the island fortress of Corregidor, President Manuel Quezon of the Philippines signed Executive Order No. 1, awarding General Douglas MacArthur five hundred thousand dollars—about ten million dollars today—along with smaller sums for his aides, an order the historian Carol Petillo uncovered in 1979 (Petillo 1979). That spring, in Oakland, the city council stitched old streets into a single thoroughfare by council actions on March 19 and April 8, 1942, and put up the signs for MacArthur Boulevard on April 11 while the general was still in the Pacific (the Interstate 580 freeway would not receive the name until a 1968 resolution).

Seventeen days after the Oakland street signs went up, on El Camino Real in San Bruno, a horse track became a prison.

The Tanforan Assembly Center operated from April 28 to October 13, 1942. The track had been named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, the grantee of Rancho Buri Buri; the stalls stood on land from that Mexican grant, beside the old royal road. While the buses from Palo Alto had gone south to a different racetrack at Santa Anita, Tanforan was the northern assembly center. Densho, the Japanese American history project, counts 8,033 people

held behind its fences. Sixty-four percent of them were citizens of the United States. About 3,700 people lived in horse stalls, three to six people in a stall built for one horse.

The artist Miné Okubo was held there. "I was given a number and placed in a horse stall," she wrote (Densho).

Fred Korematsu was held there too. Down the peninsula, Stanford sent books to the camp library. Converting the track had cost $1,188,000, about twenty-four million dollars today. Feeding the prisoners, Densho records, cost thirty-seven cents a day for each person—about seven dollars and sixty cents today—while the Army spent fifty cents a day to feed a soldier. In the same year an American general was paid a fortune on a fortress, Bay Area families were numbered into stalls at thirty-seven cents a day. Both were the war. They were not the same experience of it.

At Stanford, the war was recorded from a shaft of stone that had just gone up. Arthur Brown Jr. had designed the Hoover Tower at 285 feet, and in 1938 Herbert Hoover had settled on six hundred thousand dollars to build it, about fourteen million dollars today. Its bells came from the Belgian pavilion at the 1939 World's Fair, the great bell cast with the line: "For Peace Alone Do I Ring." The university dedicated the tower on June 20, 1941, with Hoover in the room, less than six months before Pearl Harbor. During the war the tower served as a listening post, recording broadcasts from the Far East.

In January 1945, while the war in Europe was still on and Frederick Terman was still the man Harvard had borrowed to run its Radio Research Laboratory, Terman talked Stanford's president, Donald Tresidder, out of twenty-five thousand dollars—about $465,000 today [@harris-palo-alto]. Malcolm Harris is particular about the name Terman chose. Not a microwave laboratory. Not the Stanford microwave laboratory. The Microwave Laboratory. Twenty-five thousand dollars was a drop, Harris writes, and Terman knew where the spigot was: the Office of Naval Research [@harris-palo-alto]. A university that had spent the war lending its people to other people's buildings was about to have a building of its own, aimed at the microwave tubes that Russell and Sigurd Varian had already taught the military to want.

That August, on V-J Day, a young Navy radio technician named Douglas Engelbart shipped out for the Philippines. On the way, he read Vannevar Bush's *Atlantic* essay, reprinted in *Life*, about a machine that could store and link human records [@harris-palo-alto].

Across the Pacific, Douglas MacArthur began the occupation of Japan. His father, Arthur MacArthur Jr., had been military governor of the Philippines from May 5, 1900, to July 4, 1901, had clashed with William Howard Taft, and had left the landlords where they were; the son had served there as a military adviser and field marshal in the 1930s, not a civil reformer. In Japan, as supreme commander, the younger MacArthur broke up the estates. Between November 1945 and August 1950, tenant-farmed land in Japan fell from 45.9 percent to 9.9 percent

(Williamson 1951). By July 1950, about 4.76 million acres had changed hands—an area on the order of six hundred Stanford farms.

On March 17, 1946, at LaGuardia Field in New York, Herbert Hoover boarded a military C-54 (Whyte 2017, 575–585). He was seventy-one years old. President Harry Truman had asked him to survey the postwar famine. Hoover told reporters he had been called back like an "old family doctor" (Whyte 2017, 575–585). Over five weeks he flew about thirty-five thousand miles across twenty-five countries. In Germany he walked the ruins of the chancellery offices and the bunker; in Vienna he rode in Hitler's Mercedes (Whyte 2017, 575–585). That same year, the Hoover Institution was offered what it describes as "the only original portions" of the Joseph Goebbels diaries: handwritten pages from 1925 and 1926, and typed pages from 1942 and 1943 (the larger Nazi Party archive at the Hoover Library is a microfilm copy; the paper originals went to the German Bundesarchiv). A bell for peace, and the diary of the man who ran the opposite, sat in the same shaft.

That autumn, back on the peninsula, Terman and the trustees founded the Stanford Research Institute [NEEDS SOURCE]. Terman wanted the institute to take the big military contracts [@harris-palo-alto]. Sally Denton's history of the Bechtel Corporation puts Steve Bechtel Sr., the builder, among SRI's founding directors (Denton 2016, ch. 10). Tresidder, a Hoover man, feared a university that lived on government money. Harris records that Tresidder fired the institute's director rather than let one Navy contract turn SRI into a public charge, and then died before he could name a replacement [@harris-palo-alto].

The man who had feared the government's check was gone.

02. Across the Creek (1948–1951)

San Francisquito Creek is a ditch with a county line in it. Palo Alto, in Santa Clara County, sits on the Stanford side. East Palo Alto, in San Mateo County, sits on the other, across the water and across Highway 101. The name says east; the map says north.

The name itself had been a compromise. East Palo Alto's general plan records an informal election on December 12, 1925, in an area that had been called Ravenswood and Runnymede. "East Palo Alto" won the vote. There was talk, even then, of annexing to Palo Alto, but the enabling law never arrived. The community kept a name that sounded like a neighborhood of the rich city, and a government that was not.

After the war, the federal government, through the Federal Housing Administration, became a superbank for private mortgages (Rothstein 2017; Harris 2023). The loans were designed to cost the Treasury nothing, which meant the government only backed borrowers and neighborhoods it called solid. The old Home Owners' Loan Corporation maps had colored districts by investability, using red for Black neighborhoods—the practice that took the name redlining. A red line on a map was a refusal of a loan, drawn in advance of the applicant.

Restrictive covenants did the same work in the deed, binding a buyer to sell later only to white buyers. In 1948 the Supreme Court ruled that courts could not enforce racial covenants. The FHA did not therefore start insuring Black buyers. As Harris writes, following Richard Rothstein's *The Color of Law*, the agency did not insure Black buyers under any circumstances; it brought the red line along [@harris-palo-alto].

In the late 1940s, the Peninsula Housing Association of Palo Alto, a group of middle-class and working-class residents, bought an undeveloped 260-acre ranch near Stanford and planned a cooperative subdivision: hundreds of houses and common spaces [@harris-palo-alto]. Of the first 150 families, Harris writes, 2 percent were Black [@harris-palo-alto]. Two percent of 150 is three families.

Three families were enough. The federal insurers would not back the construction loans. Banks, including Bank of America, would not make the loans the FHA refused [@harris-palo-alto]. The association gave up and sold the land to a developer who accepted the condition: whites only. The subdivision is called Ladera. It is still there. The cooperative that tried to include Black families is not.

Even the new electronics payroll could not cross the line. Art Fong, a Chinese American engineer recruited to Hewlett-Packard, came home from the war and could not find a rental in Palo Alto [@harris-palo-alto].

In 1950, a few miles down the bay at Moffett Field, the former Navy radio technician Douglas Engelbart was working as an engineer at the Ames Aeronautical Laboratory [@harris-palo-alto]. Thinking back to the essay he had read on the troopship, he decided the rest of his life would go toward building a screen that made hard problems solvable. Pretty much everyone, Harris notes, thought the idea was nuts [@harris-palo-alto].

On April 17, 1951, at the San Francisco airport, Douglas MacArthur came down the stairs of his plane into the floodlights. Six days earlier, President Truman had relieved him of command in Korea. MacArthur had not stood in the continental United States in fourteen years. Governor Earl Warren met the plane and took his arm; signs in the crowd read "MacArthur for President" (*Oregon Statesman*, April 18, 1951; *Los Angeles Times*, April 19, 1951). The next morning, on the steps of San Francisco City Hall, MacArthur renounced any run for political office (though on July 7, 1952, he would deliver the keynote address at the Republican National Convention and allow his name to be placed in nomination before Dwight Eisenhower won on the first ballot).

Five months later, on September 8, 1951, the peace of the Pacific was signed a short walk from City Hall without him. At the War Memorial Opera House, Japan and forty-eight other nations signed the Treaty of San Francisco, while the Soviet Union, Poland, and Czechoslovakia refused. That same day, at the Presidio, the United States and Japan signed the security treaty that kept American troops in Japan. MacArthur's name was on neither document.

The Korean War was already remaking the federal budget and the Stanford campus. In 1951, the Defense Department's budget rose from thirteen billion dollars to fifty-eight billion—in today's money, from about $167 billion to about $747 billion [@harris-palo-alto]. While American bombers leveled cities in North Korea, Harris writes, military support for electronics research at Stanford tripled [@harris-palo-alto]. In the early 1950s, with money from the Office of Naval Research, Stanford put up the Applied Electronics Laboratory to house the postwar offspring of the wartime Radio Research Laboratory that Terman had run at Harvard [@harris-palo-alto]. Its director was Bill Rambo, one of Terman's men from Cambridge.

Meanwhile, across Highway 101 from Palo Alto, speculators found a second tool to remake the map: blockbusting. Into the 1950s, after a white resident on the eastern side of the highway sold a house to a Black family, real estate speculators went door to door warning white neighbors that the block was changing and prices would fall, urging them to sell immediately (Rothstein 2017; Harris 2023). The speculators bought at fire-sale prices and resold at a premium to Black workers starved for houses they were allowed to buy. Within six years, Rothstein and Harris record, East Palo Alto went from exclusively white to 82 percent Black [@harris-palo-alto]. Palo Alto zoned less than 10 percent of its housing land for multifamily buildings and set a minimum size for a single-family lot [@harris-palo-alto]. Density, which is how working people share a mortgage, was planned out of the city next to the jobs.

03. The Lease (1953–1955)

In 1953, along a old lumber road that William Page, a New York timber man, had cut in the early 1860s out to a mill near today's California Avenue in Mayfield, a company signed a piece of paper with Stanford University [NEEDS SOURCE]. The paper was not a deed.

The founding grant of 1885 had forbidden the trustees to sell or convey the Palo Alto farm. It did not forbid a ground lease. Under a ground lease, the university kept the title; the tenant paid rent, built a laboratory, and hired the graduates, and when the term ended the buildings and the ground remained the landlord's to lease again. A sale would have been a goodbye. A lease brought the tenant back, and the deed never changed hands.

The first lease at the Stanford Industrial Park went to the research department of Varian Associates in 1953 [@harris-palo-alto]. It was no surprise, Harris writes, because Frederick Terman was both an investor in Varian and a member of its board [@harris-palo-alto]. The man telling the university what its land was for also owned a piece of the tenant.

In 1954, Stanford's president, J. E. Wallace Sterling, promoted Terman to provost [@harris-palo-alto]. As the historian Rebecca Lowen noted, Sterling was sufficiently anticommunist and sufficiently anti-Roosevelt to please Herbert Hoover [@lowen-creating-cold-war-university; @harris-palo-alto]. The man who knew how to write the defense proposal was now in charge of the campus.

After Varian, Harris writes, the park along Page Mill Road filled with companies Terman already knew [@harris-palo-alto]. Hewlett-Packard—whose founders had started in an Addison Avenue garage with $538 and a drill press—put its headquarters on the leased land. General Electric came. Eastman Kodak came. The park could recruit a professor the way it recruited a tenant: Carl Djerassi, a co-inventor of the birth control pill, came to teach chemistry at Stanford and brought his company, Syntex, along behind him [@harris-palo-alto]. Even the federal government took a piece of the farm's edge when Stanford ceded 87.5 acres on Miranda Avenue in 1955 for the Palo Alto Veterans Administration hospital.

In 1954, Terman added a door for the people inside the park's buildings: the Honors Cooperative Program [@harris-palo-alto]. A company in the park could send its employees to Stanford for advanced classes, ahead of the ordinary admissions line, if it paid double tuition. Harris records that the companies paid it [@harris-palo-alto]. A student finished a class in Terman's engineering school, crossed Page Mill Road, and badged into a firm that had paid double her tuition, where her professor might sit on the board or consult on weekdays. As Harris summarizes the loop: the government sponsored microwave research at Stanford, Stanford professors consulted for Varian, Varian hired Stanford students trained to make microwave tubes, and Varian sold the tubes to the government [@harris-palo-alto].

Arthur Rock, an investment banker who arrived from New York a few years later, put the arrangement in two sentences during a 2007 interview at the Computer History Museum.

"They allowed their professors to work part-time, and this was unknown in the East," Rock said. "And also, they had all this land, and they allowed these companies to rent or lease space from them" (Rock 2007).

At MIT and Princeton, in Rock's account, professors were not allowed to start companies on the side. At Stanford, talent was allowed to leave the classroom on Tuesdays, and the land was allowed to leave the orchard without leaving the deed.

04. The Eight (1955–1965)

In 1955, William Shockley left Bell Laboratories to start a company [@harris-palo-alto].

Three decades earlier, as a boy in Palo Alto, Shockley had been tested by Frederick Terman's father, the psychologist Lewis Terman, whose genius study had passed the boy over with an IQ score of 129—slightly subgenius [@harris-palo-alto]. During the war, Shockley had helped move the Navy's depth-charge fuse setting from seventy-five feet to thirty-five; after it, his treatise on electrons and holes became the semiconductor field's textbook [@harris-palo-alto]. His widowed mother still lived in Palo Alto. When Beckman Instruments of Los Angeles agreed to finance his new venture, Shockley put the Shockley Semiconductor Laboratory near her, opening in February 1956 in Mountain View, the next town down the peninsula [@harris-palo-alto].

In 1956, Mountain View was orchards, wartime housing, and the edge of Moffett Field. There was no chip industry to join and no supply chain. The supply chain was the young physicists and engineers Shockley recruited one by one from Eastern laboratories, who boarded trains because the man who wrote the textbook had asked them.

In November 1956, the Nobel Prize in Physics was awarded to Shockley, John Bardeen, and Walter Brattain for the transistor. Where a vacuum tube switched current inside hot glass with a filament that burned out, a transistor switched current inside a solid crystal, and while Bardeen and Brattain remained identified with the first point-contact device, Shockley's name sat on the junction transistor that could actually be manufactured.

He had the standing to recruit the best young people in the trade. He did not have the temperament to keep them. Jay Last, one of the recruits, recalled later that Shockley had no business sense at all (Last 2007). With the Nobel Prize secured, Harris writes, Shockley's ego was at its height; he compared the young scientists to one another and to himself, and treated the staff as lackeys [@harris-palo-alto]. Then he changed the work. Obsessed with a four-layer diode of his own design, he forbade the staff from growing the silicon transistors they had been hired to make [@harris-palo-alto].

By the summer of 1957, seven of the young men—Julius Blank, Victor Grinich, Jean Hoerni, Eugene Kleiner, Jay Last, Gordon Moore, and Sheldon Roberts—had decided to leave together [@harris-palo-alto]. Robert Noyce, a preacher's son from Iowa, joined them at the last moment to make eight; Last was twenty-seven, Kleiner a Jewish refugee of about thirty-two, and Moore the only one from Northern California (Harris 2023; Last 2007).

At first, Last recalled, they simply hoped an existing company would hire them as a group so they could keep making silicon transistors (Last 2007). Eugene Kleiner wrote to his father's contacts at the Wall Street firm Hayden, Stone & Co. The letter reached a junior banker named Arthur Rock, who flew out to California with his boss, Bud Coyle. Rock's advice was the one that felt foreign: do not join a company; start one (Last 2007).

"There wasn't any venture capital really at that time," Last remembered (Last 2007).

Rock and the group went down a list of established corporations, looking for a backer. In Rock's memory, they approached thirty-five companies and all thirty-five turned them down, unwilling to set up a subsidiary that might upset their own employees; in Last's memory, the list was twenty or thirty companies, all saying no, with corporate lawyers stopping the ones that came close (Rock 2007; Last 2007). They were about to give up when someone suggested Sherman Fairchild, an inventor, pilot, and the largest individual shareholder in IBM (Rock 2007).

Fairchild Camera and Instrument agreed to put in $1.5 million—about $17.9 million today (Rock 2007). In Rock's recollection, the eight founders took about 80 percent of the equity and

Hayden, Stone took 20 percent, a split Rock later told a room of venture capitalists they could thank him for inventing (Rock 2007).

In September 1957, the eight men left Shockley [@harris-palo-alto]. A few weeks later, in October 1957, the Soviet Union launched Sputnik, and the United States suddenly wanted tough, reliable silicon transistors for a space race [@harris-palo-alto]. Shockley's business plan had just been vindicated, and his building was empty of the men who could execute it. Fairchild Semiconductor set up shop in Mountain View, in the same town as the laboratory they had left, and within a year was shipping circuits to IBM [@harris-palo-alto].

Harris writes that Shockley tarred the group as "the traitorous eight" [@harris-palo-alto]. Shockley's wife, Emmy, later insisted her husband never used the phrase, and Jay Last agreed: "This was some newspaper guy way after the fact," Last said, noting he had not heard the term until ten or fifteen years after the walkout (Last 2007).

Whatever they were called, their walkout was legal because of a California statute written eighty-five years earlier. California Business and Professions Code Section 16600, descended from Civil Code Section 1673 of 1872 and recodified in 1941, voids any contract that restrains a person from engaging in a lawful profession. On the East Coast, along Boston's Route 128, a noncompete clause could tie a departing engineer up in court. In Mountain View, eight men could walk out on a Friday, keep their houses and their children's schools, and open a competing shop down the street on Monday.

While the chipmakers settled into Mountain View, the missile builders arrived next door. In 1955, Lockheed Aircraft had laid plans to move its expanding Missile Systems Division from Los Angeles to Sunnyvale, next to the Ames wind tunnels at Moffett Field [@harris-palo-alto]. In 1956, with prodding from Terman, Lockheed opened a research complex inside the Stanford Industrial Park and a factory in Sunnyvale [NEEDS SOURCE] [@harris-palo-alto]. According to Steve Blank's history of the division, Lockheed's local missiles payroll grew from zero in 1956 to more than 28,000 employees by 1965 (Blank). It was a contract economy that could fill a county and empty a profession in the same decade: when a defense cycle turned in 1963 and 1964, Harris records, some 30,000 California engineers were laid off statewide [@harris-palo-alto].

Inside the university, in 1959, the trustees set the Hoover Institution apart as an independent center within Stanford, under Herbert Hoover's charge that its purpose was to demonstrate that private enterprise could best serve the people. On January 17, 1961, leaving the presidency, Dwight Eisenhower went before the cameras to name the machinery that now paid the bills on both sides of El Camino Real.

"In the councils of government, we must guard against the acquisition of unwarranted influence, whether sought or unsought, by the military-industrial complex," Eisenhower said.

By 1967, as Rebecca Lowen and Malcolm Harris record, Stanford trailed only MIT and Caltech among American universities in total defense contracts [@lowen-creating-cold-war-university; @harris-palo-alto].

05. The Demo and the Contract (1959–1968)

Around the turn of the 1960s—published accounts split the date between 1959 and 1961—a Stanford creative writing fellow named Ken Kesey took a night job and volunteered for a government drug study at the Menlo Park Veterans Administration hospital on Willow Road [@harris-palo-alto]. He was paid about seventy-five dollars a session, on the order of eight hundred dollars today. Historians tie the Willow Road trials to CIA funding, though careful accounts do not file those trials under the name MKUltra.

A few miles away, at the Stanford Research Institute in Menlo Park, Douglas Engelbart was building his screen. A small Air Force grant had allowed him to write down his framework for augmenting the human intellect, and in 1964 he and Bill English built the prototype of a wooden hand-held pointing device they called a mouse [@harris-palo-alto].

Out in the surrounding towns, the two worlds kept crossing the same streets. On May 5, 1965, a band called the Warlocks played at Magoo's Pizza in Menlo Park; on December 4, 1965, at an Acid Test in San Jose, the same musicians played for the first time under the name the Grateful Dead, and the following month, in January 1966, a Stanford-educated biologist and photographer named Stewart Brand co-produced the Trips Festival in San Francisco. At SRI, meanwhile, a veteran of the wartime Office of Strategic Services named Al Hubbard turned up at an institute project called Alternative Futures, wearing khakis, a badge, and a pistol, guiding LSD trips and gathering opposition data on the New Left—though as Harris notes, drawing on historians Martin A. Lee and Bruce Shlain, Hubbard was not a CIA operative as such [@harris-palo-alto]. Despite later theories, mostly written about Laurel Canyon in Los Angeles, that intelligence agencies manufactured the counterculture, nobody has shown that the Grateful Dead were a government project. On the peninsula, the free-love surface and the Pentagon payroll simply shared a zip code.

In 1965, according to SRI's own institutional history, the Stanford campus first learned that the institute held two chemical-warfare contracts; in 1967, students learned of SRI contracts tied to counterinsurgency in Southeast Asia and began picketing. In 1967, too, the economist Milton Friedman, who would later make the Hoover Institution his home, first met California's new governor, Ronald Reagan. By 1968, Harris writes, SRI had helped develop the CIA's strategic-hamlet plan in Vietnam, and its counterinsurgency work ran into the millions of dollars [@harris-palo-alto].

On December 9, 1968, in Brooks Hall in San Francisco, Engelbart walked onto the stage of the Fall Joint Computer Conference.

He had asked for the slot in March and been given ninety minutes in a hall that held two thousand people. He said later he was "nervous as hell" (Smithsonian). A truck had hauled his gear over Skyline Boulevard from Menlo Park, and a projector borrowed from NASA threw his screen onto the wall behind him [@harris-palo-alto]. The stage lights were so bright he could not see the audience. His console was linked over telephone lines to a computer thirty miles south at SRI, and to colleagues in Menlo Park whose faces joined the text on the screen. With the mouse under his hand, he typed a word, typed it again, and showed a room of engineers accustomed to paper punch cards that text could be pointed at, edited, copied, and rearranged in real time.

"If in your office, you, as an intellectual worker, were supplied with a computer display backed up by a computer that was alive for you all day, and was instantly responsive to every action you have, how much value could you derive from that?" Engelbart asked the hall (Smithsonian).

Behind one of the cameras stood Stewart Brand. Bill Paxton, one of Engelbart's engineers, remembered the video feedback loop on the big screen as psychedelic.

"I've never seen anything that was so dangerous, and undertaken with such bravado," Brand said later (Fisher 2018).

Harris puts the cost of staging that single day at $175,000, about $1.7 million today [@harris-palo-alto]. The Defense Department's Advanced Research Projects Agency and the Air Force paid the bill; an ARPA contract cover from the following spring lists an order of about $1.5 million—roughly $14.5 million today—for the study of computer-augmented management. In Menlo Park in 1968, the landlord of the mouse and the landlord of the counterinsurgency study shared a letterhead.

Four months later, the students stopped asking from the sidewalk.

06. April Third (1969–1970)

On April 3, 1969, Stanford students met on campus and took the calendar date for their name. According to the university news service's report the next day, they voted to demand an end to all classified research, all research in chemical and biological warfare and counterinsurgency, and all studies used against the people of Vietnam, Laos, and Thailand, calling on the Board of Trustees to halt the work at the Stanford Research Institute when the board met on April 8.

By late 1968, SRI's history records, many campus critics had wanted the university to cut SRI loose, only to realize that an independent institute would simply keep the military contracts with nobody left on campus to answer for them. The students now demanded closer university control to abolish the war work, while SRI's president, Charles Anderson, and much of his staff wanted independence so no campus meeting could tell them which contracts were moral.

On the evening of April 8, 1969, students walked into the Applied Electronics Laboratory and occupied it [@harris-palo-alto]. On April 10, students packing Dinkelspiel Auditorium voted the trustees' response unacceptable and endorsed the sit-in.

Bill Rambo, the laboratory's director, protested that the building had a sinister reputation that was not fully deserved, and that most of its rooms were freely accessible [@harris-palo-alto]. The students were not interested in the doors; they were interested in the files. They held the building for nine days—more than a week, in Harris's phrase (Berlin 2017; Harris 2023). Using the laboratory's own presses and hundreds of thousands of its blank sheets, they printed a daily newsletter: the first issue, on April 10, was titled *Unclassified*, and the issues that followed were titled *Declassified* [@harris-palo-alto]. (One early flyer misnamed the building the "Atomic Energy Lab," an accidental joke on a campus where atomic energy was one of the few military sciences Terman had not led [@harris-palo-alto].)

On April 14, when faculty and staff walked through the corridors to identify students for disciplinary hearings, every demonstrator gave the same name: "The April Third Movement." Stanford's new president, Kenneth Pitzer, sent written notice that the occupiers were violating university policy, though unlike a subsequent sit-in at Encina Hall at the start of May, he did not call the sheriff to clear the laboratory. On April 15, after an internal debate, the occupiers reaffirmed that they would not tamper with classified files; they had come to stop the work, not to steal it.

Inside the campus argument over whether the electronics work was merely "defensive," the Stanford English professor H. Bruce Franklin answered a provost in words Harris records:

"Those are electronic countermeasures, used by our bombers to jam enemy radar. If you think that's defensive, you're out of your mind" [@harris-palo-alto].

In the occupation newsletter, a song written in the voice of an institute researcher put the distance between the bench and the war into four lines:

"I have never killed a man. I have never seen him die. These formulas: they could be for jelly. I don't think I know" [@harris-palo-alto].

On April 17, the day before the students walked out of the laboratory, one occupier told the university news service where the circuit led next:

"If we are successful here the next step will be to move to the Stanford Industrial Park."

A poster from those weeks called for a "Strike Against the Stanford Empire" [@harris-palo-alto]. On Friday, May 16, 1969—the morning after police shot demonstrators at People's Park in Berkeley—a flyer went out under the banner "SMASH WAR RESEARCH." It printed a map linking the Stanford Research Institute, the Hoover Institution, the Graduate School of Business,

Varian, Hewlett-Packard, Lockheed, and IBM, captioned: "Since World War II—How to Run an Empire from Palo Alto." That morning, about three hundred protesters blocked the entrance to the Stanford Industrial Park at Oregon Expressway and Hanover Street, parking a school bus across the road and letting the air out of its tires (Berlin 2017; Harris 2023). About one hundred fifty police officers met them. Around eleven o'clock in the morning, the police fired tear gas and made fifteen arrests.

Before the year was out, on December 12, 1969, two young filmmakers named Francis Ford Coppola and George Lucas opened a company called American Zoetrope in a warehouse on Folsom Street in San Francisco, a northern outpost of the new screen culture that would soon meet the peninsula's engineers.

At Stanford, the trustees answered the occupiers not by stopping the contracts, but by severing the link. On March 31, 1970,according to SRI's institutional history, the Stanford Research Institute was formally separated from the university—sold, in effect, to itself [NEEDS SOURCE]. Under the divestiture agreement, SRI agreed to pay Stanford about 1 percent of its gross receipts until twenty-five million dollars had been paid, and half a percent thereafter, while retaining the right to use the Stanford name for a term of years. (By 1989, more than twenty-five million dollars had gone back to the university; SRI staff took to calling the annual payment "alimony," and in some years it exceeded the institute's net profit.)

The morning after March 31, 1970, the same scientists sat at the same desks in Menlo Park, working on the same contracts. A student could no longer point at SRI and tell the trustees, *You own this.* The trustees had traded governance for a royalty. Down on Page Mill Road, Varian, Hewlett-Packard, and Lockheed remained tenants on the leased orchard, and the Honors Cooperative Program still collected double tuition.

The rebellion did not stop the machine. It made the machine look less like a weapons lab.

Three years later, in 1973, the Hoover economist Milton Friedman spent a day in a small private plane hopping across California with Governor Ronald Reagan, barnstorming for Proposition 1, a ballot measure to cap state spending in the California Constitution. The proposition narrowly lost, but by the last stop of the flight Friedman had made up his mind about the man beside him.

"I was able to give an enthusiastic yes to a reporter's question as to whether I would support Reagan for president," Friedman wrote [NEEDS SOURCE].

Part 7: The Road

1969–2002

The Ticket Booth (1969–1974)

December 12, 1969. San Francisco. In a warehouse at 827 Folsom Street, south of Market, Francis Ford Coppola and George Lucas open a film company and call it American Zoetrope.

The word is Greek before it is English. *Zoe*, life. *Tropos*, a turn. A zoetrope is a nineteenth-century toy, a drum with slits, spun so that a strip of still drawings looks alive. They were naming a machine that turns still pictures into a world. They had driven up from Los Angeles with a crew of film-school friends, among them the editor and sound designer Walter Murch. None of them, Murch said, had family in the industry [NEEDS SOURCE]. They wanted American movies made under the influence of the films they loved from everywhere else.

Down the Peninsula, in the hills above the Stanford Quad, an economist named George Shultz had just finished a fellowship year. He had taught at M.I.T. and then at the University of Chicago, where he became dean of the business school. In 1968 and 1969 he was a fellow at the Center for Advanced Study in the Behavioral Sciences on the Stanford campus. Then Richard Nixon put him in the cabinet three times: secretary of labor, the first director of the Office of Management and Budget, and secretary of the Treasury.

Down on the flat of the campus, the university was building a different kind of office. The story of it begins with a man named Niels Reimers. In the late 1960s he ran a pilot: take what a professor invents, file a patent, and license the patent to a company in exchange for a royalty. By the summer of 1970, Stanford made the office permanent (Berlin 2017). Reimers and an assistant worked out of a trailer behind Encina Hall. On the wall, in capital letters, a sign said MAKE IT HAPPEN. On Friday afternoons they filed the week's paper over popcorn and a beer called Henry Weinhard's. They called the ritual Management by Beer (Berlin 2017).

A patent is a temporary monopoly. The government gives the holder the exclusive right to an invention for a term of years, and in exchange the inventor publishes how it works, so that when the term ends anyone can use it. A license is the permission slip. The university keeps the patent and rents the right to use it. A royalty is the rent, usually a percentage of sales. Equity is different. Equity is a slice of the company itself.

On January 11, 1971, six months after Reimers's office became permanent, a weekly trade paper called *Electronic News* landed on desks from Boston to Sunnyvale. The headline, over a series by a columnist named Don C. Hoefler, read "Silicon Valley U.S.A." (Computer History Museum, catalog 102770432). It was the first time those two words appeared in print as the name of a place. Hoefler was writing about semiconductors, the chips cut from silicon, the second most common element in the earth's crust. He was not writing about social networks or search engines. Those did not exist. He was naming a factory district. The name outlived the factories.

In 1972, north in the city, Coppola and Lucas's American Zoetrope moved into the Sentinel Building at 916 Kearny Street in North Beach, the old Italian neighborhood where A. P. Giannini had opened a bank in a remodeled saloon in 1904 [NEEDS SOURCE].

That same year, in Menlo Park, a partnership opened its doors on a suburban road running west from the freeway, past the edge of the university, and up toward the coastal hills. The buildings on Sand Hill Road were low. They sat back among oak trees, with more parking than a visitor could explain. There was nothing on the road that looked like power. That is the point of a gate that wants to be mistaken for geography.

At 3000 Sand Hill Road, the partners were Eugene Kleiner and Tom Perkins. Kleiner had a private office in the building before the firm even had a name. The partnership began in 1972 with an $8 million fund—about $64 million in today's money—one of the largest in a business that was still small enough to fit in a handful of rooms (Berlin 2017). Half of it came from one man, Henry Hillman, a Pittsburgh industrialist. The other half came from individuals, trust funds, and institutions, among them an insurance company and Rockefeller University (Berlin 2017).

A gatekeeper is not the person who does the work. A gatekeeper is the person who decides who is allowed to pass. A century earlier, the railroad through this same valley decided which towns could ship their wheat. The charge was called a freight rate, and the men who set it said it was whatever the traffic would bear. Sand Hill Road was about to do a quieter version of the same job. The thing being moved was no longer wheat. It was a company that did not exist yet.

Kleiner was not a banker. He was one of the eight young researchers who had walked out of William Shockley's laboratory in 1957 and started Fairchild Semiconductor. As a boy in Vienna he had watched Nazis force his father to open the family safe. A police officer who knew the family pulled the father out of a line bound for the camps. Kleiner never finished high school. He volunteered for the American army, he said, because he had a personal thing to settle, and the G.I. Bill put him through engineering school (Berlin 2017). Perkins was different. He had made Hewlett-Packard's computer division the biggest in that company, and he had the polish of a man who had studied how a businessman was supposed to look. At Harvard Business School he had sat in the class of Georges Doriot, the French professor who taught a generation of Americans that venture investing was a craft (Berlin 2017).

The same year, 1972, Don Valentine founded a firm called Sequoia Capital. Valentine had run marketing at Fairchild, and then at National Semiconductor. Sequoia moved onto Sand Hill Road while other people's money was still learning the address (Berlin 2017). The operators had become the financiers. They already knew the people worth funding, because they had been those people's bosses.

Also in 1972, an engineer named Al Alcorn helped launch a video-game company called Atari (Berlin 2017). Its table-tennis game, Pong, would teach a generation to lose quarters in a bar.

Then Perkins did something that complicates the postcard. Shortly before a young associate named Robert Swanson joined the firm, Kleiner and Perkins left 3000 Sand Hill Road. They moved to the twenty-ninth floor of Two Embarcadero Center in San Francisco, looking out at the bay. Perkins did not want the neighbors to see whom he took to lunch.

"We didn't want to be considered part of the flock," Perkins said. "Eagles don't flock, was our joke" (Berlin 2017).

The joke lost. The other firms stayed on the road. Within a decade the low buildings among the oaks were the largest concentration of venture capitalists in the valley, and, as Leslie Berlin wrote, perhaps in the world (Berlin 2017). You can move your own office to a skyscraper. You cannot move the gate, once everyone else has agreed that the gate is a road.

In 1974 Congress passed a pension law with a stern name, the Employee Retirement Income Security Act, and a sterner idea inside it: a prudent person does not gamble with workers' retirement. Fund managers read that as a ban on anything speculative. Venture capital is speculative by design. Most of the companies fail.

That same year, George Shultz left the Treasury Department and went to Bechtel. Bechtel was a San Francisco engineering and construction company, private, vast, and in the habit of building the things governments decided to want: pipelines, power plants, cities in the desert. Shultz became an executive, and then president. While he was there, he joined the Stanford faculty part time and taught management and public policy. Shultz and his wife lived in a modest house on the campus, and he drove an Oldsmobile when his rank entitled him to a Cadillac (McCartney 1988). The professorial manner was not a costume he put on later. His colleagues told the historian Laton McCartney that in a company where engineers shouted, George spoke beautifully—logical, calm, the collected professor (McCartney 1988). It worked on them.

Back in the trailer behind Encina Hall, the pilot program was now a four-year-old official operation called the Office of Technology Licensing (Berlin 2017). In the summer of 1974, Niels Reimers sat at his desk going through the mail.

Patenting Life (1974–1980)

Two scientists had done something that would split the century. Stanley Cohen, at Stanford, and Herbert Boyer, at the University of California, San Francisco, had found a way to cut a gene out of one organism and stitch it into the DNA of another. DNA is the molecule that carries heredity, the long instruction sheet inside a cell. Their method let a person rewrite a line of that sheet. In the summer of 1974, Reimers decided the method should be patented, and he began the fight to file (Berlin 2017).

The fight started immediately. The plan was criticized for its potential to undermine the university's basic mission, which was to increase the world's stock of knowledge, not to fence it

(Berlin 2017). Faculty said the claim reached too far. The press ran nightmares. The mayor of Cambridge, Massachusetts, asked whether a strange orange-eyed creature, supposedly sighted nearby, could be connected to the experiments, and said he would be damned if his city was going to be diseased over academic freedom (Berlin 2017). In 1975 scientists met at Asilomar, on the Monterey coast, and agreed to regulate themselves. The gesture advertised the fear instead of calming it (Berlin 2017).

That same year, at Kleiner Perkins, Robert Swanson lost his job. Swanson was twenty-eight. He had talked his way through an M.I.T. chemistry degree and a management degree in five years, opened a Citicorp venture office in San Francisco, and been hired by Eugene Kleiner, who saw a version of his own nerve (Berlin 2017). Swanson lasted less than two years. The break came over a company called Cetus. At a long lunch, a Nobel laureate mentioned recombinant DNA as something Cetus might explore. Nobody else at the table wanted it, including Tom Perkins, who had come to the lunch to show how worried the firm was. Swanson still thought he had a job. Perkins told him the partners would like to have just the two of them working there. Swanson could stay on the payroll until the end of 1975 (Berlin 2017).

"Maybe I'm not very good," Swanson said quietly to himself (Berlin 2017).

He had time, a Ping-Pong table that doubled as a desk, and a stack of cheap meals. He cold-called the scientists who had been at Asilomar. He had never heard of Cohen, Boyer, or Reimers. Boyer, unusually for a biologist in 1976, was willing to talk about a company. His cats were named Watson and Crick (Berlin 2017). Most of his colleagues would not have taken the call on a university phone. Arthur Levinson, who would later run Genentech and sit on the boards of Apple and Google, remembered that as a young biochemist he walked down the street to a pay phone if he needed to speak to a company (Berlin 2017). A venture capitalist named Brook Byers compared a scientist thinking about business to Bob Dylan going electric. People wondered what was up, and whether bad would come of it (Berlin 2017).

Swanson asked Kleiner and Perkins to pay him a salary while he incubated the company, the way they had done for the founders of Tandem Computers. They said there was no room. The fund was barely making money. So in 1976 Swanson and Boyer started it themselves. Swanson's first proposed name was Herbob, a mash of Herbert and Bob. Boyer countered with a word built from genetic engineering technology. Genentech (Berlin 2017).

Across the valley, another company was forming. Apple had been started by Steve Jobs and Steve Wozniak, and saved as a business by Mike Markkula, a former Intel manager who wrote the checks, hired the grown-ups, and cared about how the thing looked (Berlin 2017).

The money that made Markkula's instructions matter came, in part, from the East. In the fall of 1977 Markkula and Jobs flew overnight to New York. They changed into new blue suits in a men's room at 30 Rockefeller Plaza and rode the elevator to the fifty-sixth floor, to the offices of Venrock, the venture firm of the Rockefeller family (Mallaby 2022). The partner they had to

convince was Peter Crisp. Jobs talked about a computer in every living room. The partners, by their own later account, barely heard the specifics. The whole territory was too speculative to take literally. They stepped into the hall.

"We went out in the hall, four or five of us," Crisp recalled, "we looked at each other, and we shrugged our shoulders and said, What the hell?" (Mallaby 2022).

They put in $300,000—about $1.7 million in today's money—for 10 percent of Apple. The deal valued the company at $3 million, or about $16 million today (Mallaby 2022). Crisp later said people gave Venrock too much credit for being smart. Arthur Rock, the financier who had backed Intel, took a $50,000 slice of Venrock's own allocation, because Crisp owed him a favor from 1968 (Mallaby 2022).

By the end of 1977, back at Stanford, Niels Reimers's clipping file on the recombinant DNA fight was three inches thick. He kept going. He kept thinking, he said later, about the potential for good: drugs, hormones, vaccines that did not depend on a weakened virus (Berlin 2017).

In 1978 Congress cut the tax on capital gains and eased the pension rules. On January 4, 1979, after years of argument, the patent application that would finally issue for Cohen and Boyer's method was filed with the United States Patent and Trademark Office (USPTO Patent 4,237,224). That same year, George Lucas's company, Lucasfilm—which he had built after leaving the Folsom Street experiment—started a computer division to drag high technology into movies (Smith).

And in 1979 the Labor Department clarified the pension point that mattered. Under ERISA, prudence would be judged on the whole portfolio, not on each wild bet inside it. A pension fund could put a small slice into companies that might go to zero. Within a year, venture capitalists had more than half a billion new dollars to invest—about $2.3 billion in today's money (Berlin 2017). The retirement savings of teachers, truck drivers, and clerks had become the raw material of a road in Menlo Park. The beneficiaries were not asked.

In December 1979, at Apple, Mike Markkula wrote a one-page creed called "The Apple Marketing Philosophy." It had three words at the top: Empathy, Focus, Impute. Under the third, he wrote: "Our image is the combined result of everything the customer sees, hears, or feels from Apple" (Berlin 2017).

It is also a description of a university. The sandstone, the red tile, the church at the center of the Quad, the tour guide walking backward: those are an image, and the image is doing work. Markkula called the work "impute." If a company presents a product in a slipshod way, he wrote, people will perceive it as slipshod. If it presents it in a creative, professional way, it imputes the qualities it wants them to feel (Berlin 2017). You do not argue them into it. You surround them.

Catholic California had put saints on the land: San Francisco, Santa Clara, the missions, the royal road. The fortunes that arrived after the gold rush were largely Protestant, and then, in the late twentieth century, frankly secular. They did not build missions. They built companies that asked for the feelings a mission used to ask for: belief, pilgrimage, a story about the future in which the customer is already late.

By 1980, at Atari, the memo had won in a different way. Al Alcorn wanted to build a new kind of cartridge system, one that would throw a hologram behind the game. The chief executive, Ray Kassar, told him to talk to marketing. The marketing manager told him to write a business plan.

"We didn't need a business plan for Pong or the VCS," Alcorn said (Berlin 2017).

He needed one now. The VCS was the box that had brought Atari's games into the living room. Both Pong and the VCS had been built by people who shipped first and wrote the memo second. Kassar nodded at prototypes and released almost none of them. Alcorn's hologram machine, called Cosmos, drew thousands of orders at a trade show and was never sold. He had four words for what had happened to the company: all creativity had ceased (Berlin 2017).

On October 14, 1980, Genentech began trading on the NASDAQ exchange under the symbol GENE. The opening price was $35 a share. In the first twenty minutes it hit $89, the fastest first-day gain Wall Street had seen. At the close the company was valued at $532 million, about $2.2 billion in today's money (Berlin 2017). Swanson and Boyer had each put in $500 at the start. Each of those stakes was worth about $65 million by the closing bell. The slice Kleiner and Perkins had eventually bought, for $100,000 and then another $100,000, was worth the same (Berlin 2017). The firm that had fired Swanson was now rich because of him.

Seven weeks later, on December 2, 1980, the United States Patent and Trademark Office issued Patent 4,237,224, "Process for producing biologically functional molecular chimeras" (USPTO Patent 4,237,224). A chimera, in the old stories, is an animal sewn together from different animals. In the patent, it was Cohen and Boyer's method. Stanford and the University of California owned it.

Ten days after the patent issued, on December 12, 1980, President Jimmy Carter signed the Patent and Trademark Law Amendments Act (Public Law 96-517). Everyone called it Bayh-Dole, after the two senators, Birch Bayh of Indiana and Bob Dole of Kansas. If a university, a small business, or a nonprofit made an invention using federal research money, the institution could elect to own the patent. Before this, the federal agency that had paid for the work generally kept the rights, and a great many inventions sat on a shelf because a company would not spend a decade turning an idea into a product it was not allowed to own. Bayh-Dole flipped the default. The university took title. The university had to file for the patent, had to try to license it, had to share the royalties with the human inventor, and had to give the government a license to use the invention for government purposes. The taxpayer funded the science. The university owned the paper. A company rented the paper and built the product.

Stanford had not waited for the law. Reimers had been running the office for a decade, and he had already fought the Department of Health, Education, and Welfare for the right to hold title on particular inventions (Berlin 2017). Bayh-Dole made the Stanford arrangement the national rule for every campus that took federal money. A trailer behind Encina Hall had become a clause in the United States Code.

The Bechtel Cabinet and the Myth Machines (1981–1995)

In 1981 Caspar Weinberger, who had served in Nixon's cabinet and then as a Bechtel executive, became Ronald Reagan's secretary of defense. In June 1982 Reagan asked George Shultz to be secretary of state. He was sworn in on July 16.

Senators asked, at Shultz's confirmation, whether a man who had just run Bechtel could conduct the foreign policy of a country in which Bechtel had business. He was confirmed anyway. Sally Denton, writing later about the company's nuclear work, recorded a line from those hearings:

"Private interests have no business with nuclear weapons," Shultz said (Denton 2016).

Denton set the sentence down in order to watch him walk away from it years afterward, when the laboratories that design those weapons were being handed to private managers and Bechtel was in line (Denton 2016). A revolving door is simpler than the scandal sheets make it. A revolving door is a career that passes through a public office and a private firm that does business with that office, in either order, more than once. The person does not have to fix a contract on the way through. The knowledge, the phone numbers, and the benefit of the doubt travel with the body. Two of the offices that buy the largest things a government buys, diplomacy and weapons, were now held by men who had just been signing Bechtel's paper. At the Bohemian Grove, a private encampment in the redwoods north of the city, both the Bechtels and Shultz were regulars (McCartney 1988). The network fit in a camp, a campus, and a company. It did not need a conspiracy. It needed a small enough number of dining rooms.

Back on the campus, in 1983, Stanford earned more from technology licensing than any other university in the country: $2.5 million, about $8 million in today's money. Almost 60 percent of that year's income came from the recombinant DNA patent (Berlin 2017). When Reimers had started, there were a handful of such offices in the country. After the chimera patent and Bayh-Dole, they were simply how a university did business. Reimers spent the 1980s arguing that Stanford should also take stock, not only cash, because a fast company might design the licensed invention out of its product before a royalty check ever arrived. He estimated that by refusing equity the office had left more than $50 million on the table (Berlin 2017).

By 1984, the fourteen investments in Kleiner Perkins's first fund showed a combined profit of $208 million, about $670 million today. Ninety-five percent of the profit came from two

companies, Tandem and Genentech (Mallaby 2022). Without those two, the fund would still have beaten the stock market. With them, it became the story the industry tells itself. A power law is a distribution in which the winners do not win a little more than the losers; they win so much, and so increasingly, that the middle barely matters. Most venture bets go to zero. A handful return the fund. On Sand Hill Road, a company that might decently double was a waste of a seat.

That same year, 1984, Francis Ford Coppola's brother August Coppola—Nicolas Cage's father—became dean of the College of Creative Arts at San Francisco State (Hamlin 2010).

Across the bay, George Lucas needed cash. He put the computer division of Lucasfilm up for sale. Steve Jobs, pushed out of Apple and looking for a next act, bought it. The popular sentence is "Jobs bought Pixar." The record is more precise, and Alvy Ray Smith, who was there, kept the paperwork. On January 30, 1986, Jobs wrote a certified check for $5 million. Smith and Ed Catmull signed it over to Lucasfilm. Jobs then put another $5 million into the new company to capitalize it: $10 million in all, or about $30 million in today's money (Smith).

The first press release, dated February 10, 1986, was coy. It said Jobs had paid an "undisclosed sum in the millions" (Pixar press release, Feb. 10, 1986). It also said the new firm would sell a graphics computer for about $125,000, a machine faster at pictures than a $6 million supercomputer. About forty employees came along, John Lasseter among them. They owned 30 percent. Jobs owned 70 percent, and he was chairman, not the founder in the way the brochures later implied. Catmull was president. Smith was vice president. Jobs was the investor (Smith).

That autumn, on November 5, 1986, the state of California renamed the tunnel where the road north out of the city ducks under a hill in the Presidio toward the Golden Gate. For years the official name was the Presidio Tunnel; before that, locals called it the Funston Avenue tunnel, after the approach. Senate Concurrent Resolution 86, introduced by San Francisco's state senator Milton Marks, added the words General Douglas MacArthur Tunnel, and Caltrans still carries both names on its list of structures (Outside Lands; Caltrans).

The first name was Douglas, not Arthur. Arthur MacArthur was the father, a general who commanded the Department of the Pacific, and who, on the night of May 14, 1903, when Theodore Roosevelt dedicated the column in Union Square, stood up and toasted the Army. That column is not his; it belongs to Admiral George Dewey and the battle of Manila Bay. The son is the one on the tunnel: Douglas MacArthur, who commanded in the Philippines, who left Corregidor in 1942 and said he would return, who ran the occupation of Japan, and whom President Truman fired in 1951. The Peninsula and the city have a habit of welding a famous last name to the wrong man.

Seven months later, on June 9, 1987, a monthly tabloid appeared on the Stanford campus, aimed at conservative readers and edited by a philosophy undergraduate named Peter Thiel

(Chafkin 2021). He had founded *The Stanford Review* that spring with Norman Book, a friend from high school who was about to enroll. The masthead of the first issue listed twelve names. Thiel was editor in chief. All twelve were men—a pattern that held for the rest of Thiel's career (Chafkin 2021).

The Stanford Review was a student paper, a place where young people tried on sentences they might later be paid to mean. Nobody was being bribed to print the *Review*. What a paper can be, even a small one, is a seedbed. It sorts people. It teaches them who the enemy is supposed to be, and it gives them the experience of publishing the sort of sentence that makes a room go quiet. Thiel was a student with a grievance against the university's culture, against what he and his writers treated as a closed, moralizing elite (Chafkin 2021). The method of answering it was to build a counter-elite in the same zip code, using the same credential. The names on those early mastheads—Keith Rabois, David Sacks, Norman Book, and a half dozen younger staffers—would turn up again.

On January 20, 1989, George Shultz finished his term as secretary of state—six and a half years, the longest stretch in the job since Dean Rusk—and returned to Stanford as a professor of international economics at the Graduate School of Business and a distinguished fellow at the Hoover Institution. That same year, 1989, Stanford finally let Niels Reimers's licensing office start taking stock in companies (Berlin 2017).

In 1990 Robert Swanson stepped down as chief executive of Genentech after the Swiss drug company Roche bought a majority stake (Berlin 2017).

In 1991, at San Francisco State, Dean August Coppola held what he called a skybreaking instead of a groundbreaking for the new Fine Arts Building. He had persuaded the state legislature to put up $15 million for the building in the middle of a budget crunch—six times what Stanford had earned from all its licenses in 1983 (Hamlin 2010; Berlin 2017). Alumni tossed beans into a hole. Coppola set a glass model of the building in the hole, poured wine over it, and said they would see how the dreams grew.

"The idea is to look up, rather than down," Coppola said. "Look up to the sky, the clouds, teach young people to dream" (Hamlin 2010).

Coppola stepped down as dean in 1992, and the building opened in 1993 with labs, a sound stage, and a 150-seat theater bearing his name (Hamlin 2010).

In November 1995 Pixar released *Toy Story*. A week later the company went public, in what *Variety* called the largest initial public offering of the year (*Variety*; Smith). The graphics computer had not been the fortune. The fortune was the nineteenth-century toy, scaled up. Hollywood had not moved to the Bay Area. The Bay Area had learned to manufacture myth in software, and was about to sell the factory back to Hollywood.

BackRub (1995–1998)

That same year, 1995, a young entrepreneur named Elon Musk was in California. As the story goes, Musk enrolled in a Stanford doctorate program in physics and materials science and dropped out after two days because the internet was more interesting (Vance 2015). When a lawyer later asked the university for the file, the graduate admissions office wrote back that, based on the information given, it was unable to locate a record for Elon Musk. Physics professors who were teaching there in 1995 did not remember him, and some of his early enemies thought the admission itself was a story told to decorate a young entrepreneur (Vance 2015). The record on the enrollment is thinner than the legend. What the record does show is that he built a first company, Zip2.

In the middle of the 1990s, a dormitory room at Stanford was an operations center. Larry Page, a graduate student, had stuffed a mismatched collection of computers into a cabinet he built out of Lego bricks (Levy 2011, ch. 1). The project running on them was called BackRub. It was a search engine. The name was a small joke about the way the system crawled the web by following links backward, looking at who pointed at whom. Page also named the ranking math after himself: PageRank. People assumed it meant a web page. It was a sly vanity (Levy 2011, ch. 1).

The problem was mechanical before it was philosophical. The campus network into the Gates building could carry about 10 megabits a second. The university's main line, a T3, could carry about 45. Scott Hassan, an early collaborator, remembered what happened when the team found a switch in the basement that had been set wrong.

"We were using all the bandwidth of the network," Hassan said. "And this was from a single machine" (Levy 2011, ch. 1).

At times the crawler consumed half of Stanford's entire internet capacity (Levy 2011, ch. 1). The university let the project run. Page's professors were using it to answer questions. So were their friends. A piece of student work was eating the institution's pipe, and the institution's response was to become a customer.

In January 1997 Page wrote a plan down in emails to a venture capitalist named Vinod Khosla, who had money in a search company called Excite. Excite would buy BackRub. Page, and only Page, would go work there for seven months, long enough to install it, and then he would leave in time for the fall term and his doctorate. He claimed the technology would lift Excite's traffic by 10 percent, which he translated into $130,000 a day in extra advertising, or $47 million a year. The price for all of this was $1.6 million—about $3.3 million in today's money. Of that, $300,000 (about $625,000 today) was for Stanford, because Page and Sergey Brin had done the work as graduate students and the university owned the patent. Another $200,000 was salary, $400,000 was a bonus if he installed it in three months, and $700,000 was Excite stock (Levy 2011, ch. 1).

Khosla came back at about $750,000 total (Levy 2011, ch. 1). The deal died in a conference room. Excite's chief executive, George Bell, a magazine executive the investors had installed as adult supervision, watched a bake-off. The query was the single word "internet." Excite's first results, Hassan remembered, were Chinese pages where the English word stuck out of a block of characters. BackRub's first results told the user how to use a browser. Bell was upset. If people found the answer immediately, they would leave the site, and his advertising depended on them staying. Stickiness, in the dialect of that year, was the virtue (Levy 2011, ch. 1).

"He told us he wanted Excite's search engine to be eighty percent as good as the other search engines," Hassan recalled (Levy 2011, ch. 1).

Hassan and the others looked at each other. The buyers did not want a better product. They wanted a product worse than the one they were being offered, because a worse product would keep the visitor confused, and a confused visitor was a longer visit. Page and Brin left with the patent still in the university's name.

In December 1997, Stanford and the University of California's recombinant DNA patents expired. Over the life of those patents, 468 companies had taken a license (Feldman, Colaianni, and Liu). Because Stanley Cohen had been consulting for a company that would compete with Genentech, an exclusive deal had been impossible, so the universities had licensed the method widely on purpose (Berlin 2017). A license, for a young biotech firm, had also been a permission slip saying a famous university had looked at its science and not thrown it out.

In 1998 NationsBank, of Charlotte, North Carolina, bought the San Francisco company that still carried A. P. Giannini's name. The merged bank kept the name Bank of America and put the headquarters in Charlotte. The bank that had lent on a face and a signature from a North Beach wharf after the 1906 fire was now run from North Carolina. Giannini Hall was still at Berkeley; his bank was not in his city.

The jobs on the Peninsula stayed, while the bedrooms did not keep up. Henry George, watching the railroad do this to wheat towns a century earlier, had a name for who captured the gain: the person who held the land [NEEDS SOURCE]. Stanford, by its founding grant, could not sell the farm. It could lease it. And on Sand Hill Road, four minutes from the campus that supplied the people being judged, a company that could not get a meeting in one of the low buildings did not merely lack money. It lacked the introductions to customers, the executives the partners knew by their first names, and the next round of capital.

On September 4, 1998, Page and Brin filed the incorporation papers for Google and moved off campus. Sergey Brin's girlfriend knew a manager at Intel, Susan Wojcicki, who had just bought a house on Santa Margarita Street in Menlo Park with her husband for $615,000, about $1.3 million in today's money. To help pay the mortgage, the couple rented Google the garage and several rooms for $1,700 a month, or about $3,500 today (Levy 2011, ch. 1). Page and Brin

hired a fellow student, Craig Silverstein, who had offered to show them how to compress the links so the system would fit in memory—an offer he later said was a way to get his foot in the door—and they kept a hand on campus that fall, coteaching a class (Levy 2011, ch. 1).

Stanford licensed PageRank to Google for stock: 1.8 million shares (Stanford OTL FY2005 report). Not a royalty on each search, but a piece of the firm. Many search companies already existed, and most of the companies the Office of Technology Licensing called had been unimpressed. The licensing associate who sat with Page and Brin thought a startup was the right licensee (Stanford OTL FY2005 report).

At a party, Kevin Kelly, the writer and editor who had helped start *Wired*, asked Page why the world needed another search engine.

"We are not really interested in search," Page said. "We are making an AI" (Fisher 2018).

The Network (1999–2002)

In 1999 Robert Swanson died of brain cancer at fifty-two. As Arthur Levinson later observed, the man who spent his life turning recombinant DNA technology into drugs that saved other people did not get a drug that saved him (Berlin 2017).

At Thanksgiving in 1999, a group of young men left an office in Palo Alto and walked to an automated teller machine. One of them put in a debit card from a company he had named X.com. He typed a number. The machine thought, and then it produced bills.

"Elon was very, very happy with that," a colleague remembered (Soni 2022).

Elon Musk was not yet thirty. He had sold Zip2 and put the money into a financial website that wanted to be a bank without the marble. The card working at a machine near the office was the product, made physical: money had come out of a wall because a database said it could. (A few years later, in a 2003 talk at Stanford, Musk compressed the idea: "Money is low bandwidth. You don't need some sort of big infrastructure improvement to do things with it. It's really just an entry in a database" [Vance 2015].)

X.com collided with another Palo Alto company, Confinity, where Peter Thiel and Max Levchin were trying to beam money between handheld computers. Levchin had come from Kiev; as a boy he wrote programs longhand, on paper, because he did not yet have a machine that could run them (Soni 2022). The merged firm took the name customers actually used: PayPal.

The internal war was bitter. In 2000 Musk was pushed out of the chief executive's chair. Thiel's circle, several of them alumni of *The Stanford Review*—including Keith Rabois, Norman Book, and David Sacks, who became chief operating officer—resented the X.com rebranding and the

software rewrites (Chafkin 2021; Soni 2022). Fraud leaked through the system faster than they could patch it (Soni 2022).

What saved the company was a network effect—when a service becomes more useful to each person because other people are already using it. PayPal found its road on eBay, the auction site, where strangers were already trading with strangers and needed a way to pay a person they would never meet. Even though eBay was pushing a payment service of its own, called Billpoint, users chose PayPal's ledger (eBay press release, July 8, 2002).

By the end of 2001, the expired Cohen-Boyer recombinant DNA patents had brought Stanford and the University of California about $255 million in licensing revenue—about $480 million in today's money—90 percent of it from royalties on actual products, and the other 10 percent from the fees companies paid just to be allowed in the room (Feldman, Colaianni, and Liu).

By the summer of 2002, about 60 percent of PayPal's business was happening on eBay, which had 46 million users. On July 8, 2002, under a merger agreement dated the day before, eBay announced it would buy PayPal. Each share of PayPal would become 0.39 shares of eBay. On eBay's stock price of July 5, the deal was worth $1.5 billion, or about $2.8 billion in today's money (eBay press release, July 8, 2002).

The sale closed on October 3, 2002. PayPal's stockholders approved it. An eBay executive took the payments job, and Billpoint would be phased out while PayPal kept its name. The same day, Peter Thiel resigned as chief executive and wrote the company a note (SEC exhibit; Soni 2022). Effective at the close of the markets, he wrote, this was his last day. In founding PayPal, he and Max Levchin had started by hiring their friends, and then their friends' friends, moving out in concentric circles. He called those friendships the lasting testimony (Soni 2022).

"I have always known that people are the most valuable part of any business," Thiel wrote, "and today I'm more certain than ever that this is the case" (Soni 2022).

Levchin was still the engineer who had built the system that fought the fraud. Reid Hoffman, whom Thiel had made chief operating officer against the advice of his own board, had already spent a weekend at his grandparents' house in Gualala trying to imagine what came after a collapse; on the third day he had floated a network of professional names that the world would later call LinkedIn (Soni 2022). David Sacks had been left in charge while Thiel traveled, and then learned he would not run the company he had been running. Musk, pushed out of that chair two years earlier, was already putting what he had into a car, and into rockets. Jack Selby, who went with Thiel, said they were trading again before the month was out. Ken Howery went too (Soni 2022). By the time Thiel had an office of his own, it was on Sand Hill Road (Chafkin 2021).

In August 2004, Google would go public, and Stanford would sell 184,207 of its 1.8 million shares in the offering for $15.7 million, disposing of the rest in early 2005 under a university

policy to sell public stock as soon as allowed; as licensing director Katherine Ku confirmed to the *San Jose Mercury News* on December 1, 2005, when the stock closed at $404.91 and the held stake would have topped $700 million, Stanford's total was $336 million—on the order of three-quarters of Leland and Jane Stanford's 1885 founding gift of $20 million (about $744 million today) (Stanford OTL FY2005 report; Krieger 2005). In January 2006 the Walt Disney Company would announce the purchase of Pixar for about $7.4 billion, or $12 billion in today's money (Smith); in 2007 *Fortune* would photograph Thiel's circle dressed as a crime family and name them the PayPal Mafia, leaving out the engineers who had stayed and argued about passwords and hated the khaki (Soni 2022); in 2009 Roche would buy the rest of Genentech for $46.8 billion, making every $35 share from 1980 worth $4,560 (Berlin 2017); and by 2017 Stanford's Office of Technology Licensing would count nearly $2 billion—about $1.91 billion through that fiscal year—in cumulative income, almost all of it from three inventions out of roughly 12,000 evaluated: recombinant DNA, a set of antibody patents, and Google (Berlin 2017; Stanford OTL overview).

On October 3, 2002, the young men who had been trying to throw one another out of the PayPal building walked out of it in the same week, holding stock in the buyer. Sand Hill had been the gate for the companies. This was the gate for the people. The product was sold. The address book was not.

The machine does not stop when the company is bought. It changes desks.

Part 8: The Open Era

2002–2026

1. The Seeing Stone (2000–2008)

February 2004. A dormitory at Harvard. Mark Zuckerberg and his roommates put a site online and call it Thefacebook. It is the one eastern beat of the origin, not of the bloodline. By summer the company is in Palo Alto. It does not stay a college directory. The product becomes the habit, and the habit becomes the crop.

The attention economy is a market in which the thing being sold is not the song, or the post, or the game. It is the time you spend, and the prediction of what you will do next. Shoshana Zuboff gave the harsher name, surveillance capitalism. In plain words: the service looks free because you are not the customer. The customer is whoever pays to know what you are likely to do, and then to nudge you toward doing it. A recommendation is not a suggestion in the old sense. It is a bid.

Why a platform can run that market without owning what its users post is a statute from 1996. In 1995 a New York court held the online service Prodigy liable as a publisher, because Prodigy

moderated its boards. Moderate, and you own the words. Leave them alone, and you might not. Representatives Chris Cox, a California Republican, and Ron Wyden, an Oregon Democrat, wrote the answer into the Telecommunications Act. Twenty-six words: "No provider or user of an interactive computer service shall be treated as the publisher or speaker of any information provided by another information content provider." That is Section 230. A railroad act, a century earlier, gave a company the right of way. This sentence gave a platform the right not to be treated as the speaker of what its users post.

In 2005 Congress built a different shield for a different industry. It passed the Protection of Lawful Commerce in Arms Act. The law makes gunmakers nearly impossible to sue for what a criminal does with a gun. It has an exception. If the maker knowingly violates a statute about how guns are sold or marketed, the shield drops.

June 12, 2005. Stanford's commencement. Steve Jobs did not go to the university. He stands up anyway, and he tells the students about the day he was fired from the company he had founded. "The focus of my entire adult life was gone, and it was devastating," Jobs said. The sentence is about Apple in 1985. The use of it, on this lawn, is the cult energy named in an earlier memo. Mike Markkula had written that a company's image is everything the customer sees, hears, or feels [NEEDS SOURCE]. A founder at a university he did not attend, telling a story of exile and return, is that memo performed as a sermon. He had the honesty, at least, to describe the firing as a devastation, not as a brand.

A different kind of company was taking shape in Palo Alto. Palantir. In Elvish, *palantír*: a seeing stone. In Tolkien's story the stones are powerful, and they are not unambiguously good. Sauron uses them to spy, to talk to conspirators, and to bend people who do not know the stones are dangerous to hold. Peter Thiel named the company after that object (Chafkin 2021).

The founding myth is cleaner than the recruiting. Thiel drove to the University Club in Palo Alto one morning to see Alex Karp, a man he knew from Stanford Law School. Karp thought of himself as a liberal in a circle of conservatives. The early team was the student paper come due. Stephen Cohen had edited the *Stanford Review*. Joe Lonsdale had edited it before him, and had interned at PayPal. Nathan Gettings had worked on fraud at PayPal. Sequoia's Michael Moritz passed. Kleiner Perkins passed. The check that mattered, in 2005, was $2 million from In-Q-Tel, the Central Intelligence Agency's own venture fund, against a rough prototype—about $3.4 million in 2026 money (Chafkin 2021). The venture firms on the road declined a product. The state's venture fund bought a way of seeing.

Call it software as state capacity. Not a social network. A tool a government uses when it wants to know who connects to whom, and it would rather rent the tool than build it. The company later moved into Facebook's old headquarters on University Avenue in Palo Alto, and it added a log, a system it called Gotham, that recorded each search, so an analyst could be audited. Karp's pitch was that the privacy trail would make the mining acceptable. Max Chafkin records a former engineer who sat in meetings where, right after the privacy talk, a government client would

suggest looking up an ex-girlfriend (Chafkin 2021). The employees would remind the client that the search was logged, and then let the search happen. A log is not a refusal. It is a receipt.

Across the Pacific, another search market was already running. It was not a Stanford export, and the record has to be exact. On May 17, 1994, a link between a Chinese physics institute and Stanford's SLAC laboratory had given that institute a full internet connection; China dates its own full-function connection to April 20 of that year, through a different line (Sheehan 2019). Stanford was in the first week of the Chinese internet. It was not the whole invention of it.

Robin Li did not go to Stanford. He did not go to Berkeley. He did not work at Google. He studied information management at Peking University, and computer science at the State University of New York at Buffalo, finishing in 1994. At a Dow Jones company in New Jersey he built a link-ranking system called RankDex, and he filed a patent on February 5, 1997. It was granted in 1999 and assigned to his employer. Stanford's patent on PageRank—Larry Page's, assigned to the Leland Stanford Junior University—discusses that system as prior art. The word is cited. It is not copied.

The person who carried Li into the Valley's mythology was his co-founder, Eric Xu, a biology doctorate from Texas A&M and a former postdoc at Berkeley. Xu made a documentary for Chinese state television, a journey into Silicon Valley, and he took Li to meet Jerry Yang, the founder of Yahoo. Xu's line, later, was that he was inspired by seeing an ethnic Chinese person build a company that powerful, and that he was sure Li was inspired too (Sheehan 2019, 68–69). At a screening of the film at Stanford, Li's wife told Xu she wanted her husband to be an internet entrepreneur. The seed money, $1.2 million, came half from a firm called Peninsula Capital and half from Integrity Partners. Bloomberg lists Greg Penner, a Stanford business-school graduate, among the seed investors. Li had trained in Buffalo and then in New Jersey. He had not trained on this bay. Xu had. They founded the company in January 2000 and built it in Beijing. The name is Baidu.

Google arrived in the Chinese language in 2000, was blocked for two weeks in September 2002, and by the end of that year had about a quarter of the market (Sheehan 2019). In 2006 it opened Google.cn, under Kai-Fu Lee, who had gone to Columbia and Carnegie Mellon, not to Stanford, and who had run research for Microsoft in Beijing. The company censored results in-house, and it put up a disclaimer. Eric Schmidt, the chief executive, described the decision as an "evil scale" (Sheehan 2019). They were not wild about the restrictions, he said, and they decided that not serving the users at all was the worse evil. Baidu's answer was an advertisement in which a Chinese poet defeats an American in a top hat, and the crowd shouts the last line: "Baidu understands Chinese better!" (Sheehan 2019).

In December 2007, back in California, a blog called Valleywag, part of Gawker, published a post by Owen Thomas. The headline was a taunt: "Peter Thiel is totally gay, people" (Chafkin 2021). Nick Denton, Gawker's founder, had suggested the story. Thiel is a Stanford graduate, twice,

bachelor's and law, and the founder of the *Stanford Review*. He did not sue over the headline. He began looking for someone else's suit to fund, in secret.

In 2008, an international businessman sat with China's deputy minister of culture (Sheehan 2019, 162–163). The official opened with a question: "Who is the CEO of Hollywood?"

There wasn't one.

That same year, on Sand Hill Road, Sequoia Capital—the firm Don Valentine had started—hired two investors away from the Stanford endowment (Mallaby 2022, 334). The job was a new business called Heritage, first to manage the partners' own wealth, then the wealth of the founders Sequoia had backed. Sebastian Mallaby does not give their names. The fact is the direction of the hire. The university's money managers were worth recruiting onto the road.

2. The Bus and the Loop (2009–2016)

In 2009, Remington, through its parent, Freedom Group, made a confidential product-placement deal with Activision. The gun was Remington's Adaptive Combat Rifle. The game was *Call of Duty: Modern Warfare 2*. No money changed hands. Both sides agreed to keep it secret. The documents came out years later, in the Sandy Hook litigation, and the *Wall Street Journal* reported them.

One memo read: "A primary means for young potential shooters to come into contact with firearms and ammunition is through virtual gaming scenarios." Another said that people seek out the brands of the guns they have seen, and that a lack of direct branding helps shield the company from looking like an endorsement. Executives wrote about winning their "fair share of these young consumers." A former vice president, John Trull, told the *Journal* that the executives did not understand multiplayer team deathmatches when they made the deal. The memo is real. The understanding may have been thin. Both of those sentences can be true.

By late 2009, in China, Baidu had about 58 percent of the search market and Google about 36 (Sheehan 2019). What pushed Google out was not the ad. It was a hack, Operation Aurora, that reached into the Gmail of human-rights activists, including a twenty-year-old Tibetan activist at Stanford, during Matt Sheehan's last year on the campus (Sheehan 2019, 80–83). On January 12, 2010, Google posted "A New Approach to China," and said it would stop censoring. In March it redirected the Chinese service to Hong Kong. By 2014 the services were blocked. The 2010 exit "cleared the way for Baidu" to own Chinese search (Sheehan 2019, 60).

In 2012, YouTube began optimizing its system for watch time. Think of a television network that is paid only when you do not change the channel. The rational program is the one you cannot leave. On December 14, 2012, at Sandy Hook Elementary School in Connecticut, twenty first graders and six educators were killed. On the radio, Alex Jones said the Sandy Hook murders were a hoax, staged with crisis actors, to tighten the gun laws.

October 2013. A parking lot on El Camino Real, in Palo Alto. El Camino Real, the royal road. A fourteen-seat Mercedes bus idled at the curb, leather seats inside, purple lights in the ceiling, the name of a brokerage, DeLeon Realty, on the side. Matt Sheehan was on it, a Palo Alto native home from China with a broken ankle (Sheehan 2019, 232–234). Between 2011 and 2013, China had poured more cement than the United States used in the entire twentieth century (Sheehan 2019, 231). Now the money was here. The men in the lot smoked and complained, in Mandarin, about business at home. At ten they stubbed the cigarettes and climbed on. Kim Heng, who ran the China work for the firm, stood in the aisle and spoke mostly Mandarin, with English dropped in for the street names.

The brochure was the document. A photograph of the Great Wall came down the page and became the Stanford campus, and Hoover Tower, the stack at the middle of the farm, became one more guard post on the wall. The wall continued, on the paper, into the Golden Gate Bridge. A university tower, drawn as a Chinese fortification, sold to families who had come to buy a house near the schools.

The second stop was a mansion in Menlo Park. Marble stair, limestone patio, a pool. One of the men joked. First the Mandarin, which Sheehan renders as an obscenity—*wo cao*—and then the English: "It's such a beautiful place. Why didn't Bo Xilai come here to buy his villa?" (Sheehan 2019, 232–234).

Bo Xilai was a senior Chinese official who had fallen, in 2012, in a corruption scandal. The joke landed because the bus knew what the house was for. Sheehan's phrase for the use is "the new Swiss bank account." A house in Palo Alto held value outside Chinese politics, outside a currency, and, for some buyers, outside the question of how the money was made. Fathers, he writes, often skipped the green card so they would not owe American tax on worldwide income. Mothers and children stayed. Or the house sat empty except for the vacations.

Between 2011 and 2015, according to the National Association of Realtors figures Sheehan cites, Chinese residential purchases in the United States more than quadrupled, to $28.6 billion—about eighty-five times what Stanford got for the Google shares (Sheehan 2019). In 2013, 53 percent of the reported Chinese purchases were in California. In Palo Alto, during 2012 and 2013, the median house price rose 50 percent, to nearly $2 million, and it had done that rise in two years. Some local brokers estimated that 15 to 25 percent of the town's sales were going to Chinese buyers. One broker told Sheehan he had just sold a house to a contest of four people from China, and none of them had seen it.

The legal door for a number of the families was a visa called EB-5. At the time, $500,000 into a project in a high-unemployment area—about $700,000 in 2026 money—if the project created ten jobs, put a person in line for a green card. In 2014, 9,128 of those visas went to Chinese applicants, more than 85 percent of the program (Sheehan 2019). The university's town had become a store of value, and the federal immigration code had become part of the financing.

Sheehan's own family is the turn. His great-grandfather, Tommy Sheehan, a San Francisco longshoreman, used to drive a priest named Peter C. Yorke around the city. In 1901 Yorke had keynoted the California Chinese Exclusion Convention: "Their thoughts are not our thoughts; their blood is not our blood; their outlook is not our outlook" (Sheehan 2019, 4–5). One century later the great-grandson sat on a bus that sold Palo Alto to Chinese families, and the brochure put the university's tower on the Great Wall. The priest's sentence and the brochure are the same road, read in opposite directions.

Hold the older photograph against a newer one. In 1869, at the golden spike, the picture of the finished transcontinental railroad has engineers and white workers. Not a single Chinese face. The men who had done the most dangerous work on the mountain are not in the frame. In 2013, when a new span of the Bay Bridge opened between San Francisco and Oakland, the mayors were Ed Lee and Jean Quan, both Chinese American, both the children of working people. Quan later said the posing was a joke between them: "This time the Chinese were in the middle" (Sheehan 2019, 267–268). Neither photograph is the bus. The bus is the money, which does not pose. It buys.

The loop closed in both directions. In May 2014 Baidu hired Andrew Ng, a Berkeley doctorate who had helped build deep learning at Stanford and had led Google Brain, the project that trained a network to recognize cats in YouTube videos. Baidu put his lab less than a mile from Google's own campus in Sunnyvale. The company, Sheehan notes, was worth about $60 billion—about $85 billion in 2026 money (Sheehan 2019, 106). The firm that had won the home market came back and hired the person who had taught the machines to see.

By August 2014, Christopher Wylie writes, two months after they launched a harvesting app built by a Cambridge researcher named Aleksandr Kogan, Cambridge Analytica had collected Facebook data on more than 87 million people (Wylie 2019). Facebook later confirmed the figure. Eighty-seven million is more than twice the population of California. The word for what the firm said it was selling is psychographics: not your age and your zip code, but your personality, inferred from what you liked, and then a message aimed at that inference. Wylie, who helped build it and then described it, wrote the sentence that belongs to this university: "And if their colleagues at universities like Harvard or Stanford were also getting interested in our work, I thought that surely we must be onto something" (Wylie 2019).

That is legitimation. A famous campus does not have to audit the work. It only has to be nearby, and the nearby is read as a permission slip. Wylie also says that staff from Palantir, Peter Thiel's data company, asked about access to the set and pointed at a possible legal loophole, at a meeting in Palantir's Soho Square office in London in the summer of 2014 (Wylie 2019). That is his account. It is not a court's.

That same year, in West Virginia, a young man named Caleb Cain, who had dropped out of college and thought of himself as liberal, went looking on YouTube for self-help videos. He found

a host named Stefan Molyneux. "He was willing to address young men's issues directly, in a way I'd never heard before," Cain later told *New York Times* reporter Kevin Roose. Recommendations pulled him toward the far right; he later said the videos made him feel as if they were giving him power and respect and authority.

In Beijing, meanwhile, an ex-Googler named Li Zhifei, a Johns Hopkins doctorate, had gone home and founded an artificial intelligence company called Mobvoi. Sheehan noticed two glass arrows set into the office floor. One read *California* and pointed at the desks. The other read *Silicon Valley* and pointed at a pair of chairs. Li told him the construction workers could not read the English, and had installed the arrows backward. Google, blocked in the country, invested about $40 million to $45 million in him in 2015—what Sheehan calls "the first direct investment by the parent company into a Chinese start-up since it left" (Sheehan 2019). The arrows pointed the wrong way. The money did not.

At the same time, the Stanford aura was failing in public at a blood-testing company. Elizabeth Holmes had started as a Stanford undergraduate. Sebastian Mallaby writes that she persuaded the "dean of the engineering school" to vouch for her, and that she recruited "elder statesmen from Stanford's Hoover Institution" onto the board (Mallaby 2022, 340–341). George Shultz sat among them. His presence was a license. By the middle of the decade the company, Theranos, was valued at $9 billion—about $12.7 billion in 2026 money. It went to zero.

Mallaby's line on the money complicates the postcard: "almost none of the money that Holmes raised came from Sand Hill Road practitioners" (Mallaby 2022, 340–341). The Walton family put in $150 million. Rupert Murdoch put in $121 million. The DeVos family and the Cox family put in $100 million each. A venture investor named Tim Draper made a modest angel bet because of a family connection. His father, William Draper III, had been at Yale with George H. W. Bush, and in Skull and Bones (Harris 2023). The Ivy roots of the Valley are another story. What the outsiders bought was the credential. Holmes held shares with 100 votes each, the founder-friendliness pushed past Google and Facebook. They "faked it until they could make it" (Mallaby 2022, 340–341). She faked a machine that did not work. The board of elders made the fake look like a Stanford result. A Hoover letterhead is cheaper to read than a lab notebook. The letterhead was the product.

In March 2016, Peter Thiel's secret project reached a courtroom in Florida. Terry Bollea, known as Hulk Hogan, had a sex tape, and Gawker had published it. A Florida jury awarded $140 million: $115 million in compensatory damages, then $25 million in punitive damages. Thiel later told the *New York Times* he had spent about $10 million funding the case. "It's less about revenge and more about specific deterrence," he said (Chafkin 2021).

Specific deterrence is a lawyer's phrase. You punish one person so that person, and the people watching, do not do the thing again. To his friends, Chafkin records, Thiel called it "the most philanthropic thing I've ever done" (Chafkin 2021). Gawker went bankrupt. Univision bought assets for $135 million. The site shut down. The company later settled the Bollea case for $31

million. The jury's number and the check that cleared are different numbers, and both belong in the record. A man who had learned to publish in a dormitory newsroom had funded a wrestler's lawsuit until a newsroom died. The railroad, a century earlier, put reporters on a payroll. This was the later version. You do not buy the copy. You buy the plaintiff.

3. Four Companies (2017–2021)

In Hollywood, Wang Jianlin of Dalian Wanda had bid $4.9 billion for 49 percent of Paramount, and had been turned down (Sheehan 2019, 162–163). In 2017 Beijing's capital controls, followed by a ban on what it called "irrational" overseas investment in entertainment, stopped the rest of that shopping. Yet in January 2018 the Chinese conglomerate Tencent bought just under 10 percent of Skydance—the production company run by Larry Ellison's son, David Ellison—at about a $1.5 billion valuation, with co-financing and the right to distribute in China.

That same year, 2018, *The Intercept* reported a secret Google project called Dragonfly: a censored Chinese search app that would require a sign-in and would share data with a Chinese partner. More than 740 Google employees signed a letter of protest. The letter said the objection was not about China; it was about tools that help the powerful oppress the vulnerable, wherever they are. Within about a month, Sheehan writes, the company "effectively ended" the work (Sheehan 2019, 123–124, 291). Also in 2018, President Donald Trump said in private that "almost every student that comes over to this country is a spy" (Sheehan 2019, 54–55).

August 6, 2018. In about a day, Apple removed most of the InfoWars podcasts, YouTube terminated Alex Jones's channel, and Facebook removed four of his pages. Spotify had already pulled the show. Twitter gave him a seven-day suspension, then banned him for good on September 6. Four companies, most of them on the West Coast, switched one man off.

On January 25, 2019, YouTube began reducing recommendations of what it called "borderline" content, at first in the United States, saying the change touched under 1 percent of videos. That December it claimed a 70 percent average drop, in the United States, in watch time of such content coming from recommendations to people who had not subscribed. YouTube denied that its system radicalizes people.

On March 19, 2019, the Connecticut Supreme Court held that the state's unfair-trade law could serve as a predicate exception to the 2005 federal shield protecting gunmakers. Remington had marketed its rifle in *Maxim* with the line: "Consider Your Man Card Reissued." Remington argued that its ads did not target at-risk youth and that nothing linked the ads to the Sandy Hook shooting. On November 12, 2019, the United States Supreme Court declined to review the Connecticut ruling. The door in the shield was open.

In July 2019, the Federal Trade Commission fined Facebook $5 billion, a record, and imposed a new privacy order. Five billion dollars is about fifteen times what Stanford received when it sold its Google shares. The fine did not move the ledger. It priced a scandal. The company kept the

users, the prediction, and the headquarters in Menlo Park. That same month, in the 2019 trailer for Skydance and Paramount's *Top Gun: Maverick*, the Taiwan and Japan flag patches on the flight jacket were swapped out while Tencent was a backer. Tencent later withdrew.

Also in 2019, at Stanford, Alex Stamos started the Stanford Internet Observatory, with Renée DiResta as research director, to study online networks and election rumors. On October 7, 2019, the university's rename of Serra Mall took effect, and on November 14 it was dedicated as Jane Stanford Way. In October 2020, David Starr Jordan's name came off Jordan Hall, which became Building 420.

On April 16, 2020, the *New York Times* released the first episode of Kevin Roose's podcast *Rabbit Hole*. Caleb Cain had handed Roose his whole YouTube history: more than 12,000 videos and 2,500 searches. In Cain's years on the platform, from 2014 to 2018, recommendations had driven more than 70 percent of watch time on YouTube. The same machine had eventually carried Cain out: a change in the recommendations brought him creators on the left who used the same style, Natalie Wynn among them, and Cain became a creator himself. The chief executive Roose interviewed in a later episode was Susan Wojcicki—the Intel manager who, in 1998, had rented her Menlo Park garage to the founders of Google for $1,700 a month. By 2020 she was running the largest video service on earth. The garage and the recommendation were the same peninsula, twenty-two years apart.

The academic research that followed complicated the story. A 2021 paper in the *Proceedings of the National Academy of Sciences* (Hosseinmardi et al.) found far-right consumption on YouTube small, stable, and matched to what people already did elsewhere, with no clear recommendation-driven rabbit hole. A 2023 paper in *Science Advances* (Chen, Nyhan et al.) found extremist channels concentrated among a small group of already resentful users, who arrived by subscriptions and outside links; people who had not subscribed rarely got those recommendations. A 2024 study in *PNAS* (Hosseinmardi et al.), using bots, found that recommendations had a mild moderating effect compared with what users chose for themselves. Most of that data came from after the 2019 changes, whereas Cain's descent was from 2014 to 2018. The rabbit hole can be real for a few and a myth for most.

The microphone that the four companies had switched off in 2018, meanwhile, found another plug. On October 27, 2020, months into a Spotify deal reported at about $100 million, Alex Jones sat as a guest on Joe Rogan's show, with Tim Dillon. Spotify employees protested on the company's internal chat. The general counsel's guidance, as reported, was that Spotify was not going to ban specific individuals from being guests on other people's shows. YouTube's position was that it bans channels, not guests. Rogan said he fact-checked every crazy thing Jones said. He pushed back on some claims, and Spotify said it did not edit him. The economic fact was plainer than the guilt: the companies that had switched Jones off were not the only companies that could carry a voice, and the one that paid the most for a single voice decided a guest was not a channel.

In 2021, a Connecticut judge found Jones liable by default in the Sandy Hook defamation suits for discovery violations. And on March 12, 2021, Peter Thiel made his first contribution—$10 million—to a political committee called Protect Ohio Values, formed to back his former employee J. D. Vance for the United States Senate.

4. The Freshman and the Sink (2022–2024)

On February 15, 2022, nine Sandy Hook families settled with Remington for $73 million, paid by Remington's insurers—the maximum the policies covered—after the company's second bankruptcy. In April 2022, Peter Thiel gave two more checks to Protect Ohio Values: $3.5 million on April 18, and $1.5 million on April 25, bringing his total to $15 million for Vance's Senate race.

May 16, 2022. Just after midnight. A young man in Uvalde, Texas, named Salvador Ramos turned eighteen. The families who would later sue alleged that twenty-three minutes later he bought a rifle, a Daniel Defense DDM4V7. Eight days after that, on May 24, 2022, at Robb Elementary School, nineteen children and two teachers were killed.

That summer, *Top Gun: Maverick* opened in theaters with the Taiwan and Japan flags restored to the flight jacket; it did not open in mainland China. In August 2022, a Texas jury awarded the parents of six-year-old Jesse Lewis about $49 million to $50 million against Alex Jones, after Jones conceded at trial that the Sandy Hook shooting was "100 percent real" and parents testified that his followers had sent death and rape threats and harassed them in person. In Connecticut, a jury subsequently awarded the families $964 million against Jones, with the judge adding $473 million more. Jones filed for personal bankruptcy in late 2022.

On October 7, 2022, the United States government blocked the sale to China of the most powerful class of artificial intelligence chips—Nvidia's A100 and H100—and of the tools that make them. An export control is a freight rate for a machine: whoever writes it decides who gets to multiply. Nvidia's answer was a new pair of chips, the A800 and the H800, drawn to sit just outside the line.

October 27, 2022. Elon Musk completed the purchase of Twitter for $44 billion. When Musk was assembling the financing, Faiz Siddiqui records, Larry Ellison had been asked by text: "Any interest in participating in the Twitter deal?" Ellison replied: "Yes, of course," and put in about $1 billion (Siddiqui 2025). On the day the deal closed, Musk walked into the headquarters carrying a porcelain sink, so that he could say, "let that sink in" (Isaacson 2023). He fired top executives the same day. A post went out: "The bird is freed." Forty-four billion dollars was a little more than Stanford's entire endowment three years later. He had bought a microphone for more than the university whose file on him, when a lawyer later asked, could not be found [NEEDS SOURCE]. The price is the part that is settled. Days later, in November 2022, J. D. Vance won his Senate seat in Ohio.

November 29, 2022. A Stanford freshman named Theo Baker published, in the *Stanford Daily*, the first article on questions about Stanford president Marc Tessier-Lavigne's scientific papers (Baker 2026). It was not Baker's first article; a piece on campus social life had run on October 24. This one was the one that moved the board. He had been tipped, on November 1, toward PubPeer, a website where scientists comment on one another's papers. He took the images to forensic readers, among them Elisabeth Bik. Then he called his editor, Sam Catania.

"I called Sam, the Daily editor," Baker wrote. "'Holy shit,' he said. 'We need to get going. How quickly can you write this?'" (Baker 2026, 94).

The Board of Trustees announced a review the same day. The report arrived on July 17, 2023, from an outside scientific panel, transmitted by a lawyer, Mark Filip, in a letter addressed to the special committee at 365 Lasuen Street. The findings have to be stated the way they were written, because the slogan is cruder than the page. After reviewing more than 50,000 documents across twelve papers, the panel concluded that Tessier-Lavigne did not personally engage in research misconduct—the federal term for fabrication, falsification, or plagiarism. It did not find that he directed the manipulation, or that he knew of it when it happened. It also found that several papers exhibited manipulation of research data, and that in at least four of the five papers on which he was a principal author there was apparent manipulation by others in his lab. Each of those five, the panel wrote, had "serious flaws in the presentation of research data." The board, two days later, added its own sentence: when the problems surfaced, he took "insufficient steps" to correct the scientific record.

On July 19, 2023, Tessier-Lavigne resigned the presidency, effective August 31. Richard Saller became interim president on September 1. Tessier-Lavigne remained a tenured professor of biology. His letter to the campus said the report refuted the allegations of fraud and misconduct made against him, and that he was stepping down for the good of the university. Both the letter and the panel report are public. A freshman and a student paper had started a process that ended a presidency. The president left the office. He kept the chair. That was what governance looked like when it worked, and the limit of what it worked to do: it did not find a forger in the president's own hand; it found a lab, a record left uncorrected, and an office that could not survive the finding.

Down the hill from the campus, on Sand Hill Road, the Rosewood Hotel remained what Baker called in his book "the modern clubhouse for Silicon Valley" (Baker 2026, 120). It sits on land Stanford owns. A night starts in four figures. Baker's verdict was that the epicenter is a row of dull buildings whose advantage is being close to the campus down the hill. Under the hotel, the tenant may own the building. The university owns the dirt. The bank A. P. Giannini founded had left for Charlotte in 1998. The clubhouse stayed on the farm.

In October 2023, the United States closed the export-control loopholes on Nvidia's chips; Nvidia answered with the H20. In February 2024, Joe Rogan renewed his deal with Spotify for about

$250 million, according to the *Wall Street Journal*, and the new agreement was no longer exclusive.

May 24, 2024. On the two-year anniversary of the Robb Elementary shooting, Connecticut lawyer Josh Koskoff—the same lawyer who had sued Remington for the Sandy Hook families—filed two lawsuits. One, *Mata v. Meta*, was filed in Los Angeles Superior Court for about 45 family members against Meta, parent of Instagram and Facebook, and Activision Blizzard, maker of *Call of Duty*. The other, *Rubio v. Daniel Defense*, was filed in Uvalde County for 31 family members against the rifle manufacturer, and was later removed to federal court. Both complaints were filings, not findings. The Los Angeles complaint alleged that the companies raked in billions from the comfort of "Menlo Park and Santa Monica."

"Just 23 minutes after midnight on his 18th birthday, the Uvalde shooter bought an AR-15 made by a company with a market share of less than 1 percent," Koskoff said. "This three-headed monster knowingly exposed him to the weapon, conditioned him to see it as a tool to solve his problems, and trained him to use it."

That was the families' theory. The complaint alleged that in November 2021 Ramos had downloaded *Call of Duty: Modern Warfare*, and that the rifle appeared on the game's opening title page. It alleged that Daniel Defense's Instagram called a rifle with a holographic sight "totally murdered out," and captioned a photo of soldiers "Hunters Hunt." It alleged that within weeks he was back on the company's website, and that Instagram's own rules prohibited firearm advertising, and that Meta enforced other rules and not this one. Activision's lawyers answered that a war game is speech, and that the company owed no duty to stop a third person's violence.

In June 2024, the Stanford Internet Observatory became the fight on campus. Having published on election rumors, it had drawn lawsuits and subpoenas from the House Judiciary Committee. *Platformer* reported that the observatory was being dismantled: Alex Stamos had left, Renée DiResta's contract was not renewed, and staff were told to look for jobs. Stanford disputed the verb, stating that the work would continue under new leadership. Both statements are on the record. The office had become politically expensive, and the expense was not in dispute.

On July 7, 2024, David Ellison's Skydance signed an agreement to acquire Paramount. In October 2024, Joe Rogan recorded a three-hour interview with Donald Trump in Austin. On November 4, 2024, after posting an interview with Elon Musk, Rogan endorsed Trump, saying Musk had made "the most compelling case." Across the 2024 election cycle, as Faiz Siddiqui tallied it in December of that year, Musk spent more than $275 million through a super PAC called America PAC toward Trump and other Republicans—about eighteen times Thiel's $15 million for Vance (Siddiqui 2025). Trump and Vance won.

Later in November 2024, the satirical newspaper *The Onion*, backed by Sandy Hook families, won a bankruptcy auction for the company that owned Alex Jones's InfoWars. A federal

bankruptcy judge threw the result out over process and sent the families to state court. And in December 2024, federal export controls reached high-bandwidth memory, packaging, and chipmaking equipment.

5. The Arithmetic and the Studio (2025–2026)

January 20, 2025. Washington. The presidential inauguration was supposed to be outdoors, but the cold moved it inside, into the Capitol Rotunda, and the move scrambled the seating. What the photographs show, and what the Associated Press and the *Guardian* reported, was a row of technology executives seated with the president's family, ahead of several cabinet nominees: Elon Musk, Mark Zuckerberg, Jeff Bezos, and Sundar Pichai of Google. Tim Cook was in that company of guests. The head of TikTok was in the room, the Associated Press noted, beside the nominee to lead the intelligence community. The weather is why the chart changed. The pictures are what remain. Proximity is the evidence. It does not require a theory of anyone's heart.

That same day, an executive order renamed the United States Digital Service as the United States DOGE Service—the Department of Government Efficiency. It was not a cabinet department. It was a renamed office in the Executive Office of the President, plus a temporary organization scheduled to end on July 4, 2026. Agency heads were told to give it access to unclassified records and software. Musk's role was a special government employee, a category the law limits to about 130 days in a year.

The chokepoint now was the arithmetic. You could have the data, the graduate students, and the model on a whiteboard, and still be stopped by whoever owned the machines that did the multiplying. For its fiscal year ended January 26, 2025, Nvidia reported audited revenue of $130.5 billion, up 114 percent from $60.9 billion the year before. The data-center line—the machines sold for artificial intelligence work—was $115.2 billion. One hundred thirty point five billion dollars was a little more than three times Stanford's endowment. Larry Page, at a party in the late 1990s, had said he was not really interested in search; he was making an AI. The training ground he described now had a toll booth, and the booth was a chip. A headline number in the hundreds of billions announced for new data centers was a plan, not money spent. The gate had moved from the road to the chip.

Also in January 2025, OpenAI told the *Financial Times* it had seen evidence that a Chinese lab, DeepSeek, had used distillation on OpenAI's models, against OpenAI's terms—training a new model on the answers of an older one, so the student copied the teacher's results without the teacher's data. OpenAI did not publish the evidence. A House committee later called unauthorized distillation "highly likely," relying on what OpenAI had provided.

In April 2025, selling Nvidia's H20 chip into China required a federal license, which Nvidia's own Form 8-K filing treated as a ban in practice. On May 28, 2025, the White House confirmed to Reuters that Elon Musk's off-boarding from the government would begin that night. He had led

an advisory wrecking crew, and the 130-day clock ran out. The companies remained his. The job did not. Untouchability is the distance between the person and the instrument: when the temporary chair ends, the corporations are where he left them. The statute makes the advisor temporary. The cap table does not.

In July 2025, the Skydance–Paramount merger cleared its path. On July 2, 2025, Paramount paid $16 million to settle Trump's lawsuit over a *60 Minutes* broadcast. On July 17, days after CBS host Stephen Colbert called the settlement, in his words, "a big fat bribe," CBS canceled *The Late Show*, announcing that its last season would end in May 2026. On July 24, 2025, the Federal Communications Commission approved the Paramount acquisition, valued at about $8 billion. It closed on August 7, 2025, as Paramount Skydance, at a figure Reuters put at $8.4 billion. That same month, federal licenses for Nvidia's H20 chip resumed.

On August 31, 2025, Stanford closed its fiscal year with an endowment of $40.8 billion—about fifty-five times the founding gift, counted in today's money. In that fiscal year it paid out $1.9 billion. Undergraduate tuition in the same report was $65,127, with nearly half of undergraduates receiving need-based aid.

On September 5, 2025, Judge William F. Highberger of Los Angeles Superior Court dismissed Meta from the Uvalde families' lawsuit. He wrote that holding the company liable would impose "limitless liability" on a creator or a disseminator for the violent acts of others anywhere in the world. Meta's defense firm, Gibson Dunn, later summarized the ruling this way: the shooter was the proximate cause, and Meta's conduct was "so far attenuated" from the harm that it was only "negligible or theoretical." The same day, Highberger denied Activision's anti-SLAPP motion to throw the case out as an attack on free speech; Activision then argued it owed no duty, on the same ground as Meta. Highberger reaffirmed Meta's dismissal in December 2025, and the families appealed. The families had argued that Menlo Park built the hallway that led a teenager to a rifle. The court, so far, said a hallway is not a cause.

On October 6, 2025, Paramount Skydance bought *The Free Press*, reported at about $150 million, and named its founder, Bari Weiss, editor in chief of CBS News, reporting directly to David Ellison. On October 14, 2025, the United States Supreme Court refused to hear Alex Jones's appeal of the Connecticut Sandy Hook judgment, which appeals had brought to about $1.25 billion. On November 17, 2025, in federal court in California, a San Jose resident named Chenguang Gong was sentenced to 46 months in prison for taking missile-detection infrared sensor technology, after a run of applications to Chinese talent programs between 2014 and 2022.

On December 5, 2025, Netflix—the company Reed Hastings, Stanford master's in computer science, class of 1988, had founded with Marc Randolph on August 29, 1997, in Scotts Valley before moving to Los Gatos—agreed to buy Warner Bros., HBO, and HBO Max at an enterprise value of $82.7 billion. Three days later, in December 2025, the president allowed Nvidia to sell a

newer chip, the H200, into China if the United States government took a 25 percent cut of the sale.

January 29, 2026. A federal courtroom in San Francisco. Linwei Ding, a former Google engineer who had come to the United States in 2010 for a master's degree, was convicted on seven counts of economic espionage and seven counts of stealing trade secrets. The government said he had uploaded more than 500 files—more than 2,000 pages—of Google's AI infrastructure. While Ding was in China, an intern back in Mountain View had swiped Ding's badge so the building would record him as at his desk. The exhibit built for the jury was Ding's own pitch for a company he called Zhisuan: "We have experience with Google's ten-thousand-card computational power platform; we just need to replicate and upgrade it." The Department of Justice called the verdict the first AI-related economic-espionage conviction.

Two weeks later, on February 12, 2026, OpenAI sent a memo to the House committee alleging that DeepSeek had continued its distillation practices through obfuscated routers and resellers. No court had found this; it remained an allegation. And on February 26, 2026, the Warner Bros. Discovery board declared Paramount Skydance's takeover bid superior to Netflix's. Netflix did not raise.

In April 2026, *The Onion* announced a new deal to license and then buy the InfoWars brand under a Texas state court receiver order; its chief executive said the Sandy Hook families had not received a penny, and a Texas appeals court stayed the transfer while Jones appealed. By May 2026, about ten Chinese firms were cleared to buy Nvidia's H200 chips.

Then, on August 20, 2026, U.S. District Judge Vince Chhabria vacated all seven espionage counts against Linwei Ding, ruling there was insufficient evidence that Ding knew or intended that what he did would benefit the Chinese government. The seven trade-secret counts stood. On September 1, 2026, Ding was sentenced to twelve months minus a day in custody, a $25,000 fine, and restitution of more than $189,000. Matt Sheehan had noted the caution that belongs next to such cases: a string of Chinese American researchers had been charged with espionage only to have the cases prove unfounded (Sheehan 2019, 54–55). The Ding case was what those anxieties collapsed into when a court did the sorting. The theft was proved. The spy charge was not. A badge swipe is not a nationality.

Which left the argument people repeated across the Valley: that China can manufacture, but cannot make California. The sources line up on both sides. The case for a gap comes from the scholar Jeffrey Ding, no relation to the engineer, who argues that China's weakness is diffusion—the spread of a technology through the whole economy—more than the moment of invention, while a RAND study finds the links from Chinese research to Chinese industry weaker than in the United States. The case against the slogan is a measurement. In September 2024 the Information Technology and Innovation Foundation put China ahead or level in nuclear power, electric vehicles, and batteries; near the frontier in robotics, artificial intelligence, and quantum; and behind in semiconductors and biopharma. Its 2026 Hamilton Index had China

leading seven of ten advanced industries. Dan Wang, in his book *Breakneck*, as rendered in a *Foreign Affairs* review, reversed the premise: China is an engineering state building at breakneck speed, set against the United States' lawyerly society blocking everything it can, good and bad. And Sheehan noted the rhyme with the 1980s, when Americans said Japan could copy and could not invent, while Japanese investment kept "generating jobs, income, and even innovation" (Sheehan 2019, 214). China builds at a scale California no longer can. Whether it can grow the loop on this peninsula—risk capital, an open university, and immigrant talent on the same road—is the question that remains open.

In Texas, on August 21, 2026, the state's Third Court of Appeals cut the Sandy Hook parents' $49 million jury award against Alex Jones to about $6 million—$4.1 million in compensatory damages plus about $1.5 million in punitive damages. Jones said he would appeal to the Texas Supreme Court; the families' lawyer, Mark Bankston, called the cut "irrelevant." As of August 2026, Jones had paid nothing on either state's judgment, and on September 9, 2026, the Texas Supreme Court refused to lift the stay on the sale of InfoWars to *The Onion*. Jones had already left the InfoWars name behind.

October 6, 2026. Justice Elena Kagan denied a last-minute petition, and Paramount Skydance's purchase of Warner Bros. Discovery closed. The combined company was renamed Skydance, with David Ellison and Ynon Kreiz as co-chief executives. The Ellisons and RedBird held all the voting shares. Forty-seven billion dollars of the new equity came from the Ellisons, RedBird, and the sovereign wealth funds of Saudi Arabia, Abu Dhabi, and Qatar.

Tie it to the start. At Ampex, Larry Ellison had worked on a failed CIA contract called Project Oracle. Three employees left and started a company in 1977, Larry Ellison with Bob Miner and Ed Oates; on the third name, they called it Oracle, and its first customer was the CIA. In the Reagan years, Malcolm Harris writes, the agency mailed checks in plain white envelopes and did not mind that the first versions did not work; an engineer recalled, "We bought this thing knowing damn well it wasn't going to work. We're buying an idea" (Harris 2023, ch. 4). A century before that, in January 1880, Eadweard Muybridge had shown Leland Stanford's horses moving on a zoopraxiscope at the Stanford mansion in San Francisco—"The age of movies began in California, not in Hollywood but in the Bay," Harris writes (Harris 2023, ch. 1.3)—and in 1914 Cecil B. DeMille had shot a western, *The Squaw Man*, in a barn at Selma and Vine for the Jesse L. Lasky Feature Play Company, which became Paramount. Now it sat in one portfolio: one family, a database bought as an idea, and then CBS, CNN, HBO, Paramount, and Warner Bros.

Rail controlled movement. Stanford controls succession.

It is. And it connects to me. I did not inherit a railroad, or a fund, or a studio. I inherited the habit of treating the names as scenery. Serra. Jordan. Terman. Page Mill. Sand Hill. Once you hear them as a succession, you cannot go back to hearing them as scenery. The walk after this is only the sidewalk, in the order the names were laid down.

6. Coda: The Street Names

Dusk on the farm. The oaks go black first. The signs are already lit, which is the fact this walk is heading toward, and it will not be hurried. Read them in the order the layers were laid down. You can do it on foot in an afternoon. Four centuries of proprietors, one sidewalk.

Gálvez. José de Gálvez, the inspector general who sent the expedition north. The administrator, not the priest who walked. The people already living on this ground do not have a sign at this corner.

Serra. Junípero Serra's name came off the mall across the front of the campus. Serra Mall became Jane Stanford Way. The street that kept his name is the one that runs out.

Lasuen. Fermín de Lasuén, Serra's successor, who carried the missions farther up the coast, including La Purísima. The scientific panel's letter, in 2023, was addressed to 365 Lasuen Street.

El Camino Real. The royal road, still the main street of the Peninsula. In October 2013 the buyers' bus left from a lot on this road, purple lights in the ceiling, a brochure in the aisle.

Alvarado. Juan Bautista Alvarado, the governor who in 1836 led a revolt against Mexico City's grip on California. David Starr Jordan, the first president, put the name on the row.

Stanford Avenue. The farm, the railroad fortune, the grant that could not be sold. The family could lease it. They could not cash it out.

Jane Stanford Way. The rename of Serra Mall took effect on October 7, 2019, and the university dedicated it on November 14. The official address became 450 Jane Stanford Way. Serra Street, from Campus Drive East out to El Camino, kept the old name. The co-founder is on the front door. The priest is on the exit.

Jordan Hall. The first president's name came off the building in October 2020. It is Building 420 now.

Hoover Tower. The archive, and the institution where a secretary of state came back to sit, and then sat on a blood-testing board. On the 2013 brochure, this tower is a guard post on the Great Wall.

Terman. Frederick Terman's engineering buildings, at the edge of the farm he helped lease to industry. The research park is the lease, made architectural.

Page Mill. William Page's lumber road, before it was the spine of the research park. It still runs from the timber to the chip.

Sand Hill. Four minutes from the Quad. The gate. The Rosewood, on the university's land, calling itself the modern clubhouse. A company that cannot get a meeting in one of the low buildings does not merely lack money. It lacks the next introduction.

Stand where the royal road meets the farm and the order is the order of the history. A mission administrator. A priest. A governor. A railroad family. A co-founder, put on the front door in 2019, while the priest was left on the road out. A tower that a brochure redrew as a wall. A lumber road that became a chip road. A venture road that became a clubhouse on leased ground. The signs do not argue. They accumulate.

The golden-spike photograph is the same kind of document, made a century and a half before two mayors stood in the middle of a new bridge and said, "This time the Chinese were in the middle." Who is in the frame. Who is not. A street sign is that choice, cast in metal and lifted out of reach.

And the names that are not here. The Muwekma Ohlone, whose land the campus occupies—little on the map, and what recognition exists is recent and small. The Chinese workers of the Central Pacific, absent from the golden-spike photograph, and absent from any building at the scale of the work they did. The people sterilized under the 1909 law: nothing. The farm workers: Cesar Chavez has streets in California, real ones, and they are mostly in the Central Valley and in city neighborhoods, not on this walk. The list is the record. What a place commemorates is what it decided to remember, and this peninsula decided to remember the people who held the title.

Read a sign as a document and it answers four questions: Who. When. For what. And who decided. British streets carry empire. Parisian streets carry writers. Most American streets carry presidents and trees. This peninsula carries the people who held the title, in the order they held it.

Subliminal does not mean hidden. It means beneath the threshold at which something is consciously registered. The street signs are twelve feet up and lit at night. It was never a secret. It was only subliminal.